AL-BALQA'A APPLIED UNIVERSITY
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جامعة البلقاء التطبيقية
كلية الهندسة التكنولوجية

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Introduction

Many control dynamic systems, whether they are mechanical, electrical, thermal or hydraulic, may be described by differential equations. These equations can be obtained d by utilizing physical laws governing a particular system, for example, Newton's laws for mechanical systems, and Kirchhoff's laws for electrical systems, etc. The setoff differential equations are called a mathematical model. The response of a dynamic system to an in put signal or to a disturbance signal can be obtained if these equations are solved using different computer tools.

1 Objectives

At the end of this course, the student should be able to:

-Simulate any control system given by a transfer function.
-Solve linear and nonlinear differential equations
-Investigate the transient response and steady state response of control systems.
-Use feedback in control systems.
-Determine the stability of control systems.
-Determine the effects of different control actions on system performance.
-Investigate the frequency response of control system

Simulink

Simulink is a software package for modeling, simulating, and analyzing dynamic Systems. It supports linear and nonlinear systems, modeled in continuous time, Sampled time, or a hybrid of the two. Systems can also be multirate, i.e., have Different parts that is sampled or updated at different rates.

It allows you to represent systems as block diagrams that you build using your Mouse to connect blocks and your keyboard to edit block parameters. Stateflow is Part of this environment. The State flow block is a masked Simulink model. State flow Builds an S-function that corresponds to each State flow machine. This S-function is The agent Simulink interacts with for simulation and analysis.

The control behavior that State flow models complement the algorithmic behavior Modeled in Simulink block diagrams. By incorporating State flow diagrams into Simulink models, you can add event-driven behavior to Simulink simulations. You Create models that represent both data and control flow by combining State flow Blocks with the standard Simulink block sets. These combined models are simulated Using Simulink.

The Using Simulink document describes how to work with Simulink. It explains how To manipulate Simulink blocks, access block parameters, and connect blocks to Build models. It also provides reference descriptions of each block in the standard Simulink libraries.

Experiment No 1

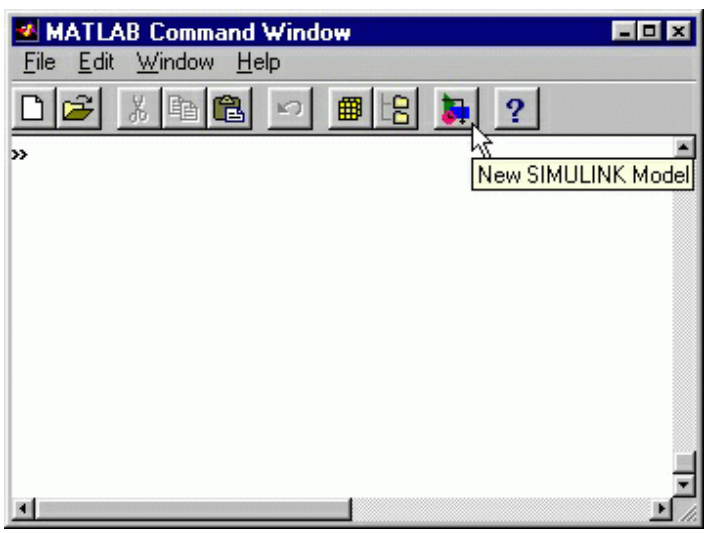
Control Tutorials for Simulink

Simulink is a graphical extension to MATLAB for modeling and simulation of systems. In Simulink, systems are drawn on screen as block diagrams. Many elements of block diagrams are available, such as transfer functions, summing junctions, etc., as well as virtual input and output devices such as function generators and oscilloscopes. Simulink is integrated with MATLAB and data can be easily transfered between the programs. In these tutorials, we will apply Simulink to the examples from the MATLAB tutorials to model the systems, build controllers, and simulate the systems. Simulink is supported on Unix, Macintosh, and Windows environments

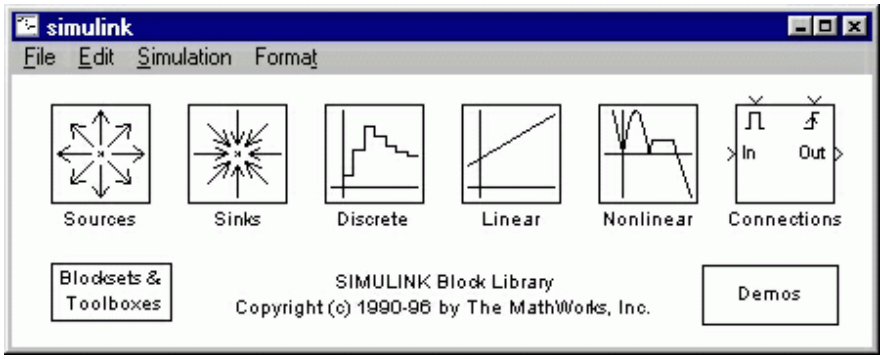
The idea behind these tutorials is that you can view them in one window while running Simulink in another window. System model files can be downloaded from the tutorials and opened in Simulink. You will modify and extend these system while learning to use Simulink for system modeling, control, and simulation. Do not confuse the windows, icons, and menus in the tutorials for your actual Simulink windows. Most images in these tutorials are not live - they simply display what you should see in your own Simulink windows. All Simulink operations should be done in your Simulink windows. **Starting Simulink**

Simulink is started from the MATLAB command prompt by entering the following command: `simulink`

Alternatively, you can hit the New Simulink Model button at the top of the MATLAB command window as shown below:



When it starts, Simulink brings up two windows. The first is the main Simulink window, which appears as:



A new model can be created by selecting **New** from the **File** menu in any Simulink window

Basic Elements

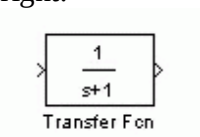
There are two major classes of items in Simulink: **blocks** and **lines**. Blocks are used to generate, modify, combine, output, and display signals. Lines are used to transfer signals from one block to another.

Blocks

There are several general classes of blocks:

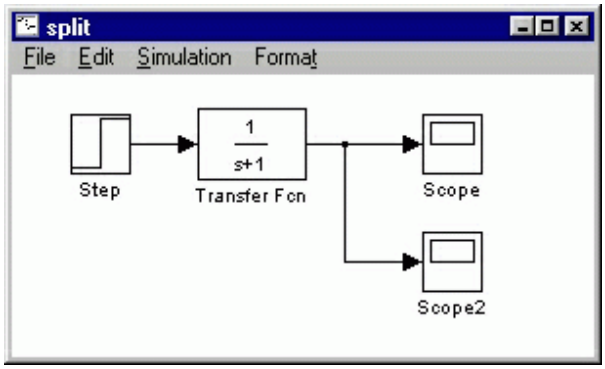
- Sources: Used to generate various signals
- Sinks: Used to output or display signals
- Discrete: Linear, discrete-time system elements (transfer functions, state-space models, etc.)
- Linear: Linear, continuous-time system elements and connections (summing junctions, gains, etc.)
- Nonlinear: Nonlinear operators (arbitrary functions, saturation, delay, etc.)
- Connections: Multiplex, Demultiplex, System Macros, etc.

Blocks have zero to several input terminals and zero to several output terminals. Unused input terminals are indicated by a small open triangle. Unused output terminals are indicated by a small triangular point. The block shown below has an unused input terminal on the left and an unused output terminal on the right.



Lines

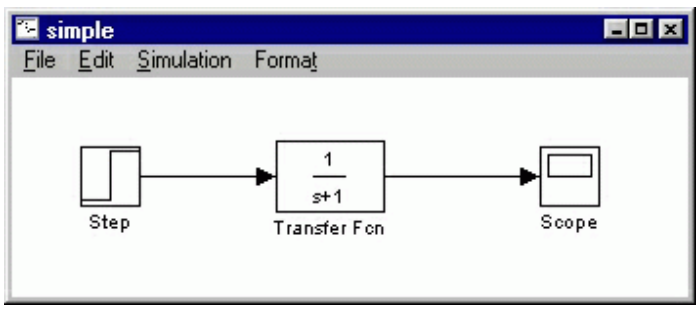
Lines transmit signals in the direction indicated by the arrow. Lines must always transmit signals from the output terminal of one block to the input terminal of another block. On exception to this is a line can tap off of another line, splitting the signal to each of two destination blocks, as shown below



Lines can never inject a signal *into* another line; lines must be combined through the use of a block such as a summing junction.

A signal can be either a scalar signal or a vector signal. For Single-Input, Single-Output systems, scalar signals are generally used. For Multi-Input, Multi-Output systems, vector signals are often used, consisting of two or more scalar signals. The lines used to transmit scalar and vector signals are identical. The type of signal carried by a line is determined by the blocks on either end of the line.

Simple Example

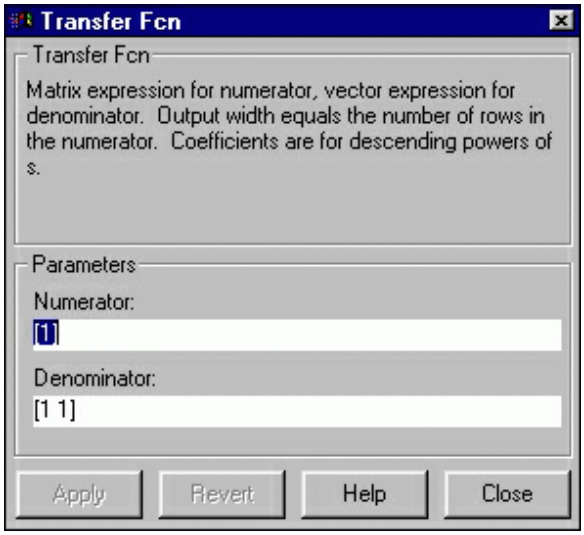


The *simple* model consists of three blocks: Step, Transfer Fcn, and Scope. The Step is a **source block** from which a step input signal originates. This signal is transfered through the **line** in the direction indicated by the arrow to the Transfer Function **linear block**. The Transfer Function modifies its input signal and outputs a new signal on a line to the Scope. The Scope is a **sink block** used to display a signal much like an oscilloscope.

There are many more types of blocks available in Simulink, some of which will be discussed later. Right now, we will examine just the three we have used in the *simple* model.

Modifying Blocks

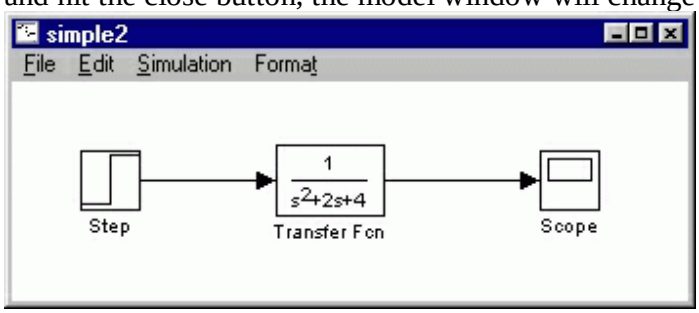
A block can be modified by double-clicking on it. For example, if you double-click on the "Transfer Fcn" block in the *simple* model, you will see the following dialog box.



This dialog box contains fields for the numerator and the denominator of the block's transfer function. By entering a vector containing the coefficients of the desired numerator or denominator polynomial, the desired transfer function can be entered. For example, to change the denominator to s^2+2s+1, enter the following into the denominator field:

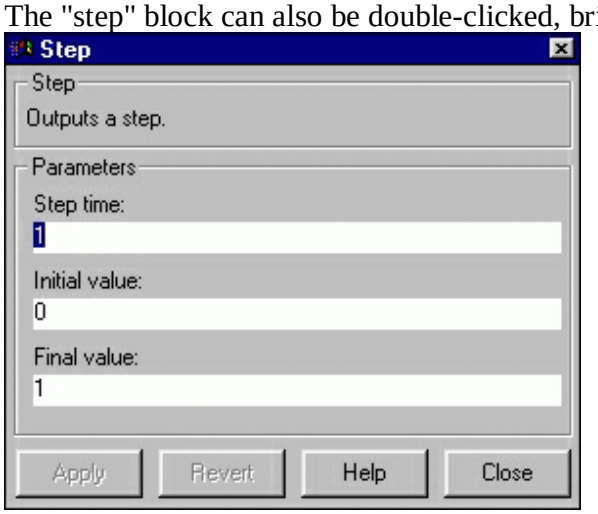
```
[1 2 1]
```

and hit the close button, the model window will change to the following,



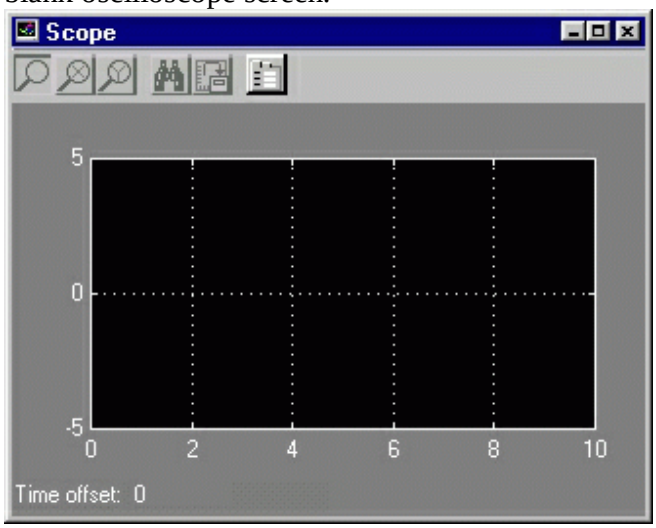
which reflects the change in the denominator of the transfer function.

The "step" block can also be double-clicked, bringing up the following dialog box.



The default parameters in this dialog box generate a step function occurring at time=1 sec, from an initial level of zero to a level of 1. (in other words, a unit step at t=1). Each of these parameters can be changed. Close this dialog before continuing.

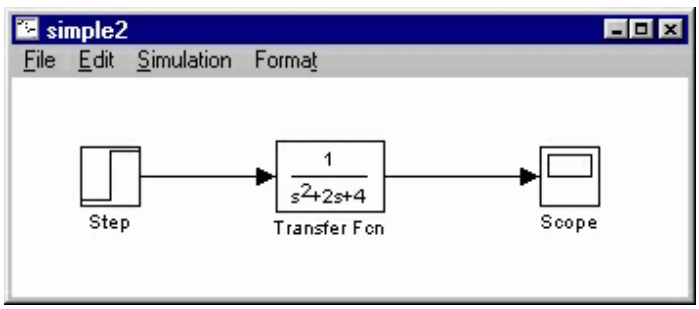
The most complicated of these three blocks is the "Scope" block. Double clicking on this brings up a blank oscilloscope screen.



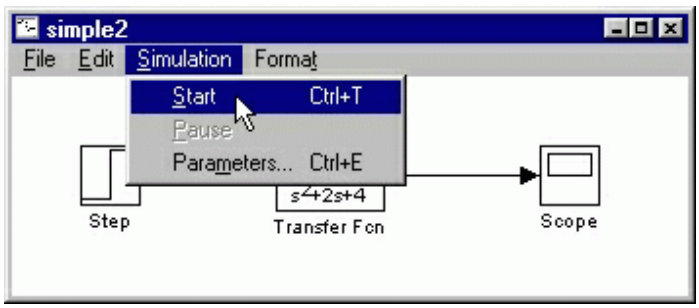
When a simulation is performed, the signal which feeds into the scope will be displayed in this window. Detailed operation of the scope will not be covered in this tutorial. The only function we will use is the autoscale button, which appears as a pair of binoculars in the upper portion of the window.

Running Simulations

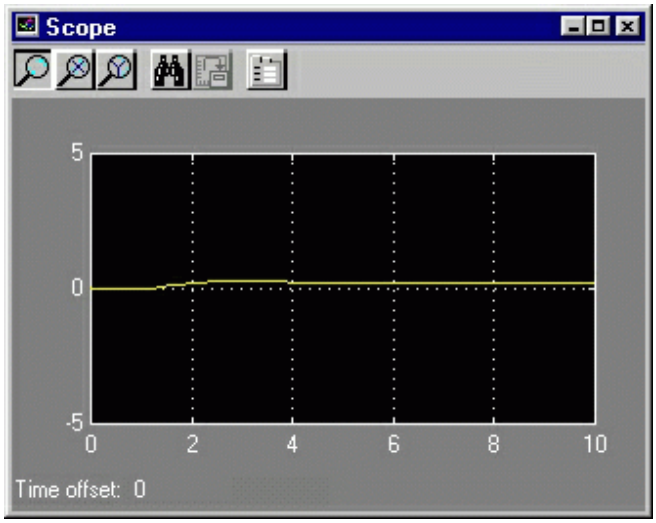
To run a simulation, we will work with the following model



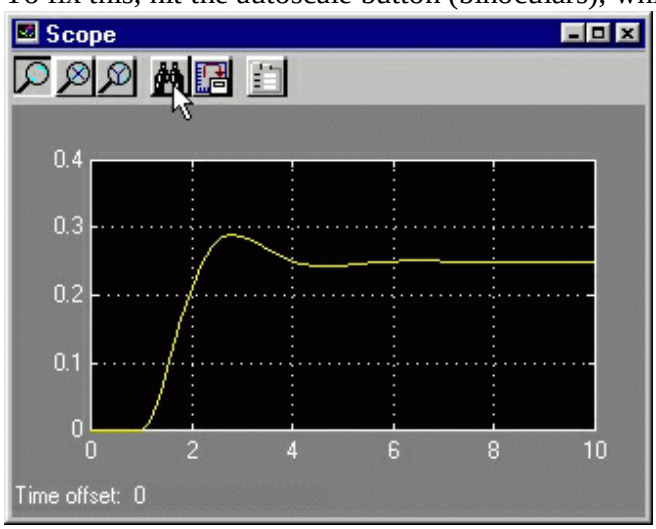
Before running a simulation of this system, first open the scope window by double-clicking on the scope block. Then, to start the simulation, either select **Start** from the **Simulation** menu (as shown below) or hit Ctrl-T in the model window.



The simulation should run very quickly and the scope window will appear as shown below.



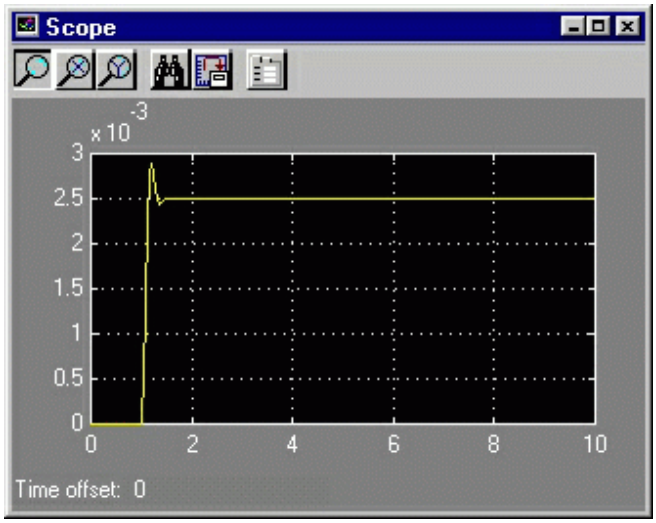
Note that the simulation output (shown in yellow) is at a very low level relative to the axes of the scope. To fix this, hit the autoscale button (binoculars), which will rescale the axes as shown below.



Note that the step response does not begin until t=1. This can be changed by double-clicking on the "step" block. Now, we will change the parameters of the system and simulate the system again. Double-click on the "Transfer Fcn" block in the model window and change the denominator to

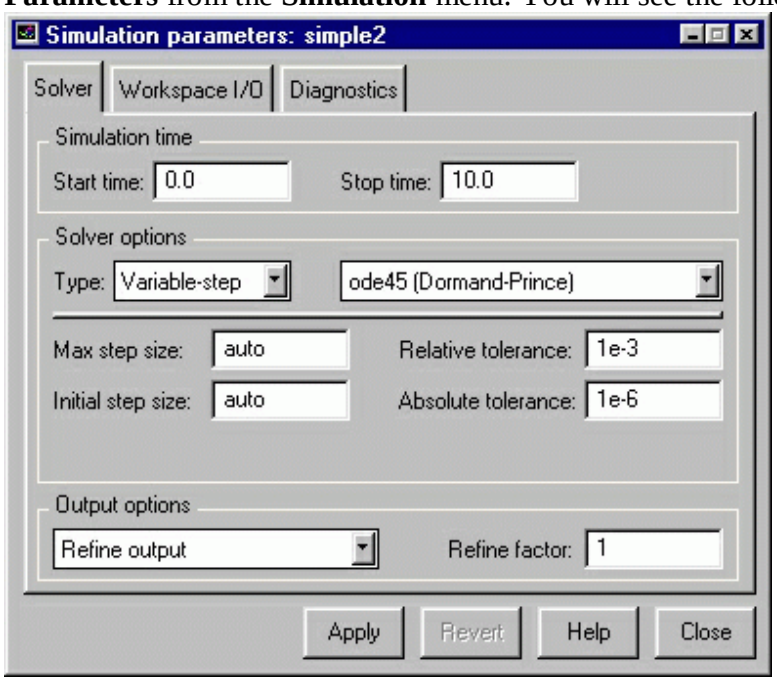
```
[1 20 400]
```

Re-run the simulation (hit Ctrl-T) and you should see what appears as a flat line in the scope window. Hit the autoscale button, and you should see the following in the scope window.



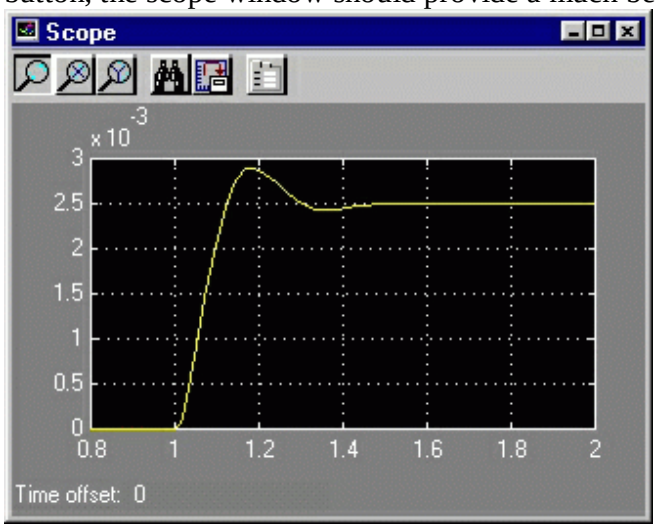
Notice that the autoscale button only changes the vertical axis. Since the new transfer function has a very fast response, it it compressed into a very narrow part of the scope window. This is not really a problem with the scope, but with the simulation itself. Simulink simulated the system for a full ten seconds even though the system had reached steady state shortly after one second.

To correct this, you need to change the parameters of the simulation itself. In the model window, select **Parameters** from the **Simulation** menu. You will see the following dialog box.



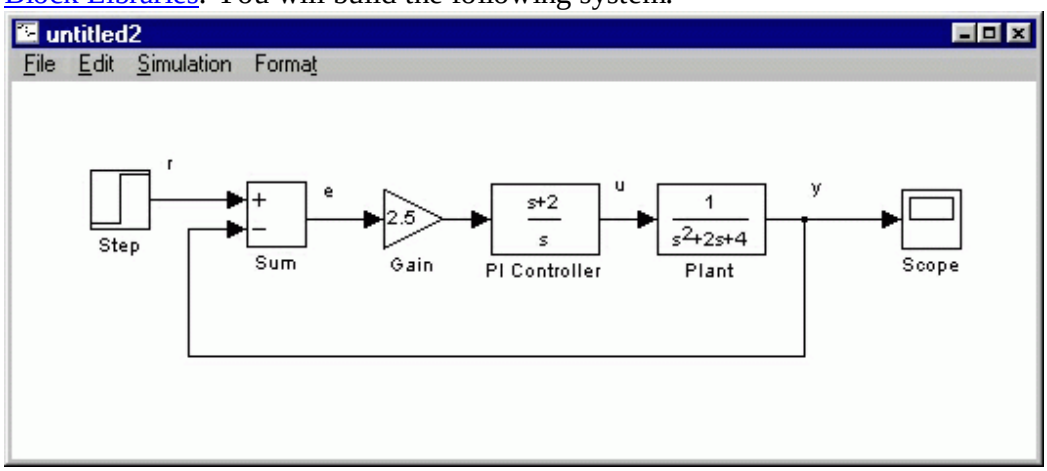
There are many simulation parameter options; we will only be concerned with the start and stop times, which tell Simulink over what time period to perform the simulation. Change **Start time** from 0.0 to 0.8 (since the step doesn't occur until t=1.0. Change **Stop time** from 10.0 to 2.0, which should be only shortly after the system settles. Close the dialog box and rerun the simulation. After hitting the autoscale

button, the scope window should provide a much better display of the step response as shown below.



Building Systems

In this section, you will learn how to build systems in Simulink using the building blocks in Simulink's Block Libraries. You will build the following system.



First you will gather all the necessary blocks from the block libraries. Then you will modify the blocks so they correspond to the blocks in the desired model. Finally, you will connect the blocks with lines to form the complete system. After this, you will simulate the complete system to verify that it works.

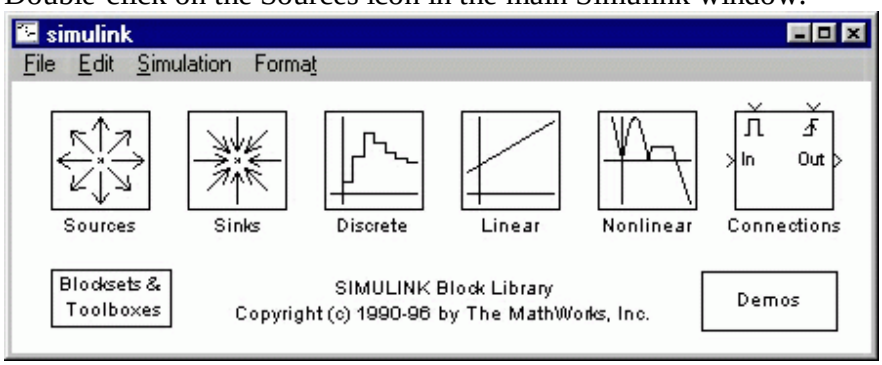
Gathering Blocks

Follow the steps below to collect the necessary blocks:

- Create a new model (**New** from the **File** menu or Ctrl-N). You will get a blank model window.

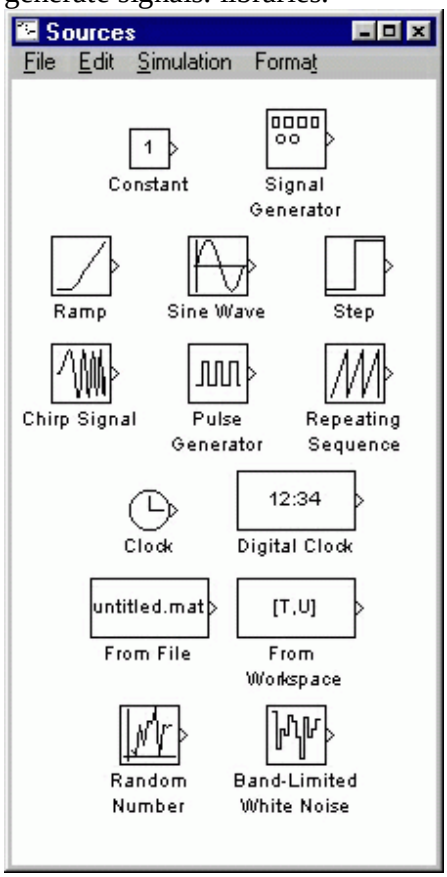


- Double-click on the Sources icon in the main Simulink window.

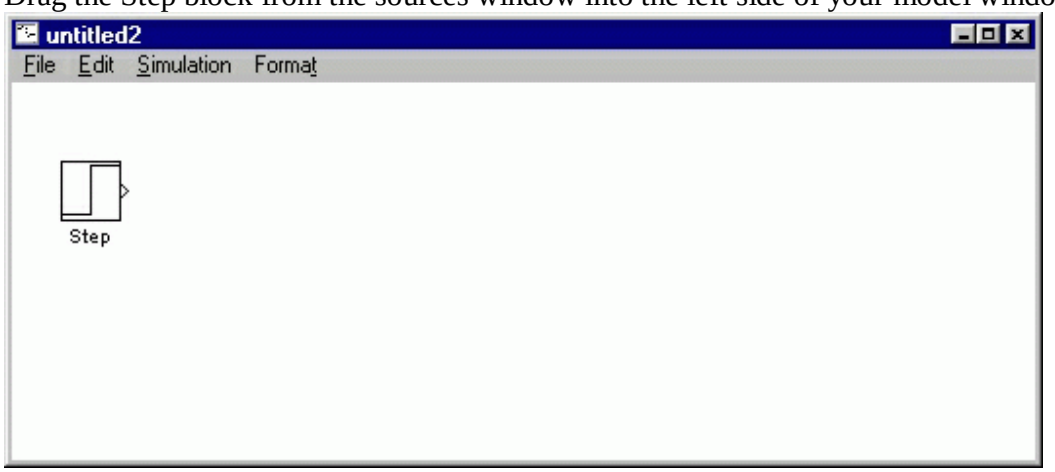


This opens the Sources window which contains the Sources Block Library. Sources are used to

generate signals. libraries.

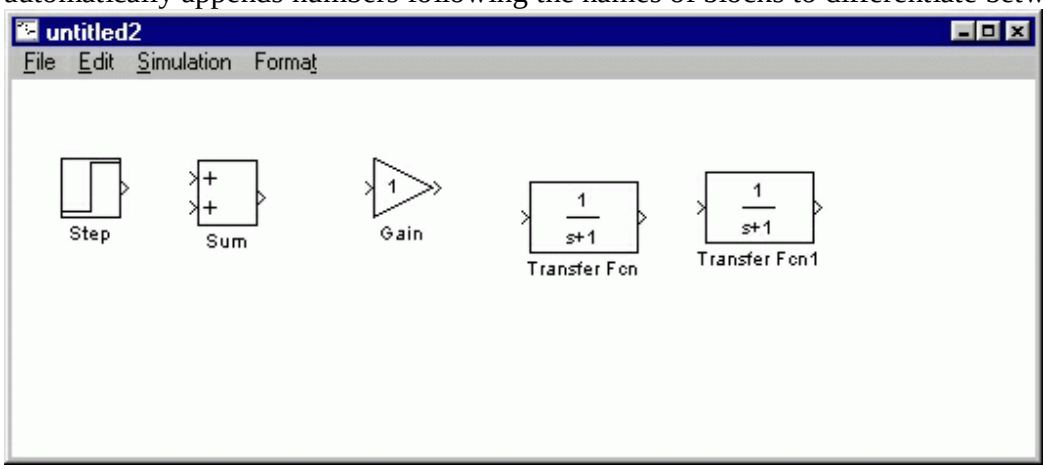


- Drag the Step block from the sources window into the left side of your model window.

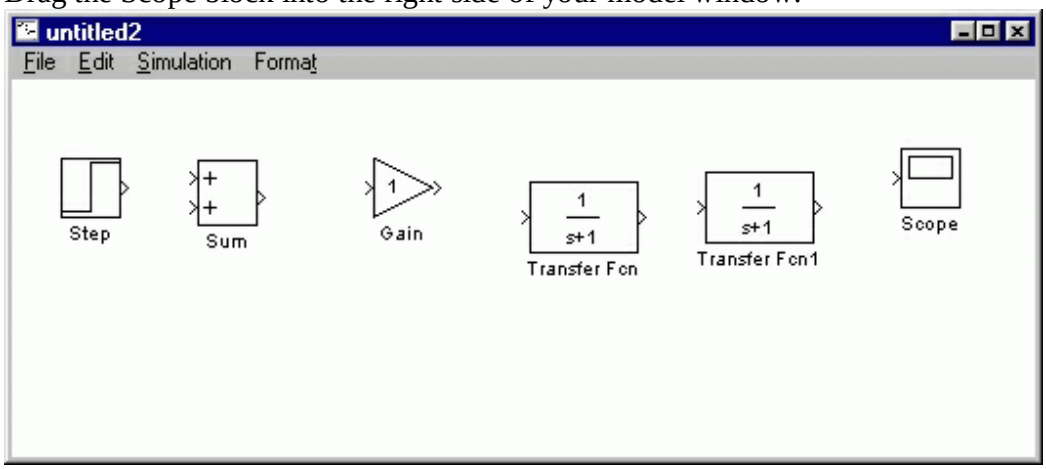


- Double-click on the Linear icon in the main Simulink window to open the Linear Block Library window.
- Drag the Sum, Gain, and two instances of the Transfer Fcn (drag it two times) into your model window arranged approximately as shown below. The exact alignment is not important since it can be changed later. Just try to get the correct relative positions. Notice that the second Transfer Function block has a 1 after its name. Since no two blocks may have the same name, Simulink

automatically appends numbers following the names of blocks to differentiate between them.



- Double-click on the Sinks icon in the main Simulink window to open the Sinks window.
- Drag the Scope block into the right side of your model window.

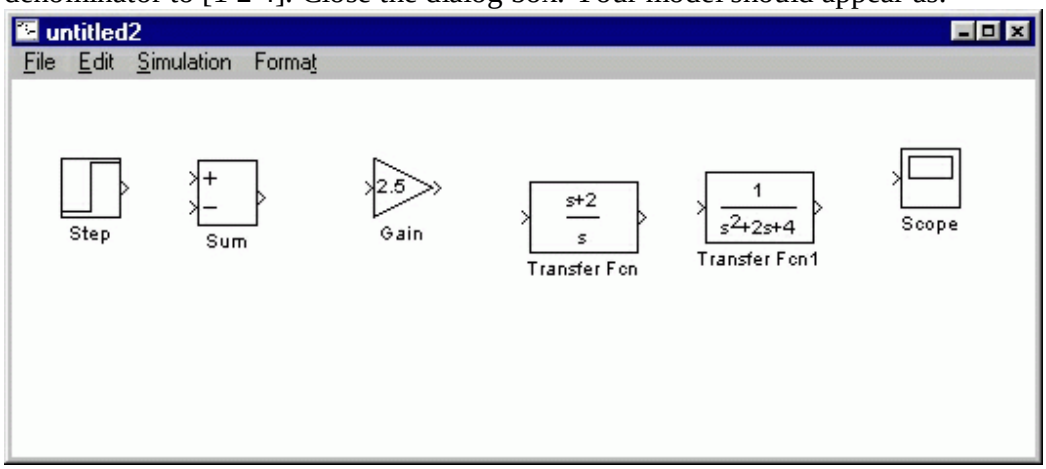


Modify Blocks

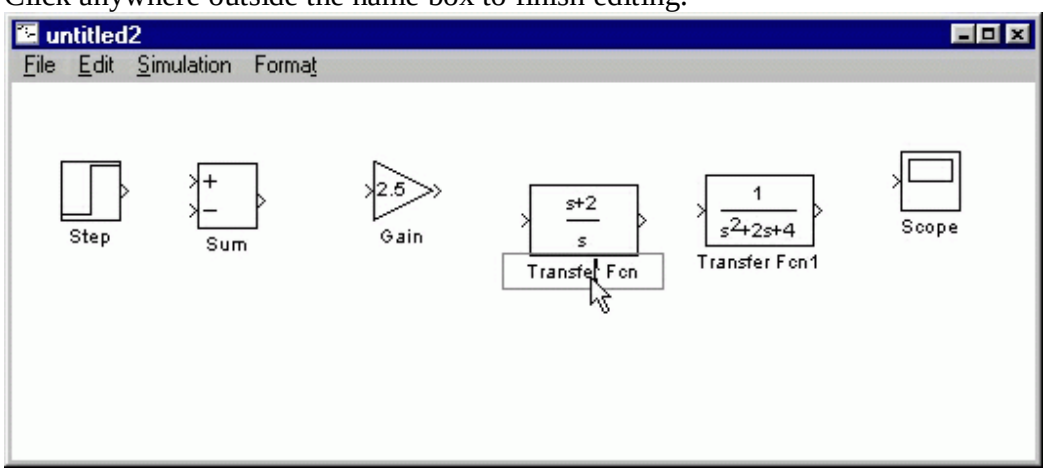
Follow these steps to properly modify the blocks in your model.

- Double-click your Sum block. Since you will want the second input to be subtracted, enter +- into the list of signs field. Close the dialog box.
- Double-click your Gain block. Change the gain to 2.5 and close the dialog box.
- Double-click the leftmost Transfer Function block. Change the numerator to [1 2] and the denominator to [1 0]. Close the dialog box.

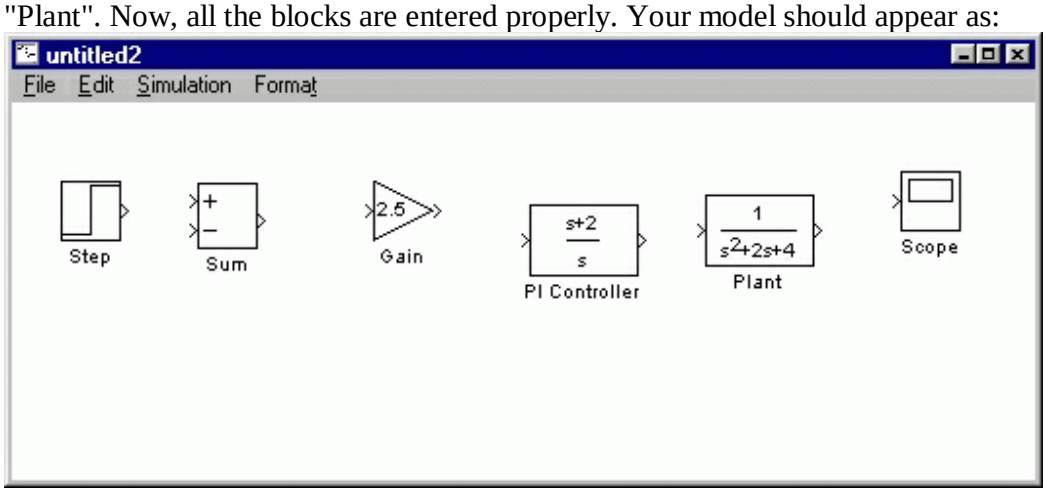
- Double-click the rightmost Transfer Function block. Leave the numerator [1], but change the denominator to [1 2 4]. Close the dialog box. Your model should appear as:



- Change the name of the first Transfer Function block by clicking on the words "Transfer Fcn". A box and an editing cursor will appear on the block's name as shown below. Use the keyboard (the mouse is also useful) to delete the existing name and type in the new name, "PI Controller". Click anywhere outside the name box to finish editing.



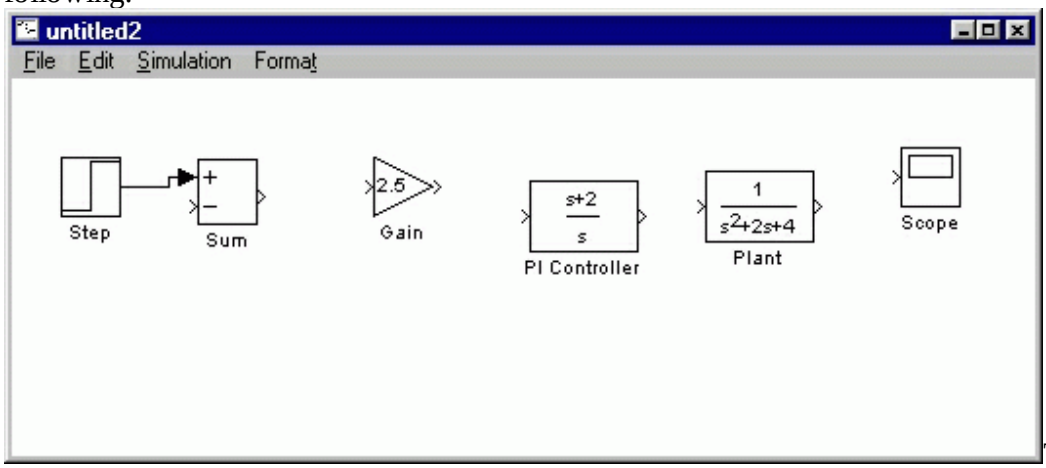
- Similarly, change the name of the second Transfer Function block from "Transfer Fcn1" to "Plant". Now, all the blocks are entered properly. Your model should appear as:



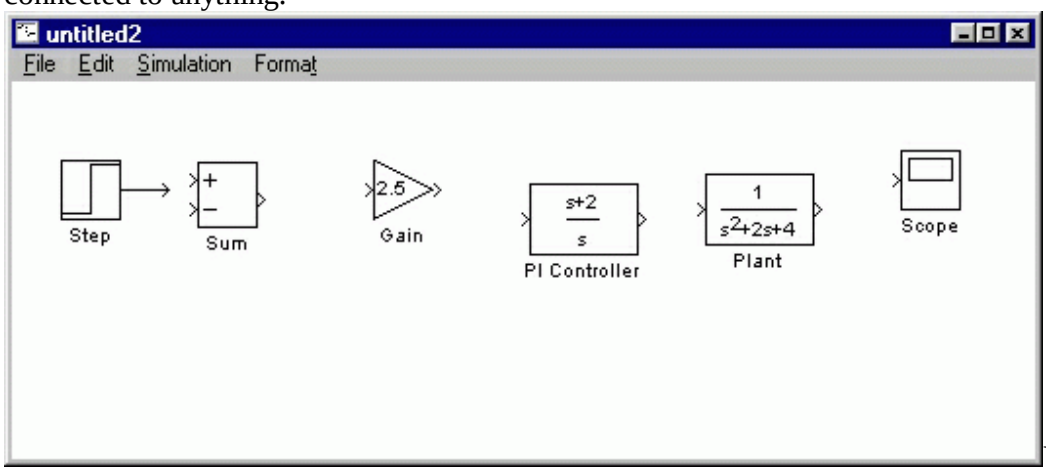
Connecting Blocks with Lines

Now that the blocks are properly laid out, you will now connect them together. Follow these steps.

- Drag the mouse from the output terminal of the Step block to the upper (positive) input of the Sum block. Let go of the mouse button only when the mouse is right on the input terminal. Do not worry about the path you follow while dragging, the line will route itself. You should see the following.



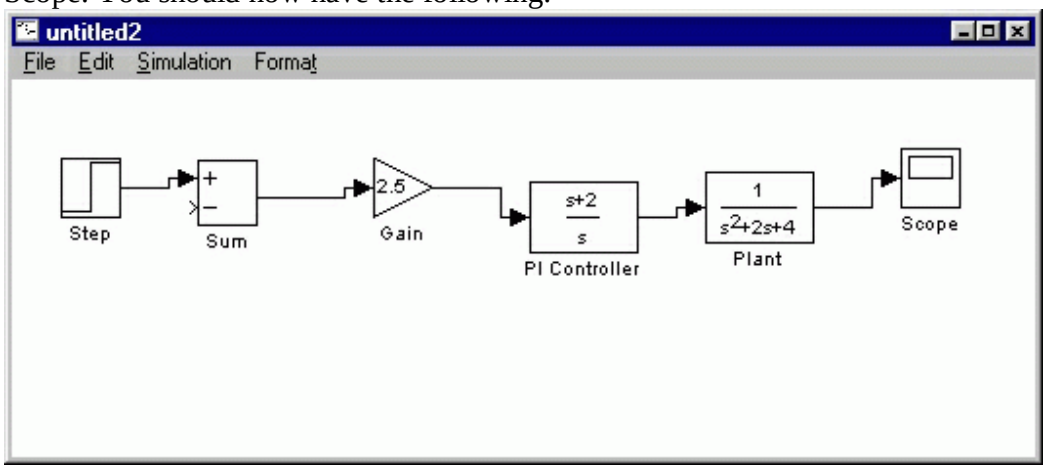
The resulting line should have a filled arrowhead. If the arrowhead is open, as shown below, it means it is not connected to anything.



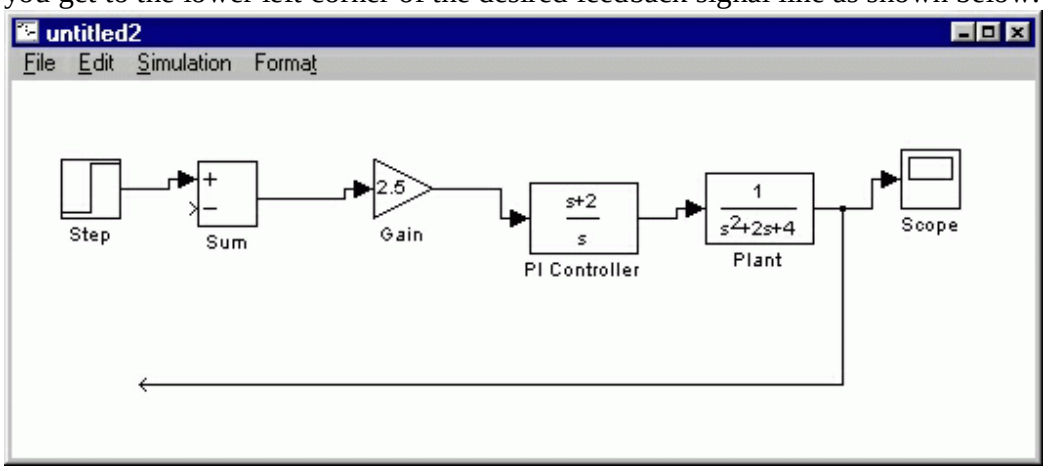
You can continue the partial line you just drew by treating the open arrowhead as an output terminal and drawing just as before. Alternatively, if you want to redraw the line, or if the line connected to the wrong terminal, you should delete the line and redraw it. To delete a line (or any other object), simply click on it to select it, and hit the delete key.

- Draw a line connecting the Sum block output to the Gain input. Also draw a line from the Gain to the PI Controller, a line from the PI Controller to the Plant, and a line from the Plant to the

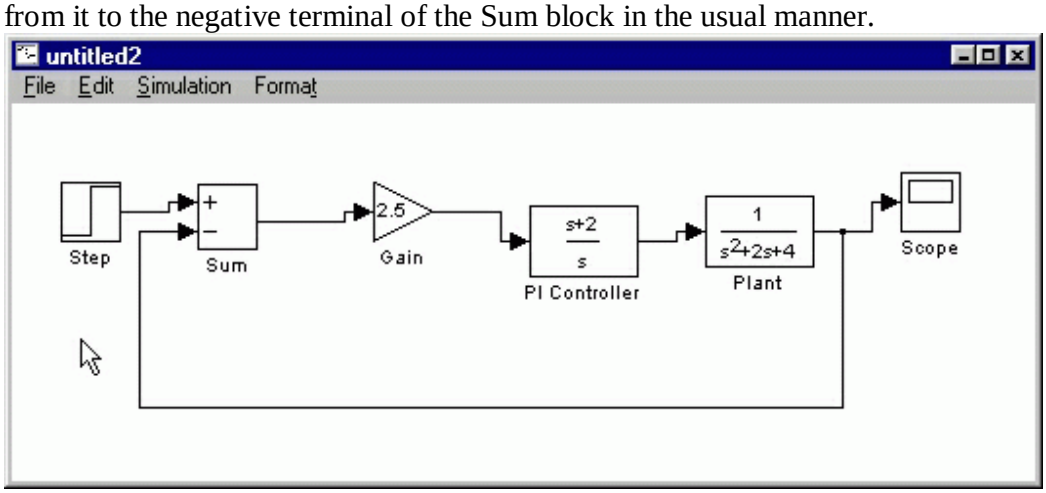
Scope. You should now have the following.



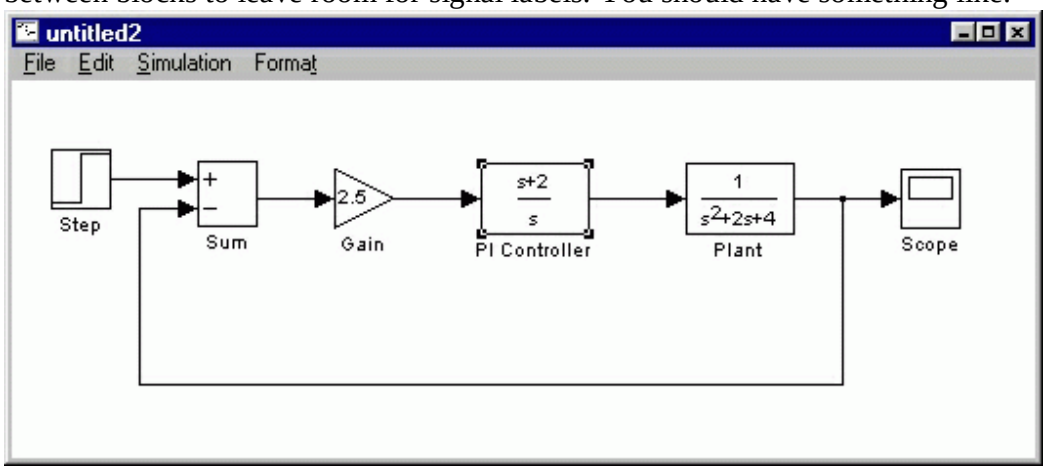
- The line remaining to be drawn is the feedback signal connecting the output of the Plant to the negative input of the Sum block. This line is different in two ways. First, since this line loops around and does not simply follow the shortest (right-angled) route so it needs to be drawn in several stages. Second, there is no output terminal to start from, so the line has to tap off of an existing line.
- To tap off the output line, hold the Ctrl key while dragging the mouse from the point on the existing line where you want to tap off. In this case, start just to the right of the Plant. Drag until you get to the lower left corner of the desired feedback signal line as shown below.



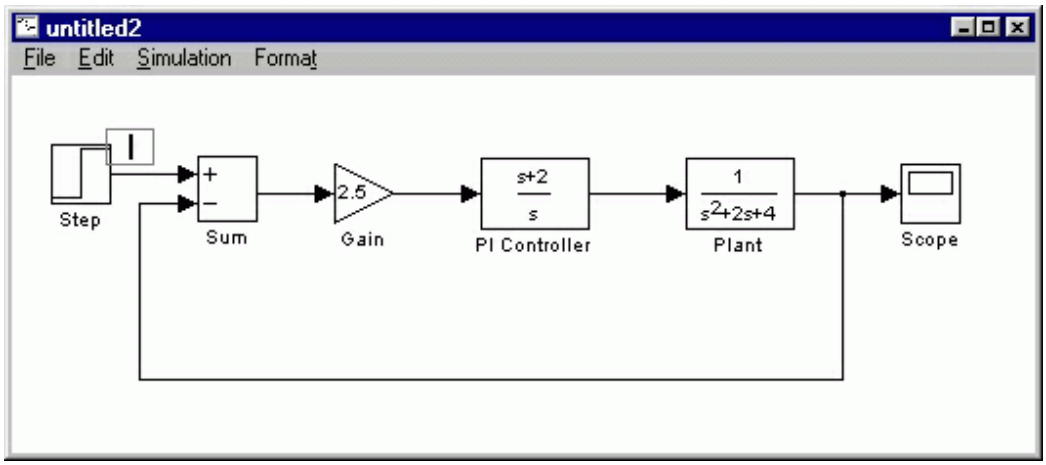
- Now, the open arrowhead of this partial line can be treated as an output terminal. Draw a line from it to the negative terminal of the Sum block in the usual manner.



- Now, you will align the blocks with each other for a neater appearance. Once connected, the actual positions of the blocks does not matter, but it is easier to read if they are aligned. To move each block, drag it with the mouse. The lines will stay connected and re-route themselves. The middles and corners of lines can also be dragged to different locations. Starting at the left, drag each block so that the lines connecting them are purely horizontal. Also, adjust the spacing between blocks to leave room for signal labels. You should have something like:

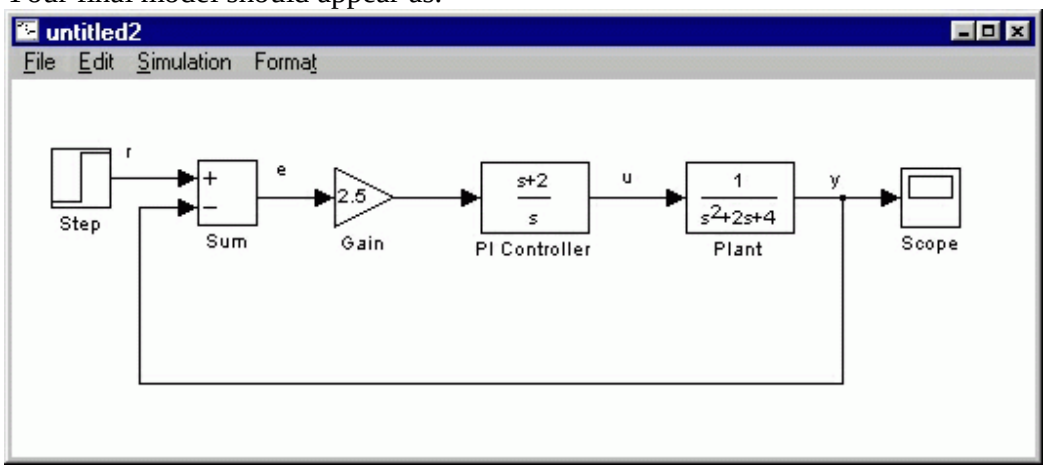


- Finally, you will place labels in your model to identify the signals. To place a label anywhere in your model, double click at the point you want the label to be. Start by double clicking above the line leading from the Step block. You will get a blank text box with an editing cursor as shown below



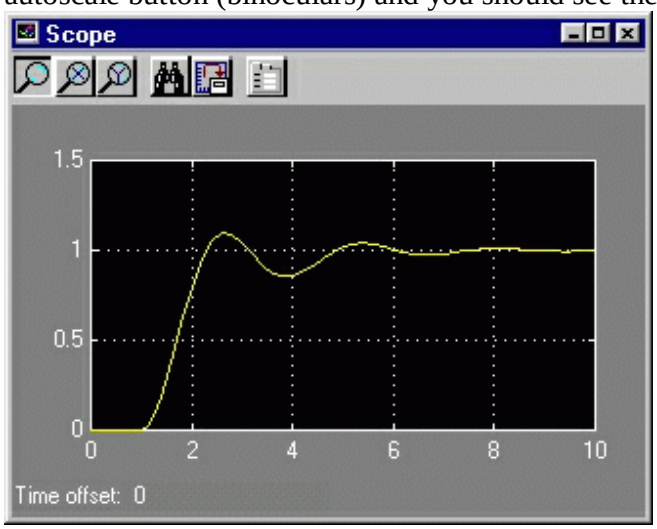
- Type an r in this box, labeling the reference signal and click outside it to end editing.

- Label the error (e) signal, the control (u) signal, and the output (y) signal in the same manner. Your final model should appear as:



- To save your model, select **Save As** in the **File** menu and type in any desired model name. The completed model can be found here.

Simulation Now that the model is complete, you can simulate the model. Select **Start** from the **Simulation** menu to run the simulation. Double-click on the Scope block to view its output. Hit the autoscale button (binoculars) and you should see the following.



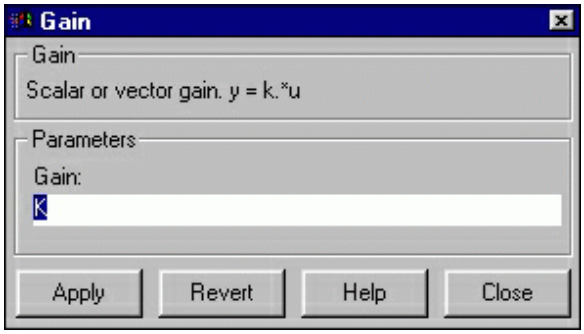
Taking Variables from MATLAB

In some cases, parameters, such as gain, may be calculated in MATLAB to be used in a Simulink model. If this is the case, it is not necessary to enter the result of the MATLAB calculation directly into Simulink. For example, suppose we calculated the gain in MATLAB in the variable K. Emulate this by entering the following command at the MATLAB command prompt.

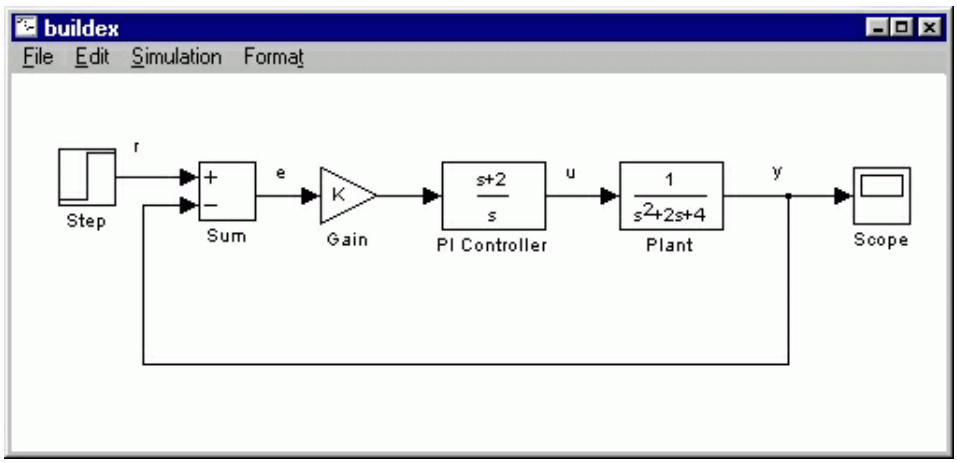
```
K=2.5
```

This variable can now be used in the Simulink Gain block. In your simulink model, double-click on the Gain block and enter the following in the Gain field.

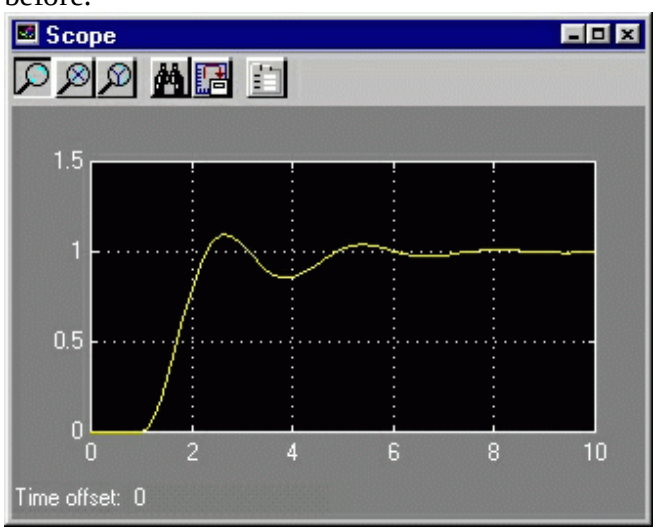
```
K
```



Close this dialog box. Notice now that the Gain block in the Simulink model shows the variable K rather than a number.



Now, you can re-run the simulation and view the output on the Scope. The result should be the same as before.

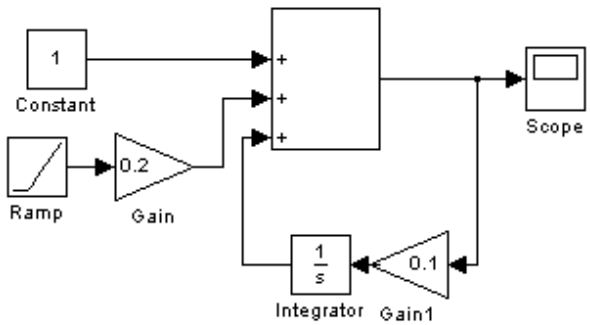


Now, if any calculations are done in MATLAB to change any of the variab used in the Simulink model, the simulation will use the new values the next time it is run. To try this, in MATLAB, change the gain, K, by entering the following at the command prompt. `K=5`

Example: simulate and solve

1- $y(t) = 1 + 0.2t + 0.1\int y(t)dt$

A- simulation diagram

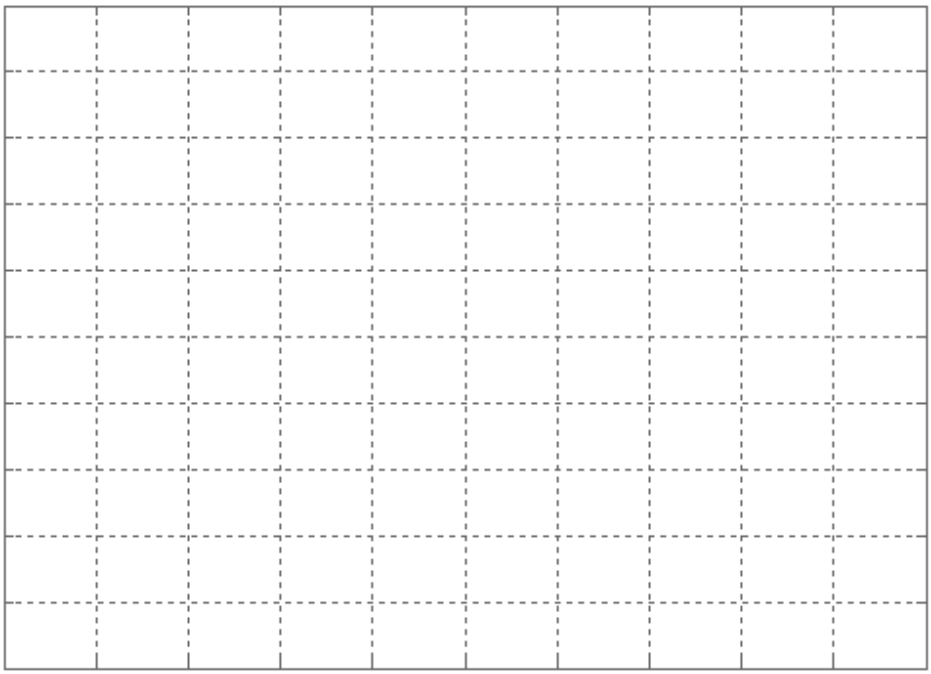


B-solution

2- $y(t) = t^2 + \sin 30t + 2\frac{d}{dt}y(t)$

A- simulation diagram

B-solution



Experiment N0 2

Solving Differential Equations and Modeling Dynamic Systems

Objectives:

- To solve linear differential equations.
- To solve a set of differential equations
- To represent a dynamic system using transfer functions
- To represent a dynamic system using state space

A: Solving linear differential equations

Consider the following linear differential equation

$$\dddot{y}(t) + a\ddot{y}(t) + b\dot{y}(t) + cy(t) = x(t)$$

Where y(t) is the output, x(t) is the input

To solve this differential equation the next steps should be done

1) Rewrite this equation as

$$\dddot{y}(t) = -a\ddot{y}(t) - b\dot{y}(t) - cy(t) + x(t) \qquad (2.1)$$

2) Use a summing element to add all terms in the right-half side
3) After the summing element use a number of integrators equals the order of the equation
4) Feed back the output of each integrator to the summing element

Instruments and Components:

- PC with MATLAB (Simulink) software.

Procedure:

1- Simulate a third order differential equation with coefficients as in table (2.1)

Simulation diagram

2- Plot y (t) and x (t) using an oscilloscope.

3- Record y (t) and x (t) with respect to the time as in table 2.1

Table 2.1 x (t) = constant=2, a= 2, b= 3, c=4

t (sec)									
x (t)									
Y(t)									

4-Plot y(t) on Fig.2.1

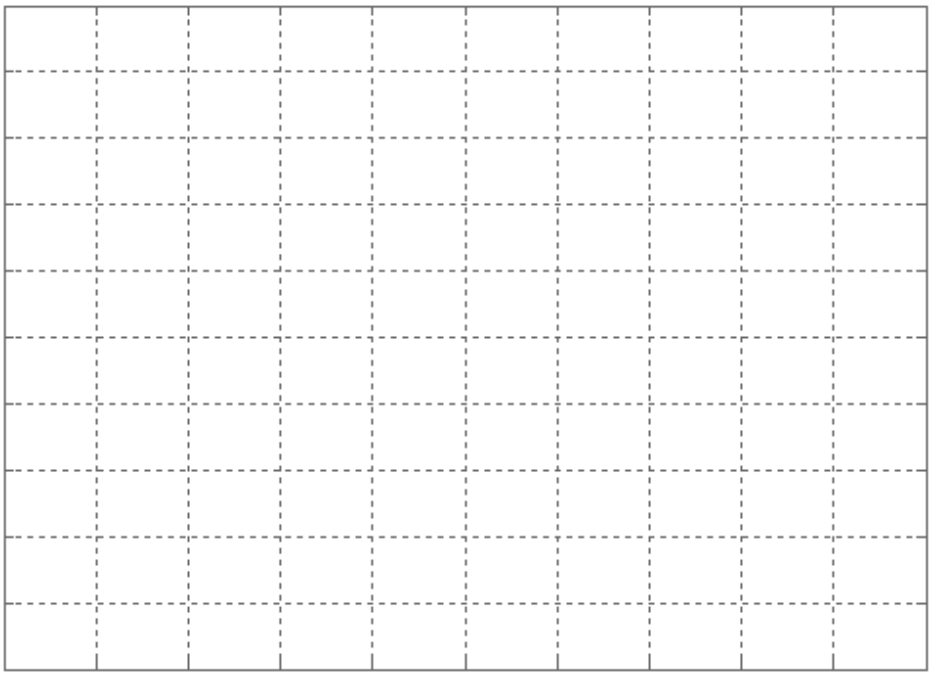


Fig.2.1

B: Solving a set of differential equations

Consider a set of two differential equations

$$\ddot{y}(t) + a\dot{y}(t) + 2z(t)y(t) = x(t)$$
$$\ddot{z}(t) + 0.1\dot{y}(t)\dot{z}(t) + bz(t) = r(t) \qquad (2.2)$$

These equations can be solved in steps as in (**A**)

Procedure:

1-Simulate these two equations with coefficients as in table (2.2)

Simulation diagram

Table 2.2 x (t) =constant=1, r (t) =constant=2, a= 4, b= 2

t (sec)											
y (t)											
z (t)											

2- Plot y (t) and z (t) using an oscilloscope.

3-Record y (t) and z (t) with respect to the time as in table 2.2

4-Plot y(t) and z(t) on Fig.2.2

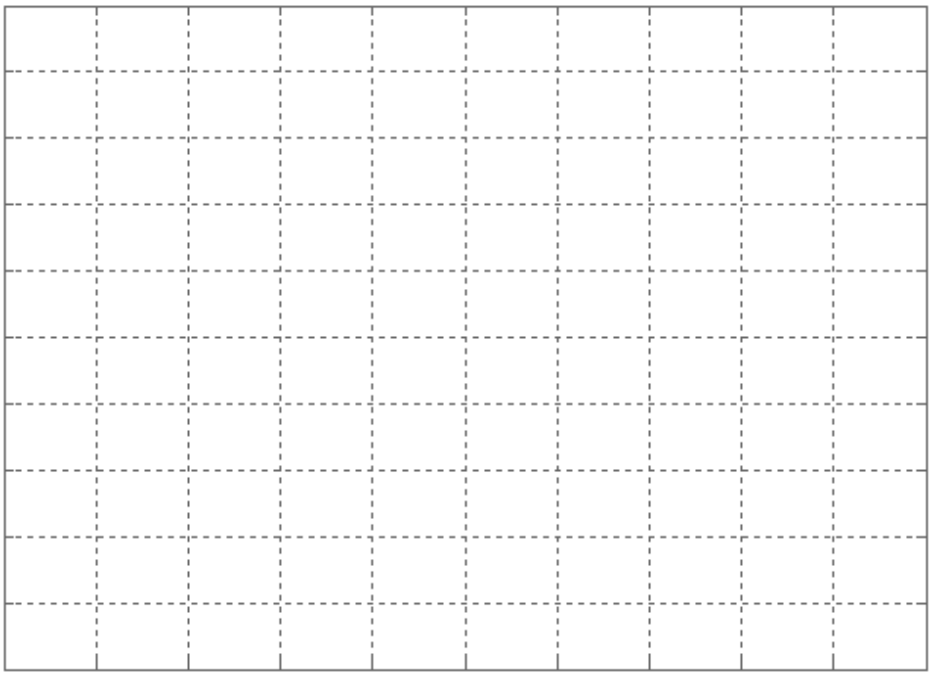


Fig.2.2

C: Transfer function approach

Consider a dynamic system given by the following differential equation

$$y^{(n)}(t) + a_1 y^{(n-1)}(t) + \ldots + a_{n-1} y'(t) + a_n y(t) = r(t) \quad \ldots\ldots 2.3$$

Where y (t) is the output (the dependent variable), and r(t) is the input (the independent variable).

This system can be represented using transfer function as

$$\frac{Y(s)}{R(s)} = \frac{1}{s^{(n)} + a_1 s^{(n-1)} + \ldots + a_{n-1} s + a_n} \quad \ldots\ldots 2.4$$

Procedure:

1- Simulate the following equation as a transfer function using 'Transfer function block in Simulink

$$y'''(t) + a_1 y''(t) + a_2\ y'(t) + a_3\ y(t) = r(t) \quad \ldots\ldots 2.13$$

2-Transfer function

3-Apply the coefficients and input as in table 2.3

Table 2.3: a_1=2; a_2 = 8; a_3=4; r (t) =constant=1

$t[s\}$												
$r(t)$												
$y(t)$												

4- Plot y (t) and x (t) using an oscilloscope.

5- Record y (t) and x (t) with respect to the time as in table 2.3

6- Plot y(t) on Fig.2.3

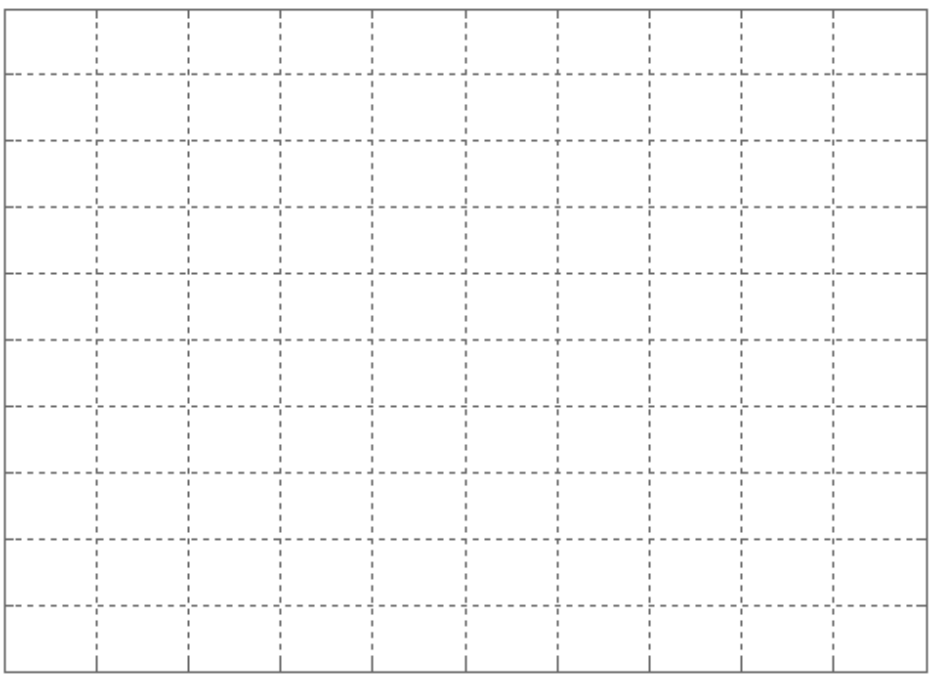


Fig.2.3

D: State Space approach

Equation 2.3 in state space form is

$$\dot{x}(t)=Ax(t)+Br(t) \qquad 2.5$$

$$y(t) =Cx(t)=+Dr(t) \qquad 2.6$$

In 2.5 and 3.6:

The state variables are;

$$
\begin{aligned}
&x_1(t) = y(t)\\
&x_2(t) = y'(t)\\
&.\\
&.\\
&x_n(t) = y^{(n-1)}(t)
\end{aligned}
\qquad 2.7
$$

The state equations are:

$$
\begin{aligned}
&x_1'(t) = x_2(t)\\
&x_2'(t) = x_3(t)\\
&.\\
&.\\
&x_{n-1}'(t) = x_n\\
&x_n'(t) = -a_n x_1(t) - \ldots - a_1 x_n(t) + r(t)
\end{aligned}
\qquad 2.8
$$

And:

$$
\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ . \\ . \\ x_n \end{bmatrix} \qquad \mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & . & 0 \\ 0 & 0 & 1 & . & 0 \\ . & . & . & . & . \\ . & . & . & . & . \\ -a_n & -a_{n-1} & -a_{n-2} & . & -a_1 \end{bmatrix} \qquad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ . \\ . \\ 1 \end{bmatrix} \qquad 2.9
$$

The output equation:

$$
\mathbf{y} = \begin{bmatrix} 1 & 0 & . & . & 0 \end{bmatrix} . \begin{bmatrix} x_1 \\ x_2 \\ . \\ . \\ x_n \end{bmatrix} \qquad 2.10
$$

Or $\mathbf{y} = \mathbf{Cx}$ 2.11

Where

$$\mathbf{C} = [1 \quad 0 \quad . \quad . \quad 0] \qquad 2.12$$

7- Procedure:

1-Write a state space equations for a system given by equation 2.13

Input equations in vector form

Output equations in vector form

2-Simulate the state space equation using state space block and the coefficients of Tab.2.4

3-Plot and record y(t)

Table2.4: a_1=2; a_2=8; a_3=4; r (t)=constant=1

$t[s\}$												
$r(t)$												
$y(t)$												

4- Draw y(t) on Fig.2.4

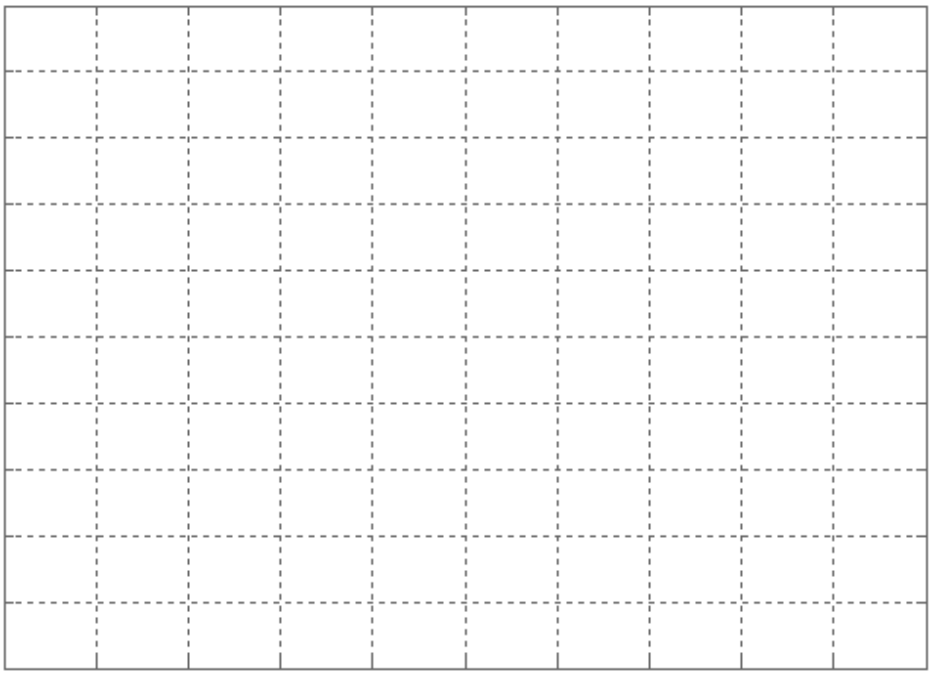


Fig.2.4

Conclusion and results

Experiment No3

Control System with Feedback.

Objectives:

- To use fixed feedback.
- To use flexible feedback

Introduction:

a) the use of fixed feedback.

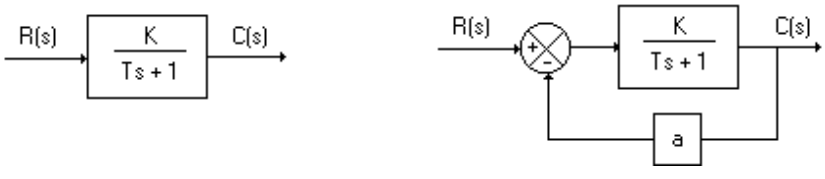
The primary purpose of using feedback in control systems is to reduce the sensitivity of the system to parameter variations and unwanted disturbances. Also the feedback can change the time response of control system.

1) Consider the system shown in figure 3.1 a. The time constant of the system is T and the gain is K. The addition of a negative feedback loop around this element reduces the time constant and the gain. Figure 3.2 b, shows the system with the same feed forward transfer function as that shown in figure 3.2 a, with the exception that a negative feedback loop has been added. The transfer function of this system is:

$$\frac{C(s)}{R(s)} = \frac{K}{Ts+1+a} = \frac{K/1+a}{\{T/1+a\, s\}+1} = \frac{K_\Sigma}{T_\Sigma s+1} \qquad 3.1$$

The time constant and the gain of this system have been reduced by (1+a).

If, instead of a negative feedback loop, a positive feedback is added, then the time constant and the gain can be increased.



A B

Fig.3.1

2) Reduction the effect of disturbance:

Figure 3.2 shows a control system subjected to a disturbance when two inputs (the reference input and disturbance) are present in a linear system, each input can treated independently of the other, and the output corresponding to each input alone can be added to give the complete output.

For the open-loop control system the effect of the disturbance is equal to:

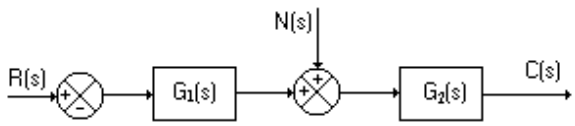
$C_N(s) = + G_2(s) \cdot N(s)$ **E**quation 3.3.

And for the closed- loop system the effect of the disturbance is equal to:

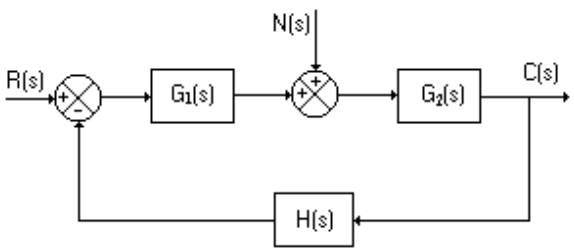
$$C_N(s) = + \frac{G2(s)}{1 + G1(s) \cdot G2(s) \cdot H(s)}$$ **E**quation 3.4.

Note that the affect of the disturbance in the closed-loop system has been reduced to:

$$\frac{G2(s)}{1 + G1(s) \cdot G2(s) \cdot H(s)}$$



(A)



(B)

Figure 3.2

Procedure:

Part A: Changing time response:

1- Simulate the closed loop system shown in fig. 3.1.B with parameters as in table 3.1.,

2- Plot and record the output signal

3- Vary the input signal as in table 3.2

Table 3.1., T=1; K=1; r (t)=unit step

a	$t\{s\}$												
0	$c(t)$												
3	$c(t)$												
6	$c(t)$												

4-plot the data from table 3.1 on Fig.3.3

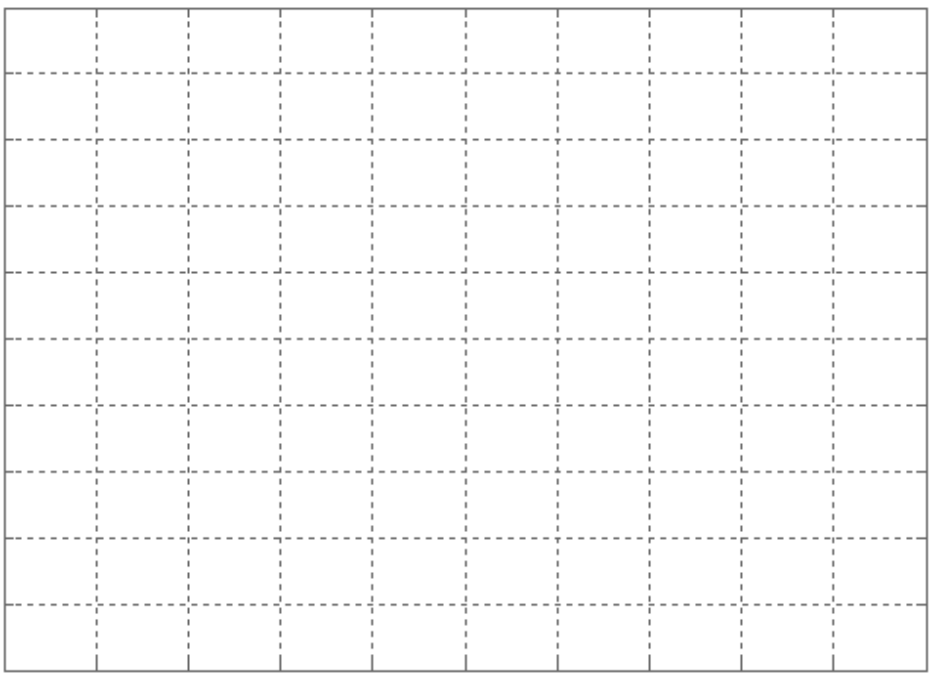


Fig.3.3

Comments:

Part B: **Reduction the time response sensitivity to disturbance:**

1-Simulate the system shown in fig. 3.2.B with $G_1(s) = \frac{1}{s+1}$ and $G_2(s) = \frac{1}{s+4}$

2- Record the output as in table 3.2.

Table 3.2.; n (t) =unit step,r(t)=0

H(s)	$t\{s\}$												
0	$c_n(t)$												
1	$c_n(t)$												
4	$c_n(t)$												

3-plot the data from table 3.2 on Fig.3.4

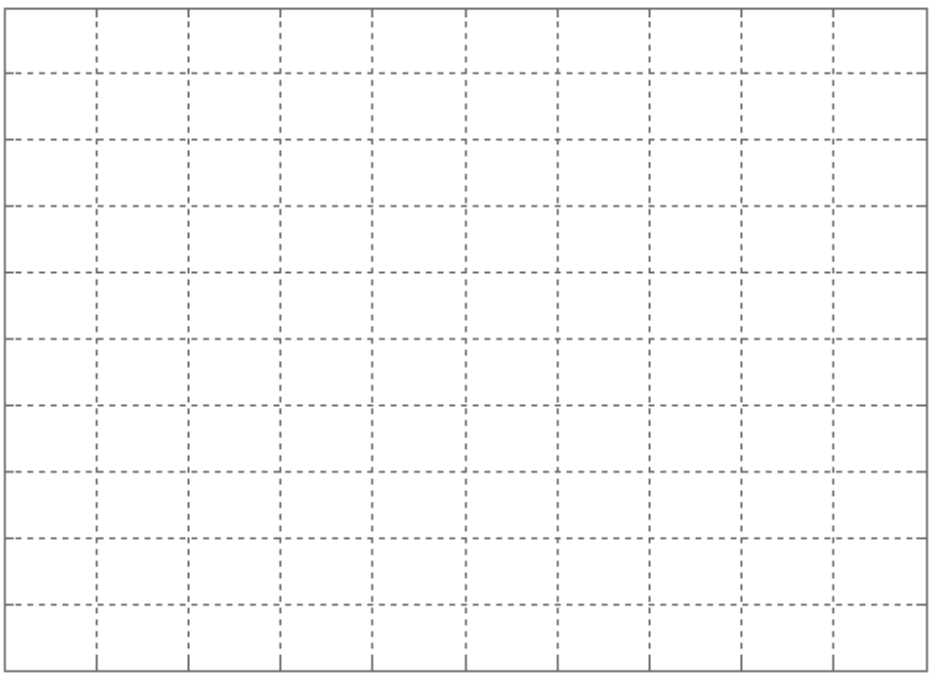


Fig.3.4

Comments:

Part C: Reduction the time response sensitivity to parameter variation:

1-Simulate the system shown in fig. 3.2.B with $G_1(s) = \frac{1}{s+1}$ and $G_2(s) = \frac{1+\Delta K}{s+4}$

2- Record the output as in table 3.3.

Table 3.3; n (t)=unit step, r(t)=0,ΔK=0.2

H(s)	$t\{s\}$												
0	$c_k(t)$												
1	$c_k(t)$												
4	$c_k(t)$												

3-plot the data from table 3.3 on Fig.3.5

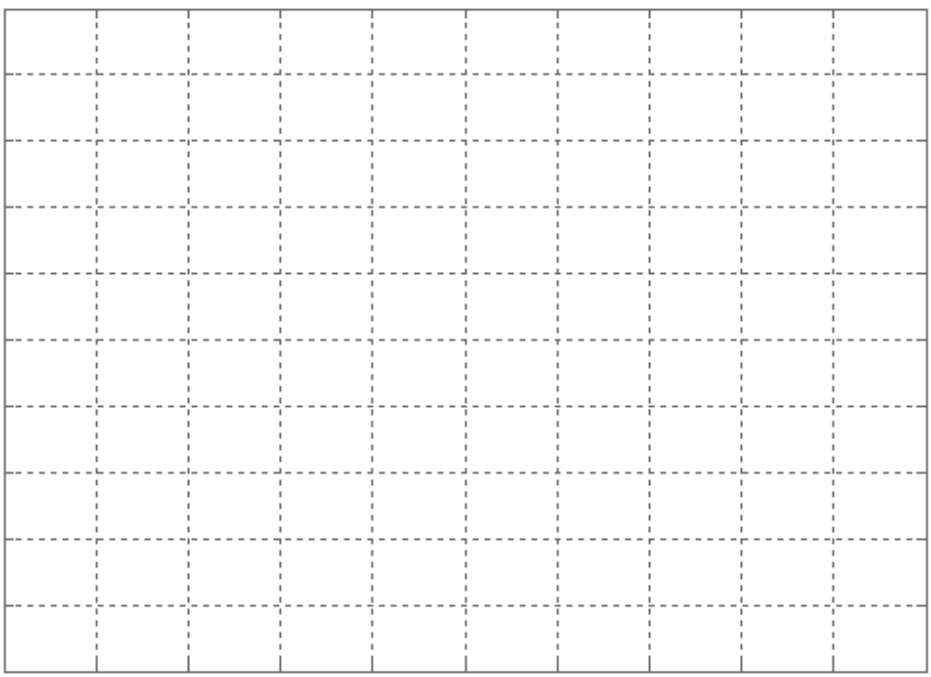


Fig.3.5

Comments:

b) to use flexible feedback

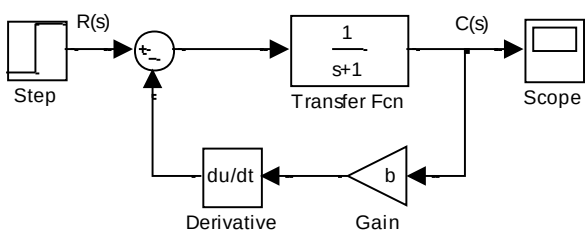
Consider a first-order system with a gain K and time constant T. The addition of a positive feedback loop, with a transfer function bs around this system reduces the time constant to T-bk, while the gain of the system remains the same. It means that this type of feedback changes the transient response only, so it is called flexible. If the constant b is set equal to T/K, then the time constant becomes zero. Note that if a disturbance causes T-bk to be negative instead of zero, the system becomes unstable. If the sign of the feedback is changed, the time constant will increase.

Instruments and Components:

- PC with Simulink.

Procedure:

1- Simulate a closed loop system with: G(s)=1/{s+1}; H(s)= bs ,negative feedback and r(t)=unit step,as shown :



2- Plot the output signal on the oscilloscope, and record the results in table 3.4.

Table 3.4

b	$t[s]$												
0	$c(t)$												
1	$c(t)$												
2	$c(t)$												

3- Plot the data from table 3.4 on Fig.3.6

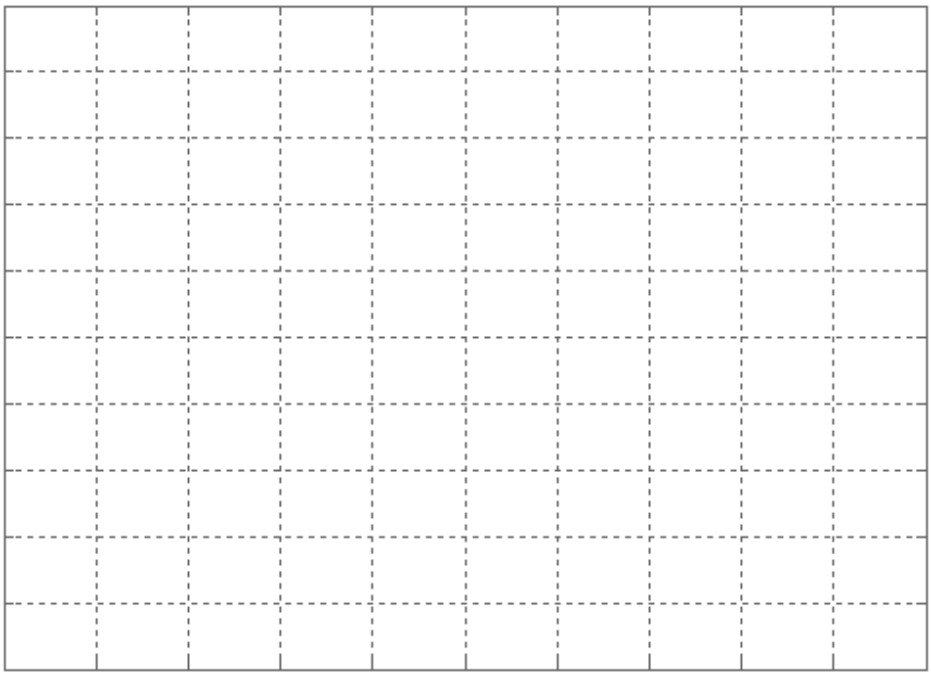
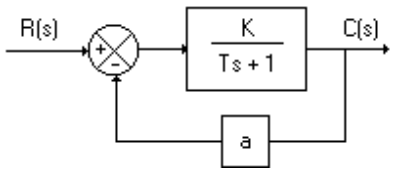


Fig.3.6

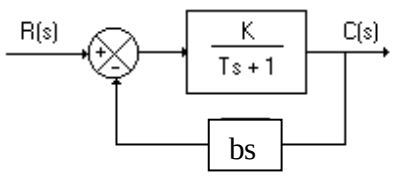
Comments

Assignment

1- obtain c(t) for r(t)=1



2- Obtain c(t) for r(t)=1



Experiment No 4

Time Response of First-Order Systems

Objectives:

- To study transient response of first-order systems.
- To study steady-state response of first-order systems.

Introduction:

The time response of a control system consists of two parts: the transient and the steady-state response. By transient response, we mean that which goes from the initial state to the final state. By steady-state response, we mean the manner in which the system output behaves as time approaches infinity.

In analyzing and designing control systems, we must have the basis of comparison of performance of various control systems. This basis may be set up by specifying particular test input signals and by comparing the response of various systems to these input values.

The commonly used test input signals are those of step function, ramp function, acceleration function, impulse function, these functions are called a periodic signals.

The first-order system can be represented by the following transfer function:

$$G(s) = \frac{1}{Ts + 1}$$

And the output signal:

$C(s) = G(s)\, R(s)$

Or $c(t) = L^{-1}\{C(s)\} = L^{-1}\{G(s)\, R(s)\}$ Eq. (4.1)

Physically, this system may represent an R-C electric circuit, thermal system, liquid-level system, etc. A block diagram of a first-order system is:

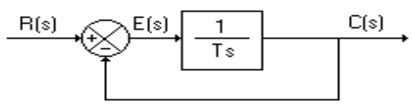


Figure 4.1: *Block diagram of First-Order System.*

Procedure:

Transient response (dynamic response)

A) Transient response to varying step input

1- Construct the circuit shown in figure 4.1. Apply an input signal to the circuit as shown in the tables.

2- Plot the output signal and the input signal on the oscilloscope.

3- Record c (t) and e(t) with respect to the time as in the following tables

Table 4.1 r(t)=unit step; T=1

$t[s]$													
c													
e													

Table 4.2 r(t) =step(final value 2); T=1

$t[s]$													
c													
e													

Table 4.3 r(t) =step(final value0.5);; T=1

$t[s]$													
c													
e													

4- Plot r(t), c(t) and form tables 4.1,4.2 and 4.3 on Fig.4.2 .a

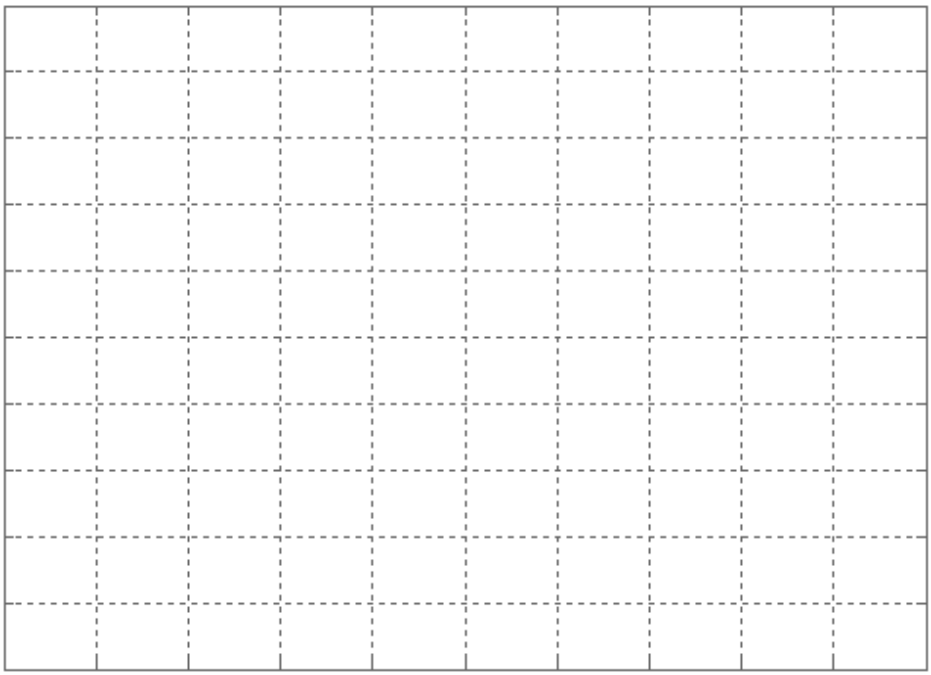


Fig.4.2.a:

1- 5- Plot e(t) form tables 4.1,4.2 and 4.3 on Fig.4.2 .b

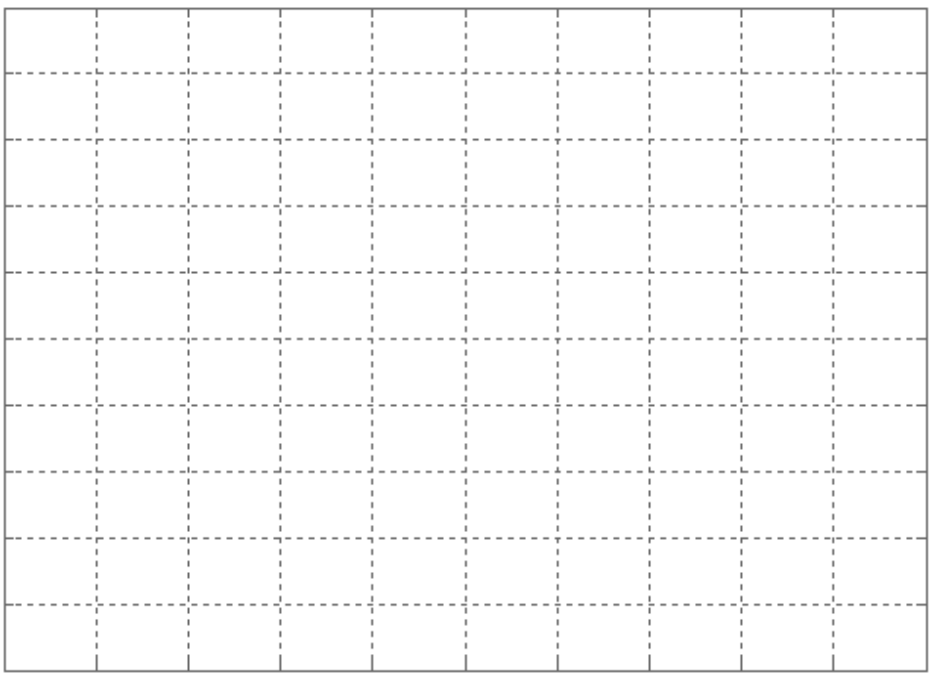


Fig.4.2.b:

Conclusion:

B) unit step transient response to varying time constant

1-Construct the circuit shown in figure 4.1.

2-Apply an input signal to the circuit as shown in the tables.

3-Plot the output signal and the input signal on the oscilloscope.

4-Record c (t) with respect to the time as in the following tables

Table 4.4; r (t) =unit step; T=0. 4

$t[s]$													
c													

Table 4.5; r (t) =unit step; T=1.

$t[s]$													
c													

Table 4.6; r (t) = unit step ;T=1.4

$t[s]$													
c													

5-Plot r(t) ,c(t) form tables 4.4,4.5 and 4.6 on Fig.4.3

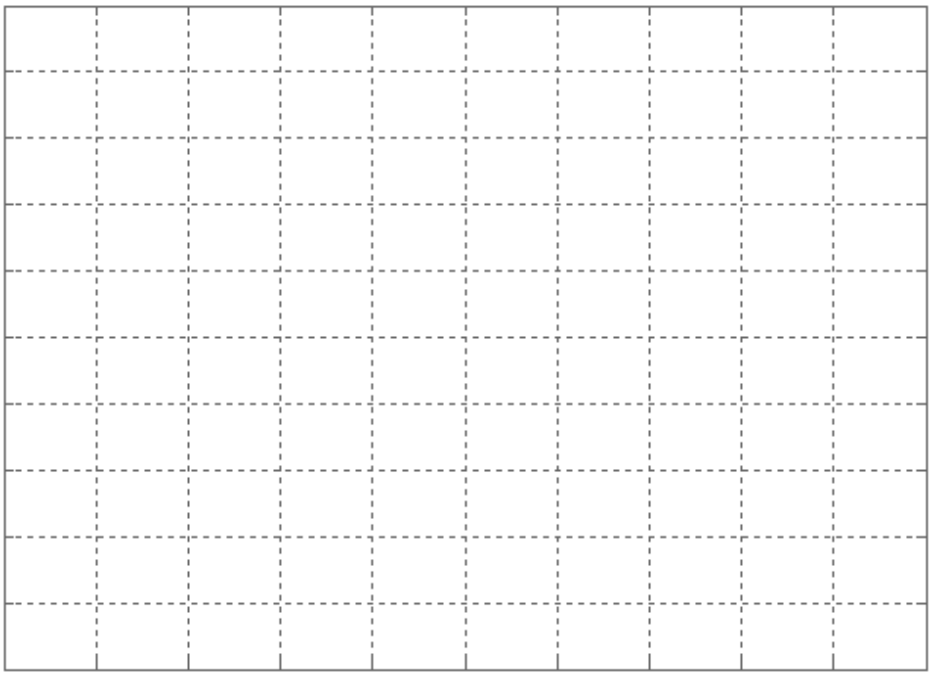


Fig.4.3:

Conclusion:

C) Unit ramp transient response to varying time constant response

1-Construct the circuit shown in figure 4.1.A

2-Apply an input signal to the circuit as shown in the tables.

3-Plot the output signal and the input signal on the oscilloscope.

4-Record c (t) and e(t) with respect to the time as in the following tables

Table 4.7; r (t) = unit ramo; T=0. 4

$t[s]$													
c													
e													

Table 4.8; r (t) = unit ramp ; T=1.

$t[s]$													
c													
e													

5-Plot c (t) and e(t) form tables 4.7,4.8 on Fig.4.4

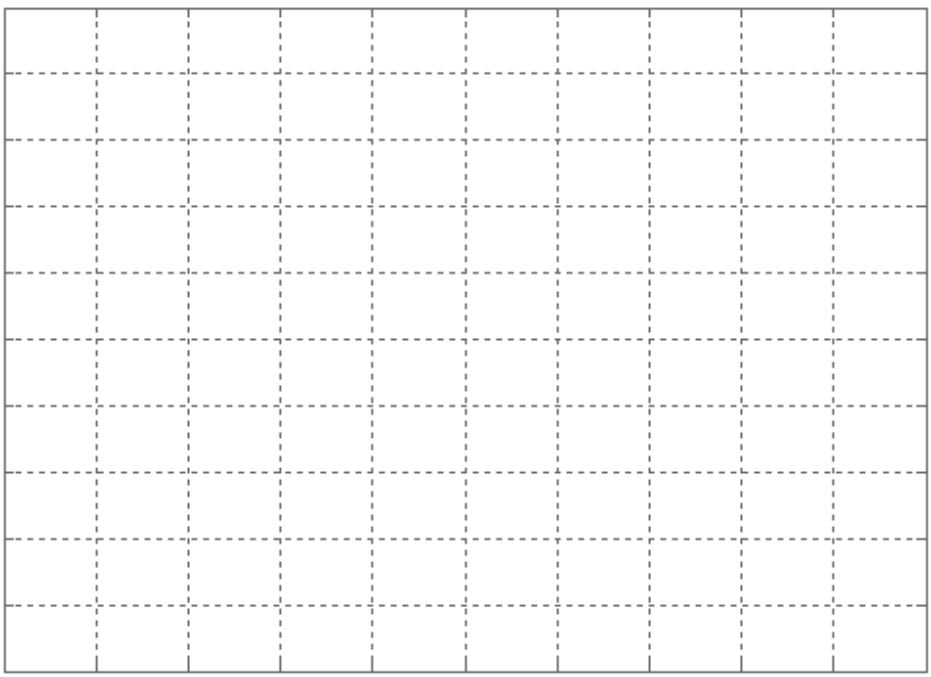


Fig.4.4:

Conclusion:

Steady state response (static response)

1- Construct the circuit shown in figure 4.1.

2-Apply an input signal(constant), to the circuit as shown in the table.

3- After the output reaches it is final value (steady state value), record C with. . . respect to the input as in the following table

Table 4.9, T=1

R	1	2	3	4	5	6	7	8	9	10			
C													

4 plot C=f (R) on Fig.4.5

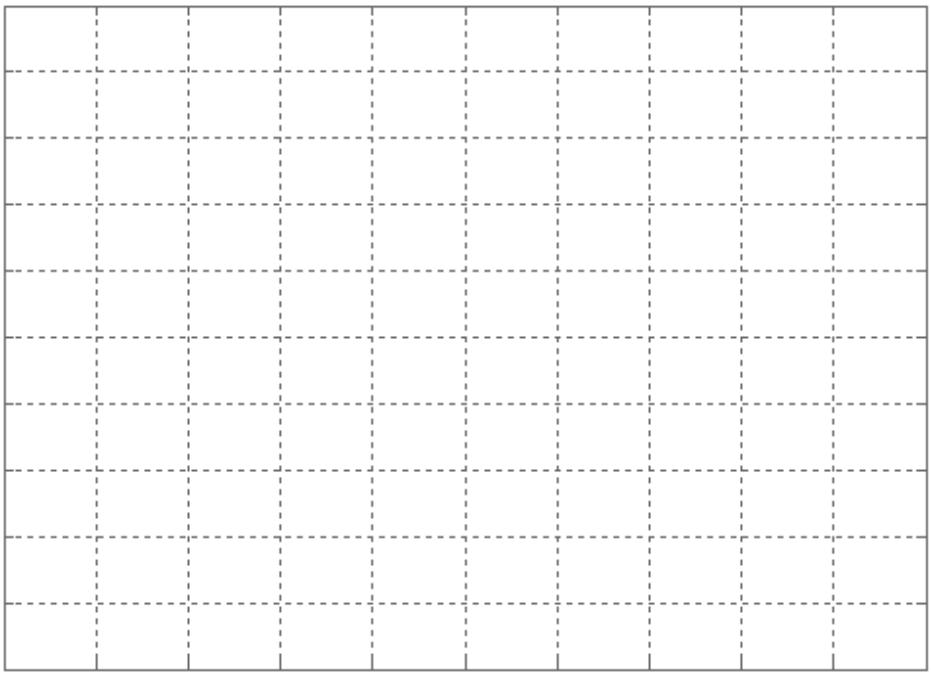
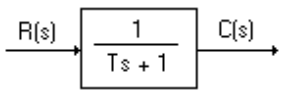


Fig.4.5

Conclusion

Assignment

Using inverse Laplace transform, obtain c (t) and e(t) for



1- **r(t)=A**
2- **r(t)=t**

Experiment No5

Modeling and Control of a DC Servo System

Objectives:

- To study transient response
- Apply speed feed back
- Apply position feed back

Introduction:

Servo drives used in machine-tools and industrial robot manipulators require four-quadrant operation and also have very demanding requirements for current loop bandwidth, velocity loop bandwidth, torque smoothness, and high static and dynamic stiffness in a closed position loop.
The primary purpose of using feedback in control systems is to reduce the sensitivity of the system to parameter variations and unwanted disturbances. Also the feedback can change the time response of control system.

Dynamic Model of DC Servo System
The position servo system consists of a permanent magnet field armature controlled DC motor Interfaced to an output shaft with a gear ratio of N. The schematic diagram of this system is shown in Fig.1

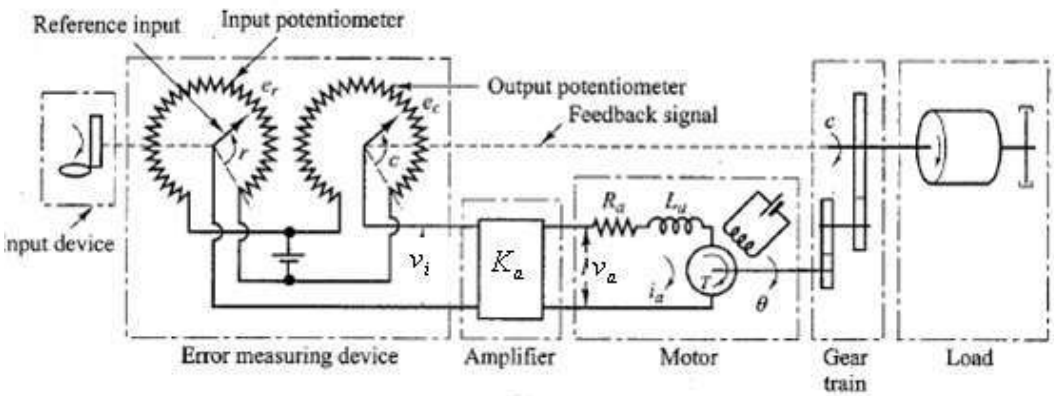


Fig.1 schematic diagram of servos system

The system dynamics may be modeled by the following equations

$$v_a(t) = v_i K_a \qquad 1$$

$$v_a(t) = L_a \frac{di_a(t)}{dt} + R_a i_a(t) + e(t) \qquad 2$$

$$T_e(t) = K_t\, i_a(t) \qquad 3$$

$$J_m \frac{d\omega(t)}{dt} = T_e - T_d - B_m \omega(t) \qquad 4$$

$$e(t) = K_e \omega(t) \qquad 5$$

$$\theta(t) = \int \omega(t)\, dt \qquad 6$$

Where:
i_a (t) = motor current (A),
R_a = motor armature resistance (Ω),
La motor armature inductance
v_a(t) = motor armature voltage (V),
e(t) = motor back emf (V),
Te(t) = motor torque (Nm),
T_d = load torque (Nm),
J m= net motor-tacho-gears-shafts inertia (Nm/radian/sec/sec),
Bm = net motor-tacho-gears-shafts damping (Nm/radian/sec),
ω(t) = motor shaft angular speed (radians/sec),
Kt = motor torque constant (Nm/A),
Ke = motor back emf constant (V/radian/sec),
N = gear ratio coupling motor to output shaft (radians/radian),
θ(t) = angular position of output shaft (radians),
Kθ = position sensor gain (Volts/radian),
Kω = tacho generator gain (Volts/radian/sec)
Ka= power amplifier gain,

Or in block diagram form

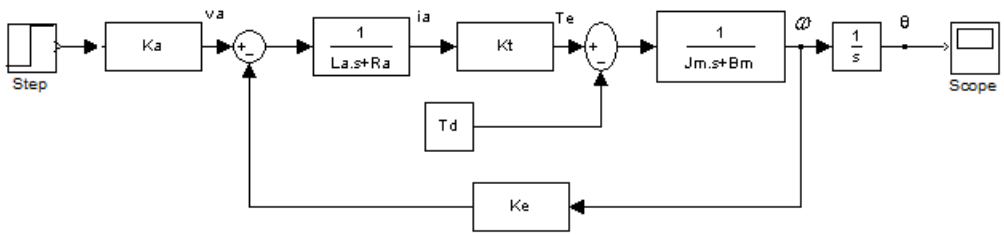


Fig.2 block diagram of servos system

A) Transient response of DC motor

1-Simulate the motor as shown in Fig.3

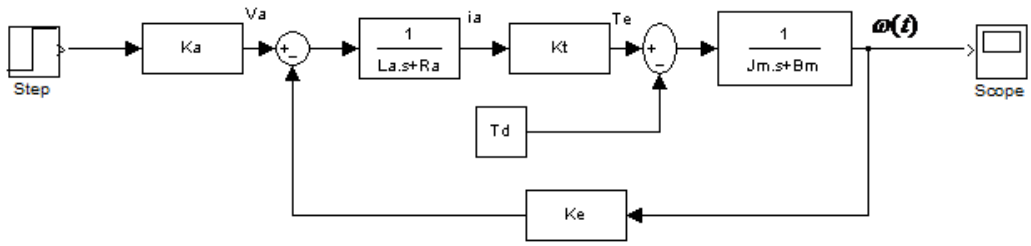


Fig.3Block diagram of DC motor

2- Inter the parameters using Table 1

¨TABLE 1: Motor rated output and nominal parameters.¨	
Ka = 40 V/V	R_a = 4.7 (Ω),
Ke = 0.45 V/(rad/s)	L_a = 0.011 H
Td = 1.147 Nm	J_m = 0.0098 K_g m^2
Kt = 0.45 Nm/A	B_m = 0.0015 Nm/(rad/s)
Kω =0.1	Kθ = 0.1

3-Run the simulation and record the output signals: ω(t), i_a (t) in Table 2

Table 2:,V_i=UNIT STEP, Td=0

$t\{s\}$													
ω(t)													
i_a (t)													

4- Run the simulation record the output signals: ω(t), i_a (t) in Table 3

Table 3: V_i=UNIT STEP ,Td=1.147

$t\{s\}$													
ω(t)													
i_a (t)													

4-plot ω(t) from Table 2 and Table 3 on Fig.4

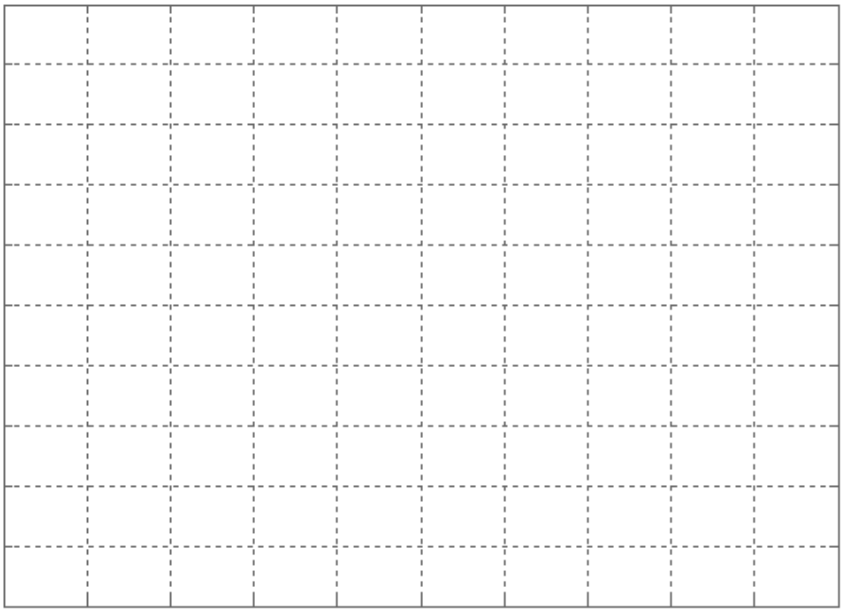


Fig.4

5-plot i_a (t) from Table 2 and Table 3 on Fig.5

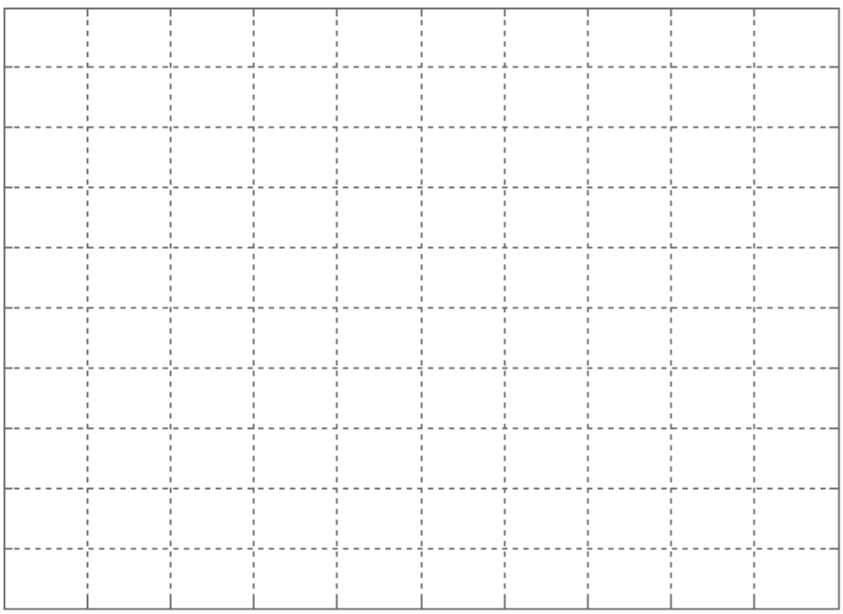


Fig.5.

Comments:

B-**Transient response of speed closed loop system**

1-Simulate the motor as shown in Fig.6

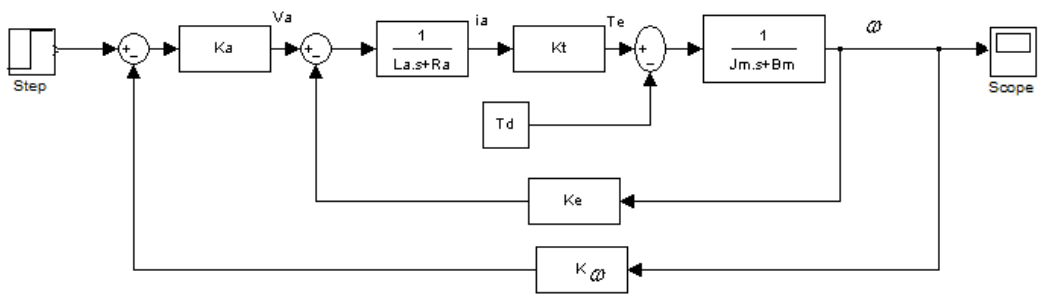


Fig.6 Block diagram of speed closed loop system

2- Inter the parameters using Table 4

3-Run the simulation and record the output signals: ω(t), i_a (t) in Table 4

Table 4:,V_i= STEP(FIN.VALUE=10), Td=0

t{*s*}													
ω(t)													
i_a (t)													

4- Run the simulation record the output signals: ω(t), i_a (t) in Table 5

Table 5: V_i= STEP(FIN.VALUE=10), Td=1.147

t{*s*}													
ω(t)													
i_a (t)													

5-plot ω(t) from Table 4 and Table 5 on Fig.7

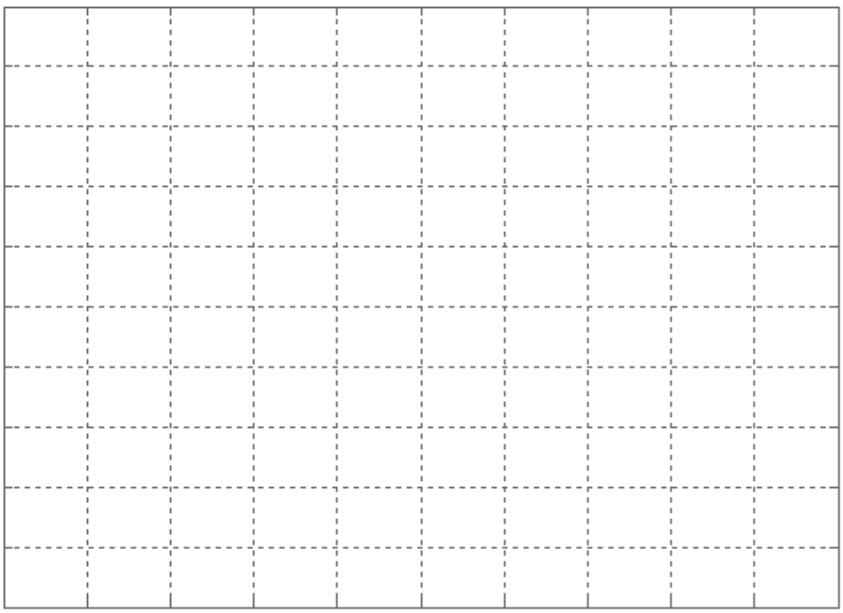


Fig.7

5-plot i_a (t) from Table 4 and Table 5 on Fig.8

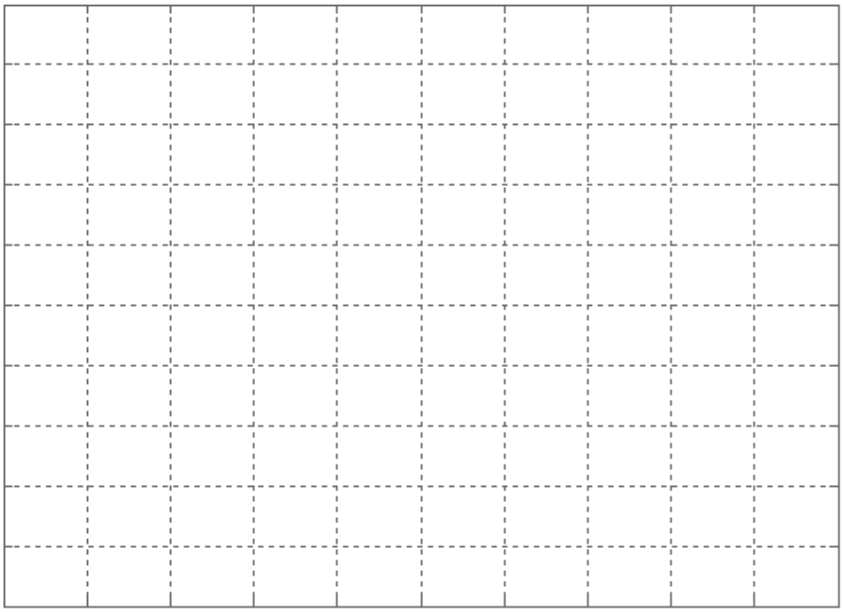


Fig.8

Comments:

C-Transient response of position closed loop system

1-Simulate the position closed loop system as shown in Fig.9

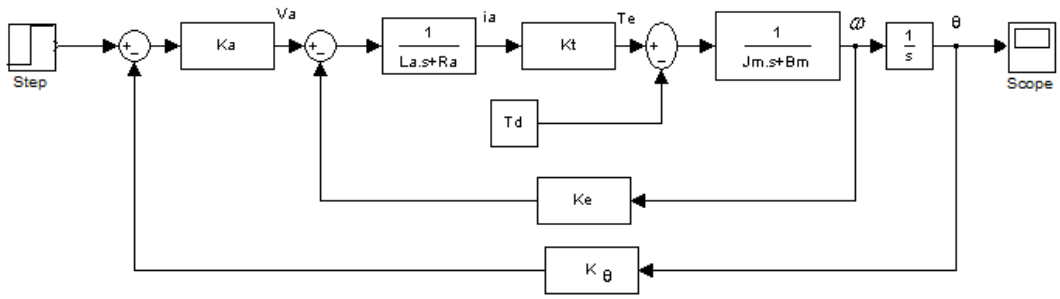


Fig.9Block diagram of position closed loop system

2- Inter the parameters using Table 1

3-Run the simulation and record the output signals: θ(t) , i_a (t) in Table 6

Table 6:, V_i= STEP(FIN.VALUE=10),, Td=0

$t\{s\}$													
θ(t)													
i_a (t)													

4- Run the simulation record the output signals: ω(t), i_a (t) in Table 7

Table 7: V_i= STEP(FIN.VALUE=10),, Td=1.147

$t\{s\}$													
θ(t)													
i_a (t)													

5-plot θ(t) from Table 6 and Table 7 on Fig.10

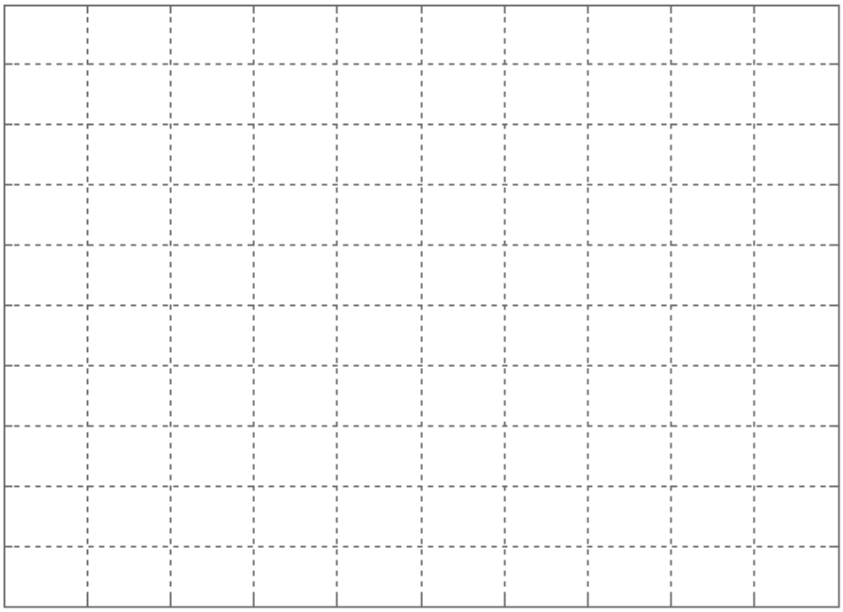


Fig.10

5-plot i_a (t) from Table 6 and Table 7 on Fig.11

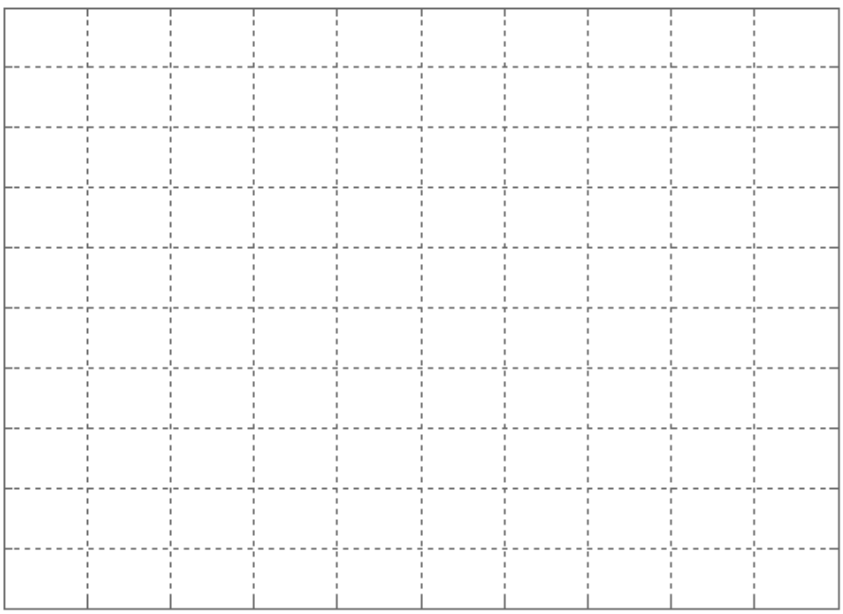


Fig.11

Comments:

Experiment N0 6

Time Response of Second Order Systems

Objectives:

- To study the transient response of the second order system.
- To illustrate the effect of damping ratio variation on the system response.
- To determine the transient response specifications.

Instruments and Components:

- PC with Simulink

Introduction:

Consider the second order system shown in Figure 6.1, the closed loop transfer function is:

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{S^2 + 2\zeta\omega_n S + \omega_n^2} \qquad \ldots 6.1$$

The dynamic behavior of this system can be described in terms of two parameters (ζ and ω).

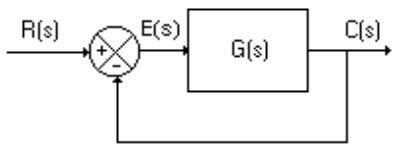


Figure 6.1.

Where G(s) = $\dfrac{\omega_n^2}{S(S+2\zeta\omega_n)}$

Procedure:

A) The effect of damping ratio variation on the system response

1. simulate the system in fig.6.1
2. Plot the output & input signals using the oscilloscope.
3. Record the results in Table 6.1.

Table 6.1: $\omega_n = 1$; r (t) =unit step

ζ	$t[s]$											
0.6	$c(t)$											
1	$c(t)$											
3	$c(t)$											

4. Set the value of damping ratio to zero
5. Plot, measure and record the output signal in Table 6.2.

Table 6.2: $\omega_n = 1$; r (t)=unit step

$t[s]$												
$c(t)$												

6. Set the value of damping ratio to -2

7. Plot the output signal as 1n table 6.3

Table 6.3: $\omega_n = 1$; r (t)=1

$t[s]$											
$c(t)$											

7- Plot the time response from Table 6.1 on one graph.(Fig.6.2)

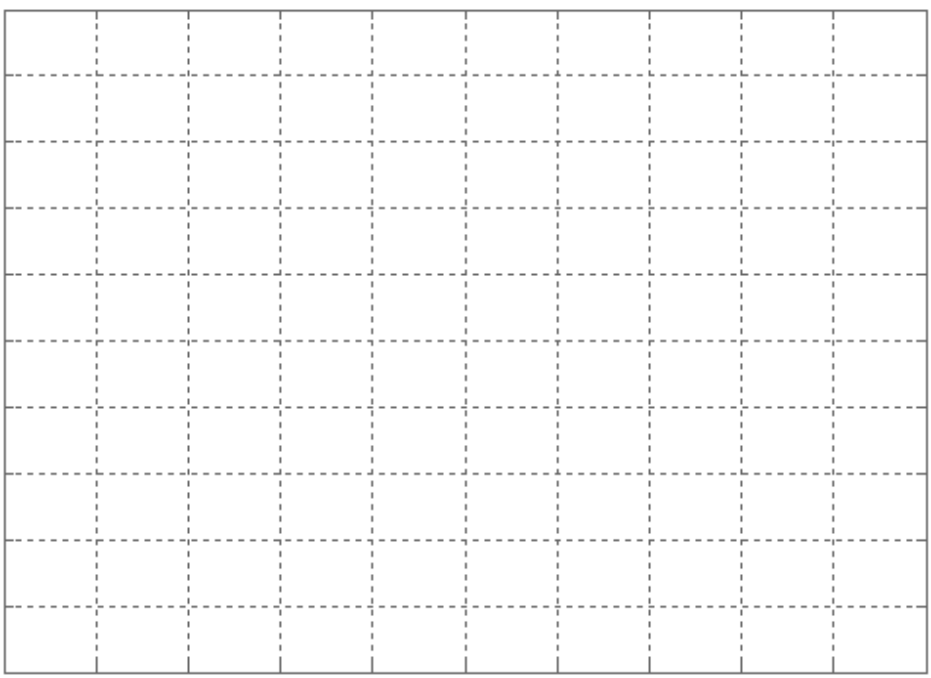


Fig.6.2

Comments:

8-Plot the time response from Table6.2 and 6.3 on the same graph.(Fig.6.3)

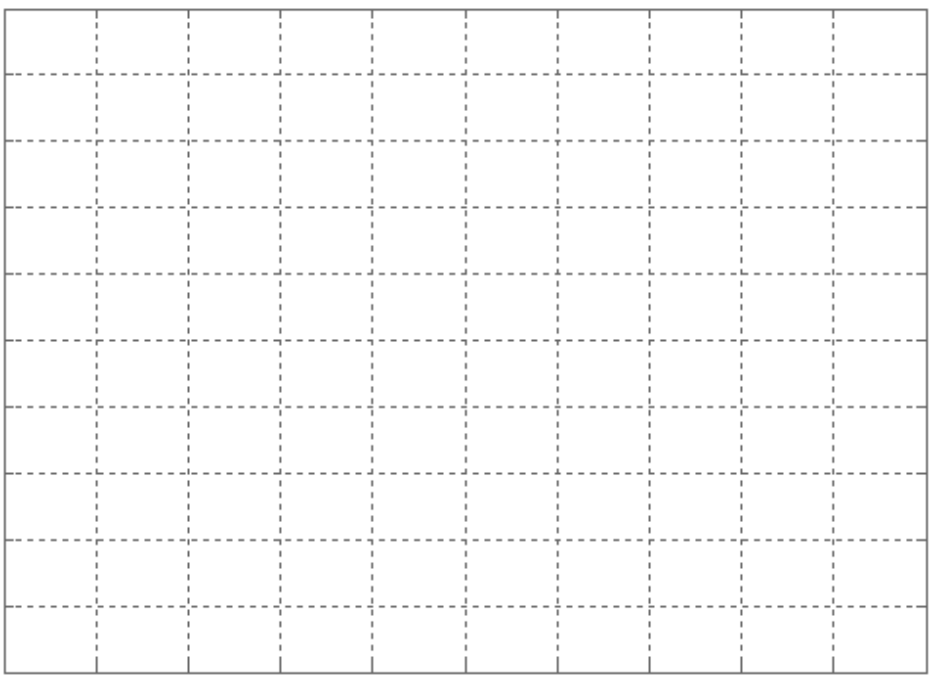


Fig.6.3

Comments:

9-From the plotted curves obtain the time response specifications and record them into Table 6.4.

ζ	t_d **(s)**	$t_r(s)$	$t_p s)$	$M_p(\%)$	$t_{ss}(s)$
0.6					
1					
3					

10- Calculate the roots of the characteristic equation and determine the type of time response as in table 6.5

feedback	Damping ratio	Values of roots	Type of roots	Type of response
negative	**0.6**			
	1			
	3			
positive	**-0.2**			
Zero	**0**			

B) The effect of frequency variation on the system response

1-simulate the system in fig.6.1

2- Plot the output & input signals using the oscilloscope.

3-Record the results in Table 6.4.

Table 6.5: ζ **=0. 5;** r (t)=unit step

ω_n	$t[s]$											
1	$c(t)$											
4	$c(t)$											

8-Plot the time response from Table 6.5 on the same graph.(fig.6.4)

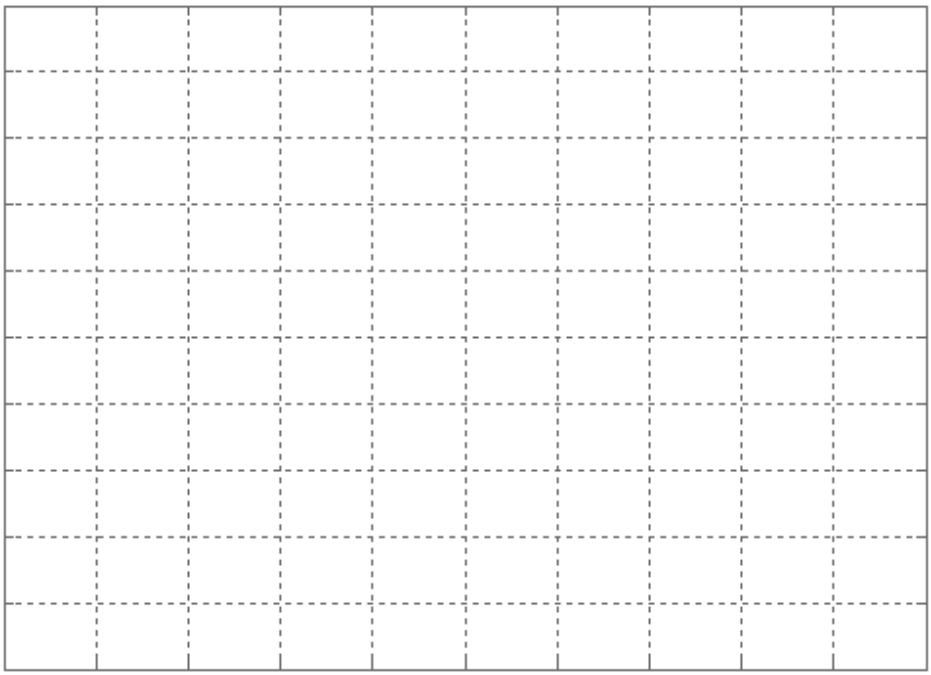


Fig.6.4

Comments:

Experiment No7

Basic Control Actions (Part One)

Objectives:

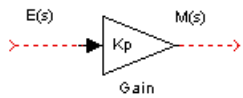
- To study the proportional control action [P control].
- To study the integral control action [I control].
- To study the PI control.

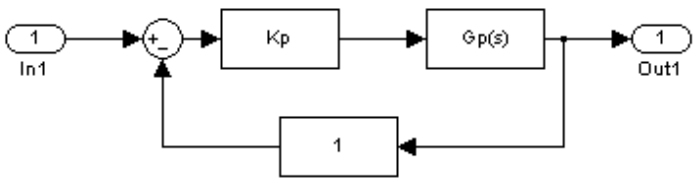
Introduction:

Proportional controller: the relationship between the output of the controller and the actuating error signal e (t) [the input of the controller]:

$$m(t) = k_P.e(t)$$

Or, in Laplace transformed quantities,

$\frac{M(s)}{E(s)} = k_P$	



Proportional control

Where k_P is termed to the proportional sensitivity or the gain. Now consider a first-order control system as shown in figure A.
If there is no control action, i.e. $G_C(s) = 1$, the steady-state error is:

$$e_{SS} = \frac{1}{1+k}$$

If there is a proportional control action with a gain: $G_C(s) = K_P$
The steady-state error becomes:

$$e_{SS} = \frac{1}{1+k.k_P}$$

The value of the steady state error becomes smaller as the gain K_P becomes larger.

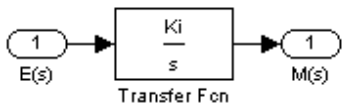


Figure A

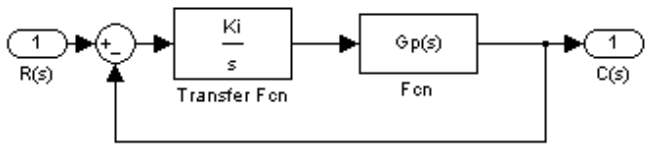
Integral controller the relationship between the output of the controller and the actuating error signal e (t) [the input of the controller]:

$$\mathbf{m(t) = k_i \int e(t)dt}$$

Or, in Laplace transformed quantities,

$\frac{\mathbf{M(s)}}{\mathbf{E(s)}} = \frac{\mathbf{K_i}}{\mathbf{S}}$	

The steady-state error becomes zero.

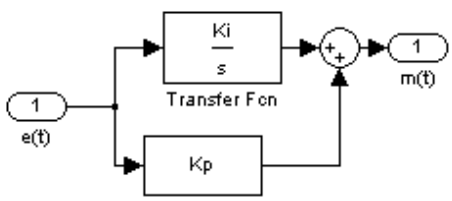


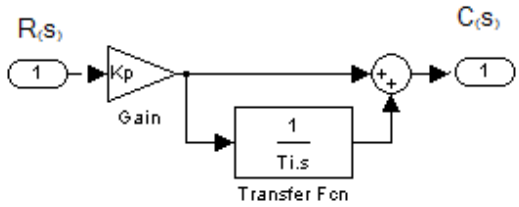
integral control

Proportional – plus – integral controller is defined by the following equation:

$$\mathbf{m(t) = k_p .e(t) + K_i \int_0^t e(t)dt = K_p [e(t) + \frac{1}{T_i} \int e(t)dt]}$$

Or, the transfer function of the controller is:

$\frac{M(s)}{E(s)} = k_p + \frac{K_i}{S}$	

$\frac{M(s)}{E(s)} = k_p(1+\frac{1}{T_i S})$	

Proportional plus integral control

Note that the integral control action has converted the order of the control system.
And the steady-state error becomes zero.

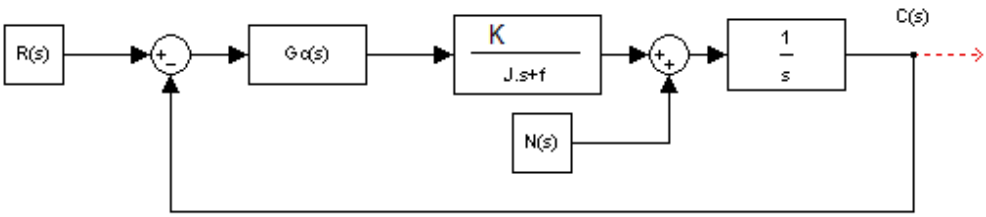


Figure B

A) First order system

Procedure:

1- Simulate the system in fig.A. For K=2: T=2

2-Plot the output signal and the error signal.

3-Record and measure the values with respect to k_P given in table 7.1.

Table 7.1 r(t)=unit step

Kp	**t(s)**									
1	c e									
5	c e									

4-plot the data from table 7.1on Fig.7.1

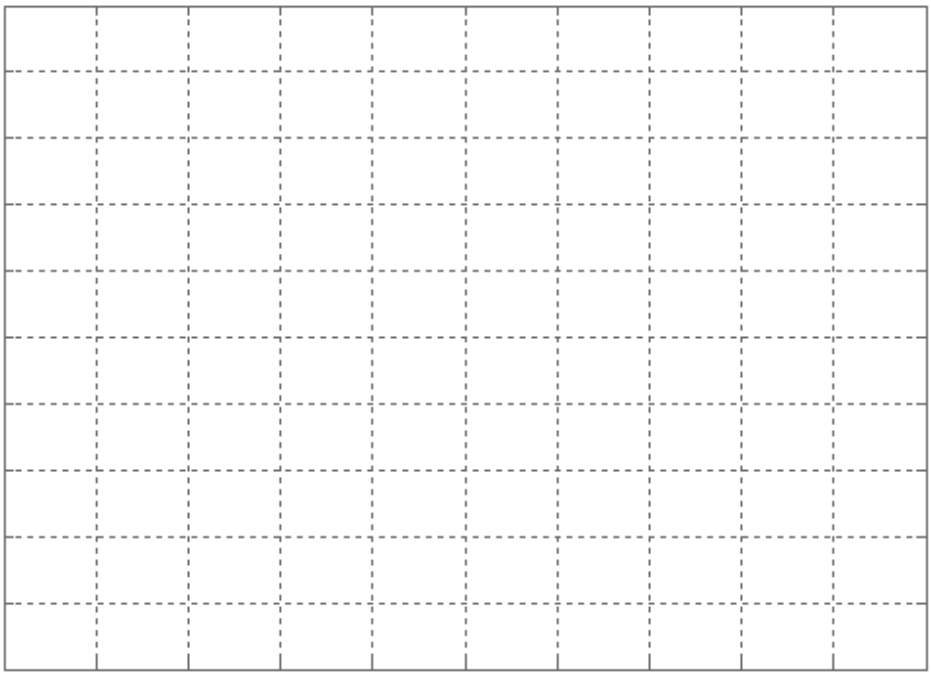


Fig. 7.1

Comments:

5-Use integral controller instead of P controller . For K=2: T=0.2

6- Record and measure the values with respect to K_i given in table 7.2

Table 7.2 r(t)=unit step; simulation stop time 2 seconds

Ki	**t(s)**									
10	c e									
20	c e									

7-plot c(t) from table 7.1 on fig.7.2.a, and e(t)on fig.7.2.b

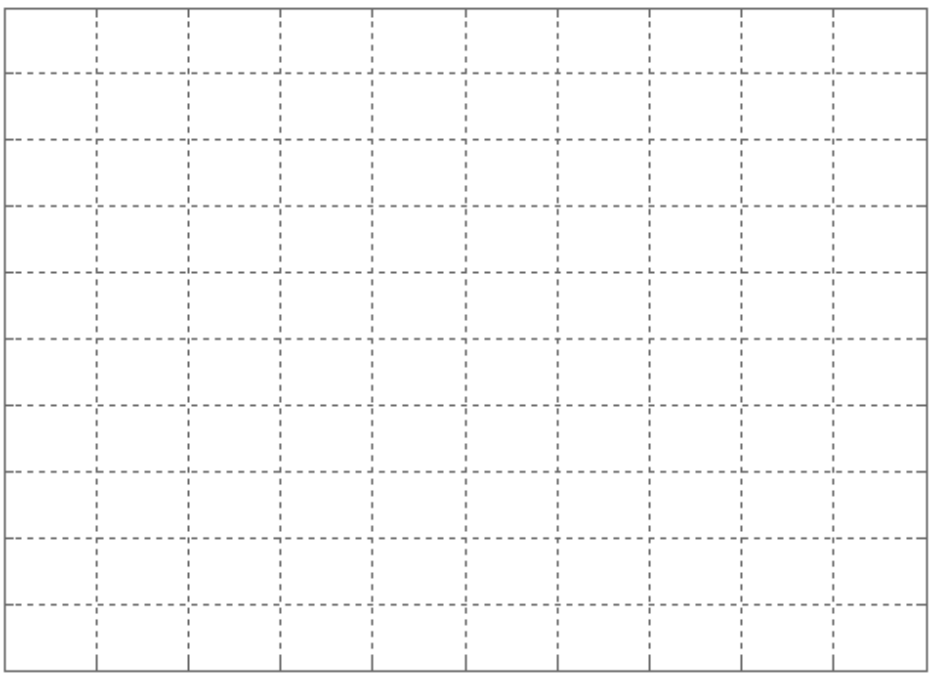


Fig. 7.2.a

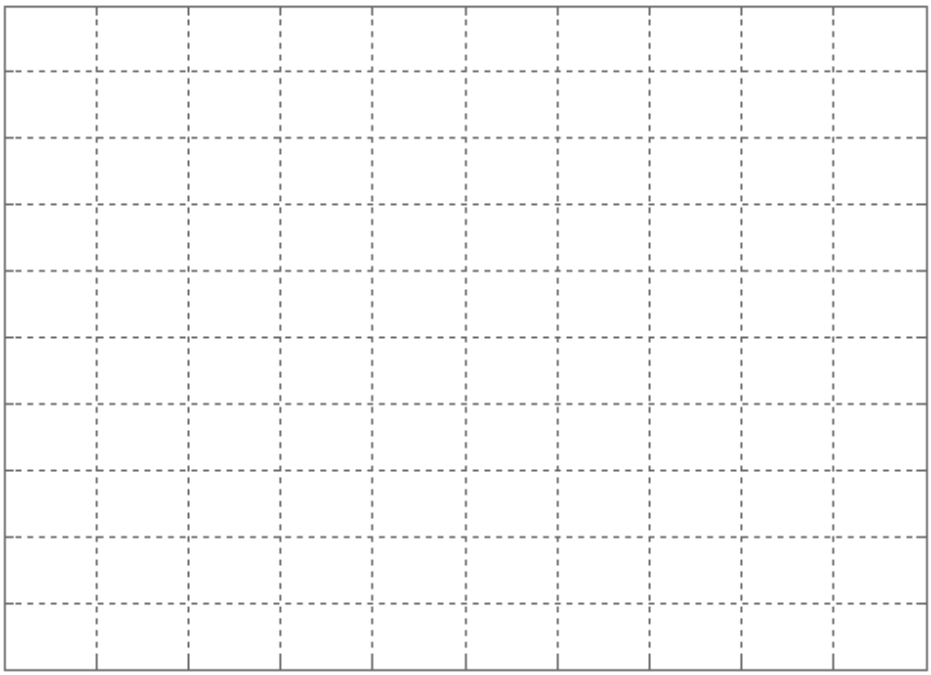


Fig. 7.2.b

Comments:

B) Second order system

Procedure:

1- Simulate the system in fig.B For K=1; j=1; f=1.

2-Plot the output signal and the error signal.

3-Record and measure the values with respect to k_P given in table 7.3.

Table 7.3 .a: r (t) =constant=5; n (t) =0

Kp	t(s)									
1	c e									
4	c e									

Table 7.3.b: r (t) =constant=5; n (t) =constant=1

Kp	t(s)									
1	c e									
4	c e									

4-plot the data from table 7.3.a on one figure 7.3.

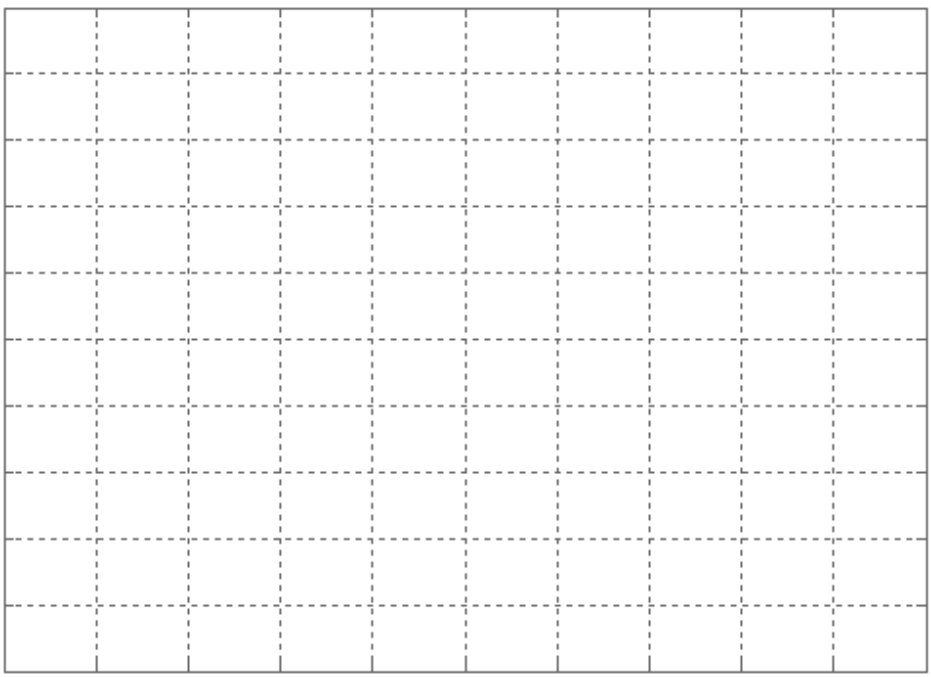


Fig.7.3.

Comments:

5-plot the data from table 7.3.b on one figure7.4

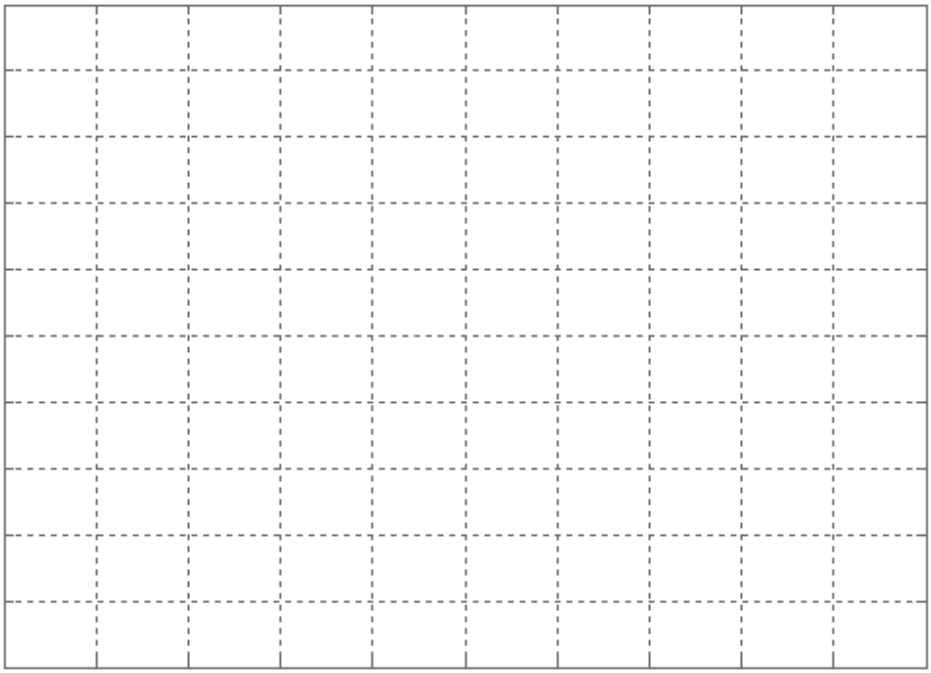


Fig.7.4

Comments:

6-Replace the proportional P controller with integral controller.

7- Note that the system is unstable for all Ki

Comments:

8-Apply a PI controller to system of fig.B

9- Vary the parameters of the controller to have a reasonable response: overshoot ≤20%, zero steady state error

10-Record the results in table 7.5.a

Table 7.5.a: K_p= :Ti = ; r (t) =constant=5; n (t) =0

t[*s*]												
c												
e												

11- Using the obtained parameters determine $c_n(t)$. Record the results in table 7.5.b

Table 7.5.b: K_p= :T_i= ; r (t) =constant=5; n (t) =constant=1

t[*s*]												
c												
e												

11-plot the data from table 7.5 on one figure 7.6

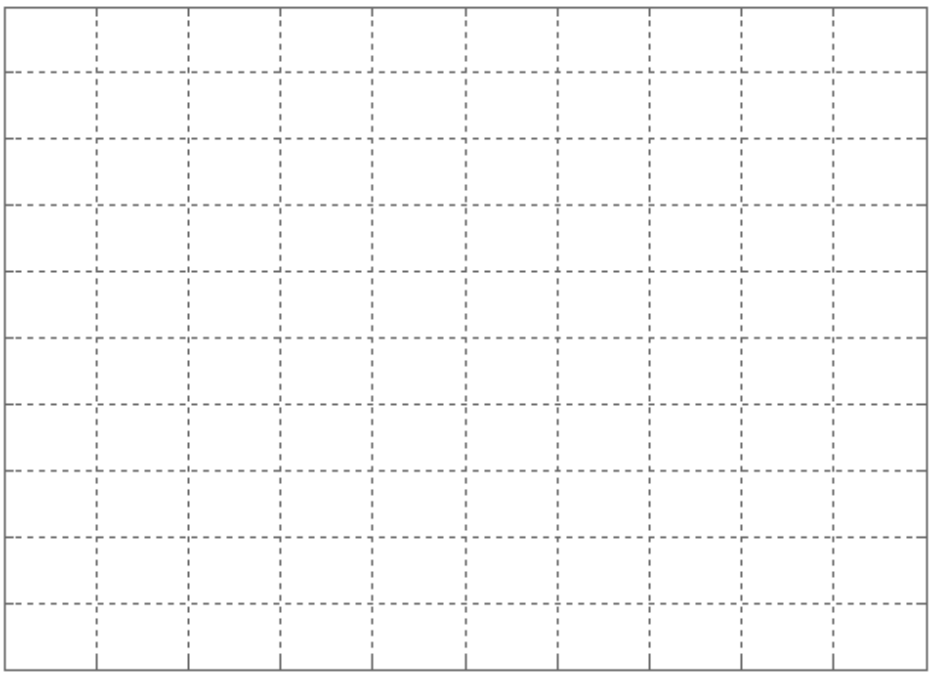


Fig.7.6

Comments:

Conclusion

Basic Control Actions (Part Two)

Objectives:

1 To study the PD control action.
2 To study the PID control action.

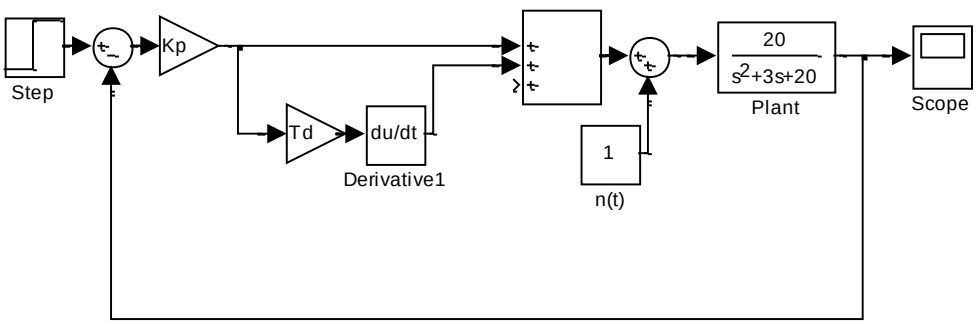
Introduction:

The control action of a proportional-plus-derivative controller is defined by the following equation:

$$\mathbf{m(t) = Kp \cdot e(t) + K_d \frac{de(t)}{dt} = K_p[e(t) + T_d \frac{de(t)}{dt}]}$$

And the transfer functions:

$$\mathbf{\frac{M(s)}{E(s)} = Kp(1 + T_d S)}$$



Proportional-plus-derivative control

Where Kp represents the proportional sensitivity and Td represents the derivative time constant. The derivative control action, sometimes called rate controller where the magnitude of the controller output is proportional to rate of change of the actuating error signal and not the actuating error itself. For this reason this model never used alone, it is always used in combination with proportional or proportional-plus-integral action. Derivative control does not affect the steady state error directly, it adds damping to the system and this permits the use of a large value for the gain.

Consider a second order control system with a small damping ratio figure 7.1, the roots of the characteristic equation are complex-conjugate, and the response to a unit-step input contains to oscillate. The addition of PD controller will reduce the oscillation of the output and reduce the steady state error .The addition of integral control will eliminate the steady state error

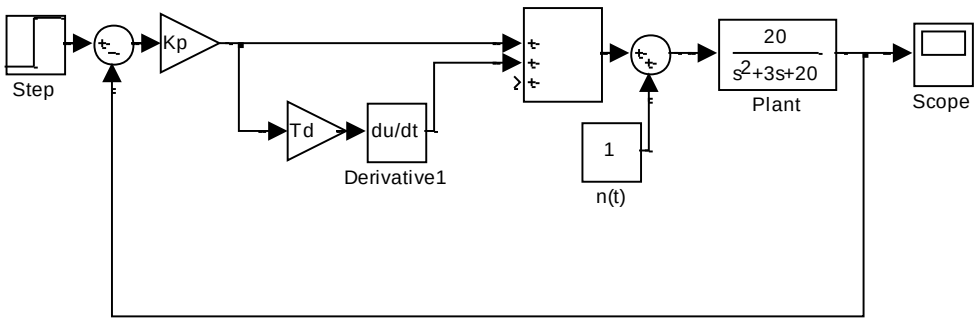


Fig.7.1.a

Procedure:

1. Connect the system shown in figure 7.1.a
2. Plot, measure and record the output signal and the error signal with respect to table 7.1.a

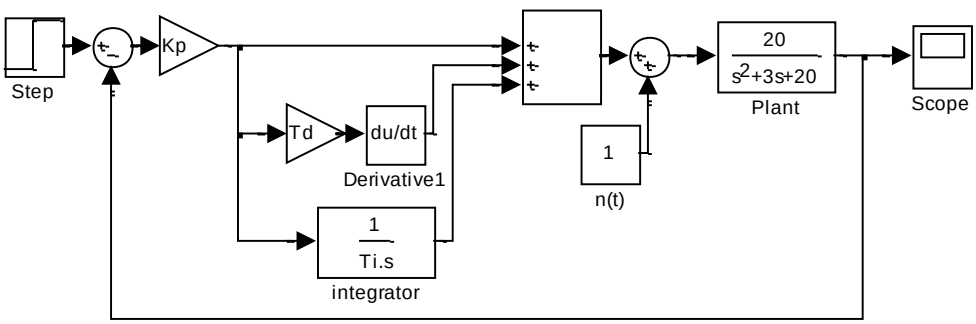
Table 7.1.a ,r(t)=unit step

Td	K_P	**t**										
0	10	C(t)										
5	10	C(t)										

3-Plot c (t) and e (t) from table 7.1.a on one graph.(Fig.7.7)

Fig.7.7

4-Add an integrator to circuit of figure 7.1 .a as shown



PID control

5-Select the time constant of integration so that the max. Overshoot is $\leq 20\%$. Record in table 7.2

Table 7.2, ir(t)= unit step

Td	K_P	t	0	.5	1	1.5	2	3	4	5	6	7	8	∞
10	5	c												
		e												

6-Plot c (t) and e (t) from table 7.2 on one graph.(fig.7.8)

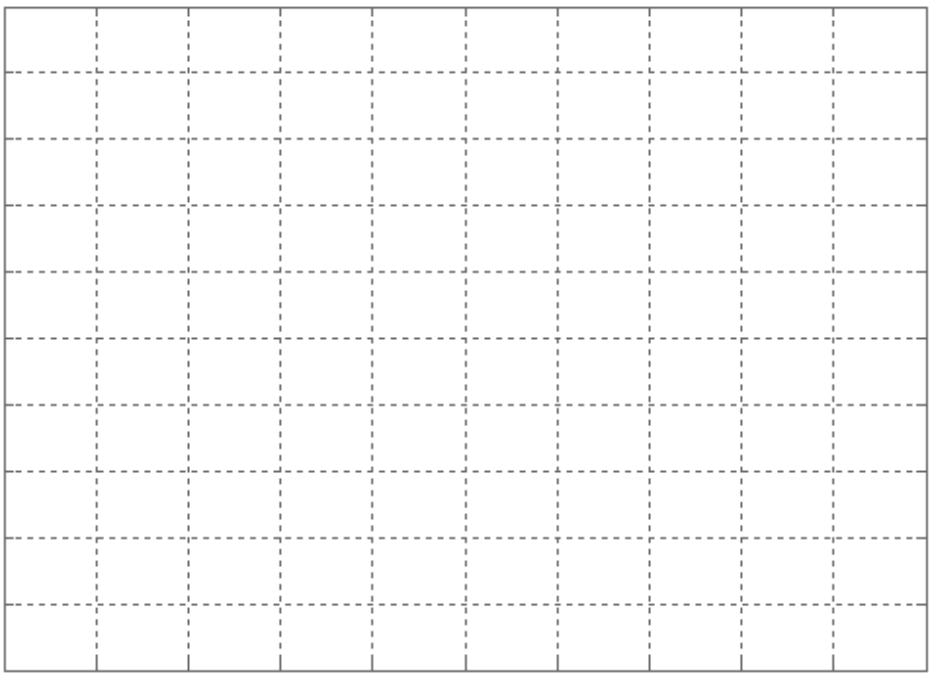


Fig.7.8

Comments

Experiment No 8
Tuning Rules of PID Controllers

Objectives:

To study the tuning rules of PID controllers for:
A) S-shaped output of a control system
B) An under damped out put of control system

Introduction:

The process of selecting the controller parameters to meet given performance specifications is known as controller tuning. Many rules are suggested to do that (meaning to set values K_p, Ti and T_d) based on experimental step response or based on the value of K_p.

Ziegler –Nichols Rules

There are two methods called Ziegler-Nichols tuning rules. In both methods, they aimed at obtaining 25%maximum overshoot (Mp) in step response as shown in figure (8-1).a

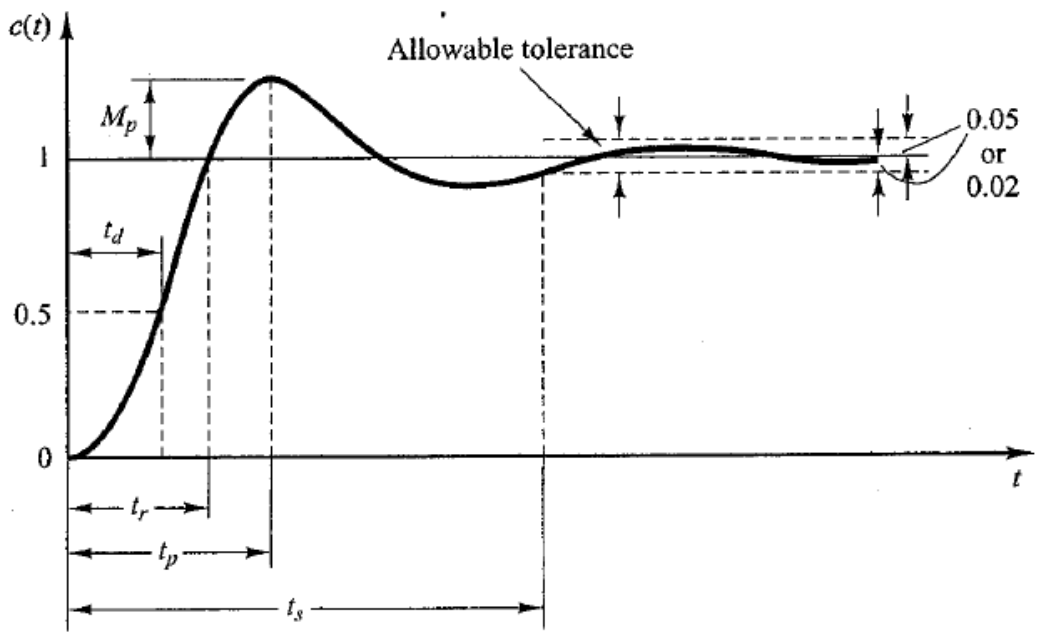


figure (8-1).a

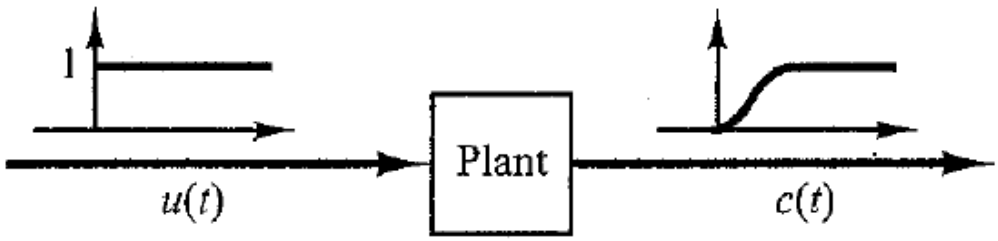


Figure 8.1

First Method : If the plant involves neither integrator(s) nor dominant complex-conjugate poles, the unit step response curve may look like an S-shaped curve (figure 8-2)

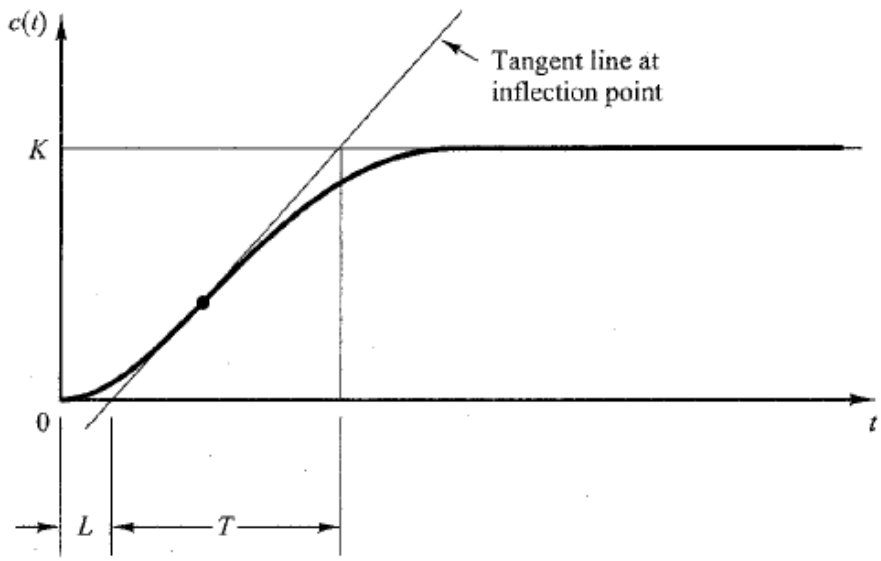


Figure 8.2

Instruments And Components:

- PC with Simulink

Procedure:

1. Simulate a plant given as: $G(s) = \dfrac{4}{(s+1)(s+4)}$
2. Plot the output signal to a unit-step input

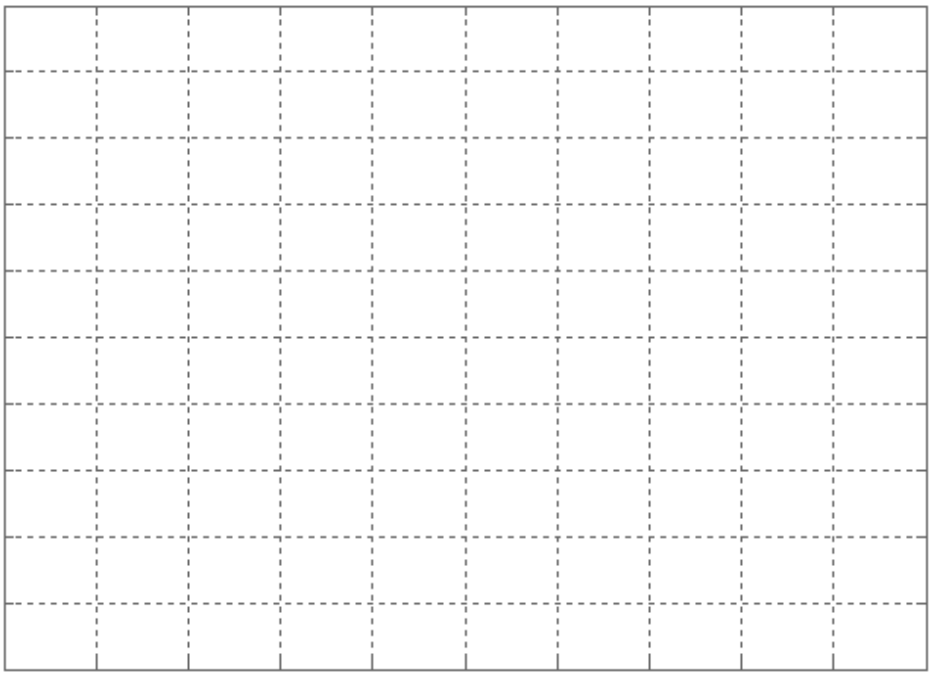


Fig.8.3

3. Draw a tangent line at the inflection point as shown in figure (8-2)
4. Determine the values L and T
5. Connect to the plant a unity feedback and a PID controller with a transfer function: $G_c(s) = K_p\{1 + T_d s + \frac{1}{T_i s}\}$
6. Set the value of K_p, T_i and T_d as in the following table

$K_p =$ $T_i =$ $T_d =$

Type of controller	K_p	T_i	T_d
PID	1.2T/L	2L	0.5L

7.Plot the output signal to a unit step input for each case. Ziegler-Nichols Tuning Rule Based on Step Response of Plant (First Method)

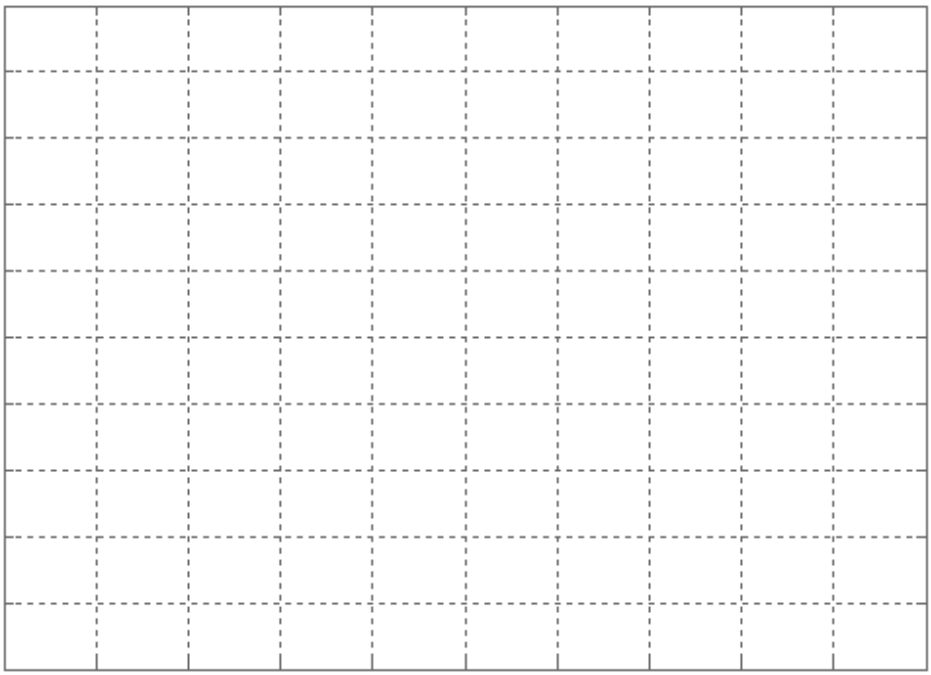


Fig.8.4

7. Tune each controller to obtain 25%maximum overshoot (Mp) in step response .Then plot the output on Fig.8.5

Tuned values of parameters

Type of controller	K_p	T_i	T_d
PID			

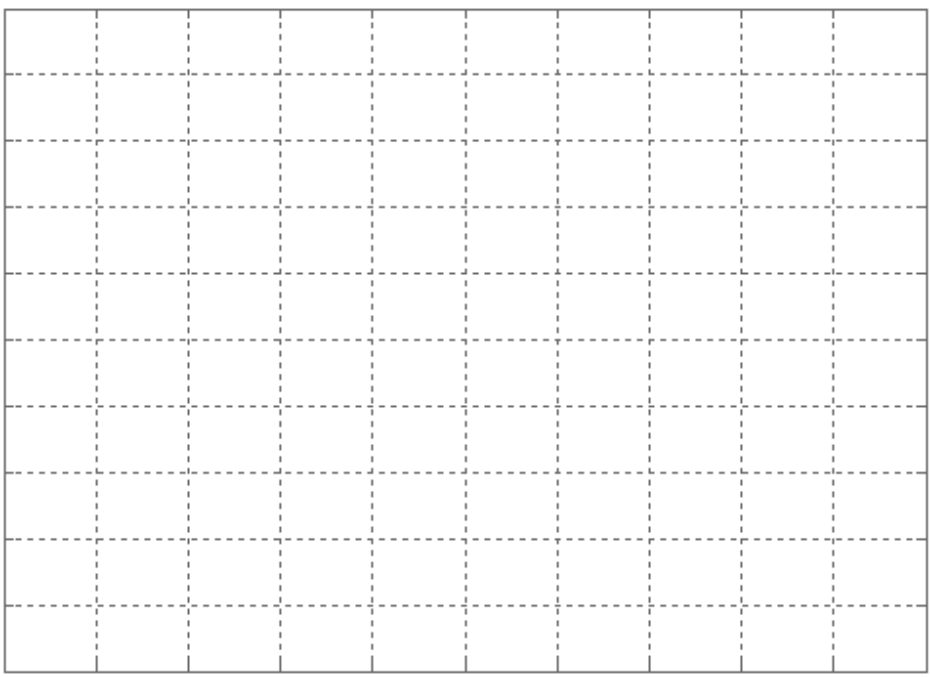
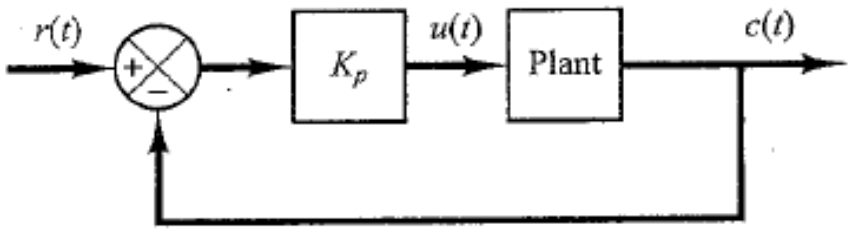


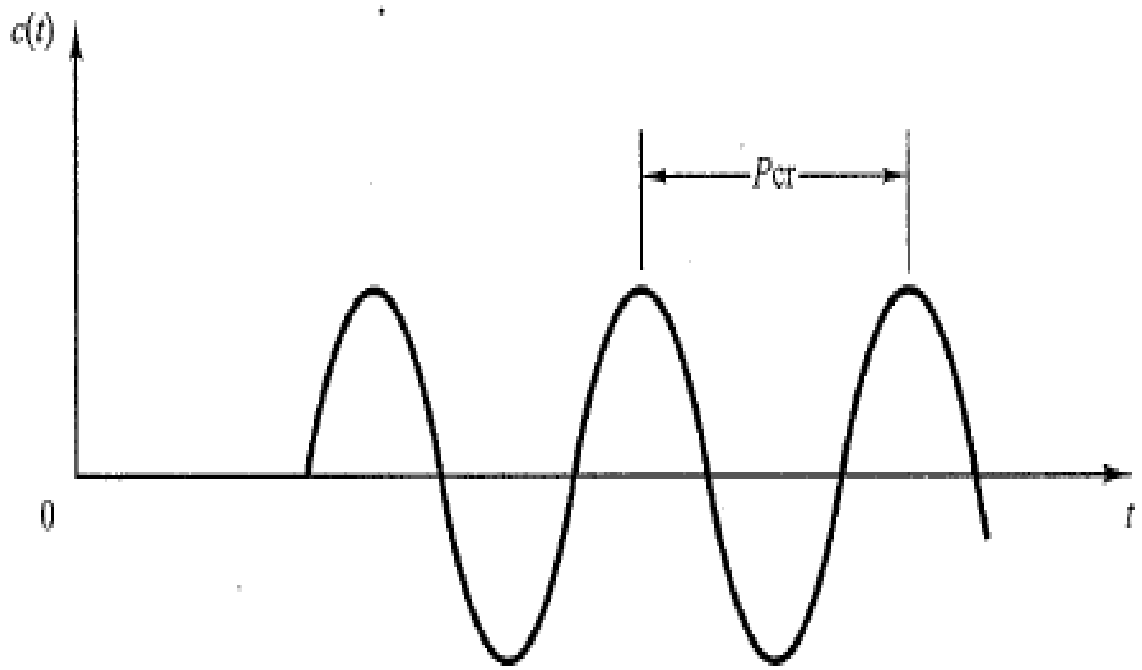
Fig.8.5

Second Method:

1) Simulate a plant given by: $G(s) = \frac{4}{s(s+1)(S+4)}$, with unity feedback and PID controller
2) Set T_i =inf. And T_d=0
3) Increase K_p from zero to a critical value K_{cr} where the output first exhibits sustained oscillations as shown

K_{cr}





4) Plot the output signal
5) Determine P_{cr} from the plot
6) Set K_p,T_i and T_d as in the following table
7) Plot the output signal for each case (fig.6.6)

Ziegler-Nichols Tuning Rule Based on Critical Gain Kcr and critical Period Pcr (Second Method)

Type of controller	K_p	T_i	T_d
PID	0.6Kcr	0.5Pcr	0.125Pcr

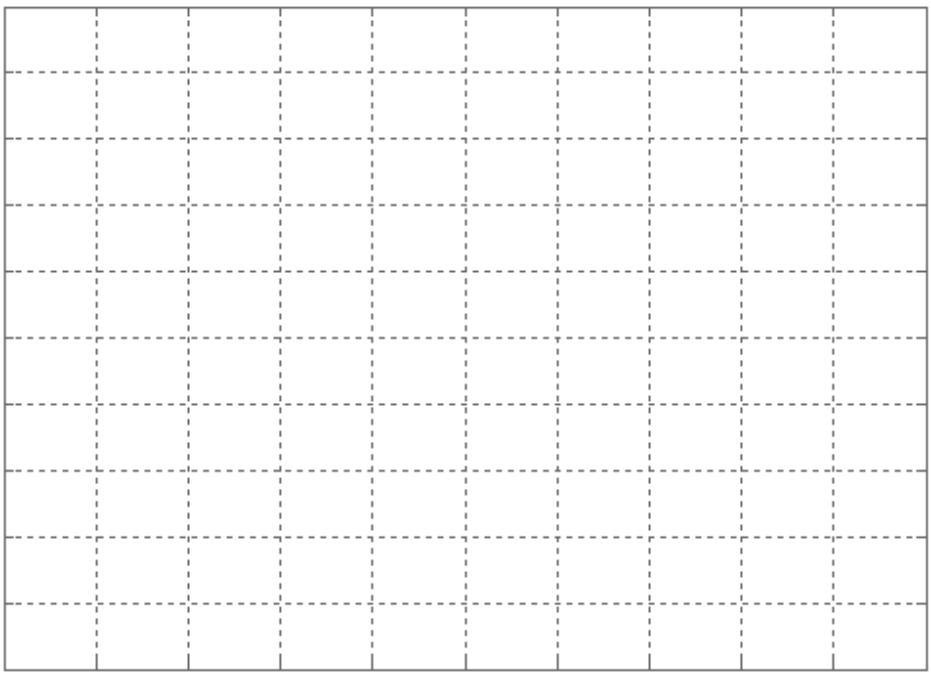
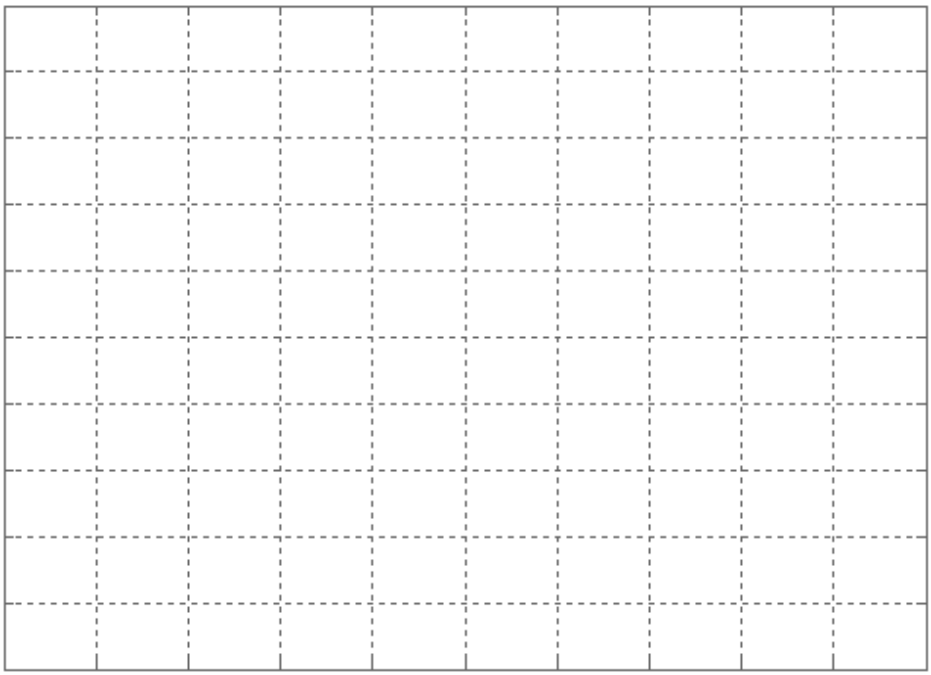


Fig.8.6

8.Tune each controller to obtain 25%maximum overshoot (Mp) in step response .Then plot the output on Fig.8.5

Tuned values of parameters

Type of controller	K_p	T_i	T_d
PID			



Summary:

Experiment No9

Frequency-Response Analysis

Objectives:

- To study the frequency-response of integral and derivative factors.
- To study the frequency-response of first order factor. And second order system

Introduction:

By the term "frequency response" we mean the steady-state response of control system to a sinusoidal input, where the frequency of the input signal is varied over a certain range the frequency-response characteristics of a system can be obtained directly from the sinusoidal transfer function, i.e. the transfer function is which s is replaced by $j\omega$, where ω is the angular frequency. If the transfer function of a control system is G(s), the sinusoidal transfer function becomes:

$$\frac{Y(j\omega)}{X(j\omega)} = G(j\omega)$$

Since G ($j\omega$) is a complex quantity, it can be rewritten if the following form:

$$G(j\omega) = |G(j\omega)|\, e^{j\phi}$$

Where |G ($j\omega$)| represents the magnitude and ϕ represents the angle of G ($j\omega$).

A stable linear time-invariant system subjected to a sinusoidal input will, at steady state, have a sinusoidal output of the same frequency of the input, but the amplitude and phase of the output will be different from those of the input. In fact, the amplitude of the output is given by the product of the input and the magnitude |G ($j\omega$)|, while the phase angle differs from that of the input by the amount ϕ.

For an integral factor:

The sinusoidal transfer function of is: G ($j\omega$) = $\frac{1}{j\omega}$.

The magnitude is: |G ($j\omega$)| = $\frac{1}{\omega}$,

And phase angle is $\phi = -90^o$.
The logarithmic magnitude Lm is: 20 log |G ($j\omega$)| = -20 log (ω).

For the derivative factor
Similarly,

|G ($j\omega$)| = ω,

$\phi = 90^o$,

And Lm = 20 log (ω).

For the first order factor:

|G ($j\omega$)| = $\left|\frac{1}{1+j\omega T}\right| = \left|\frac{1}{\sqrt{1+\omega^2 T^2}}\right|$,

$\phi = -\tan^{-1}(\omega t)$,

And Lm = -20 log ($\sqrt{1+\omega^2 T^2}$)

Procedure:

1. Simulate each element as shown in tables
2. Plot and record the magnitude and the phase

Table 9.1 integrator

$\omega[rad/\sec]$												
Lm [dB]												
ϕ[deg]												

Table 9.2 differentiator

$\omega[rad/\sec]$												
Lm [dB]												
ϕ[deg]												

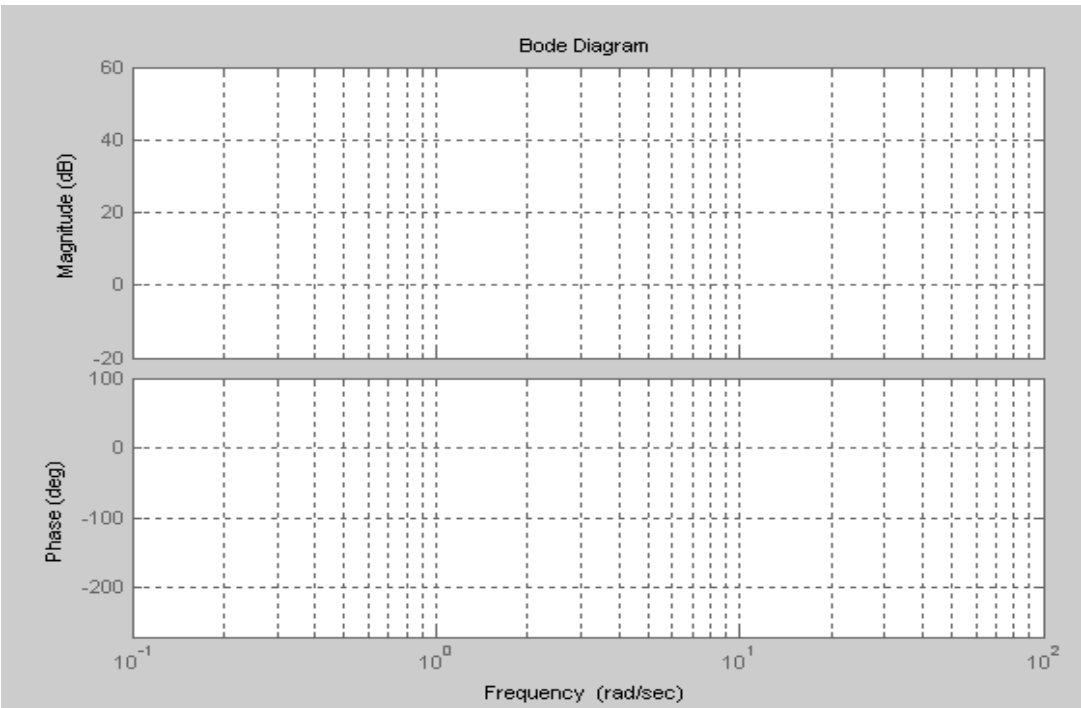


Fig.

Comments:

Table 9.3.a first order element: K= 1; T=1

$\omega[rad/\sec]$												
Lm [dB]												
ϕ[deg]												

Table 9.3.b first order element: K= 10; T=1

$\omega[rad/\sec]$												
Lm [dB]												
ϕ[deg]												

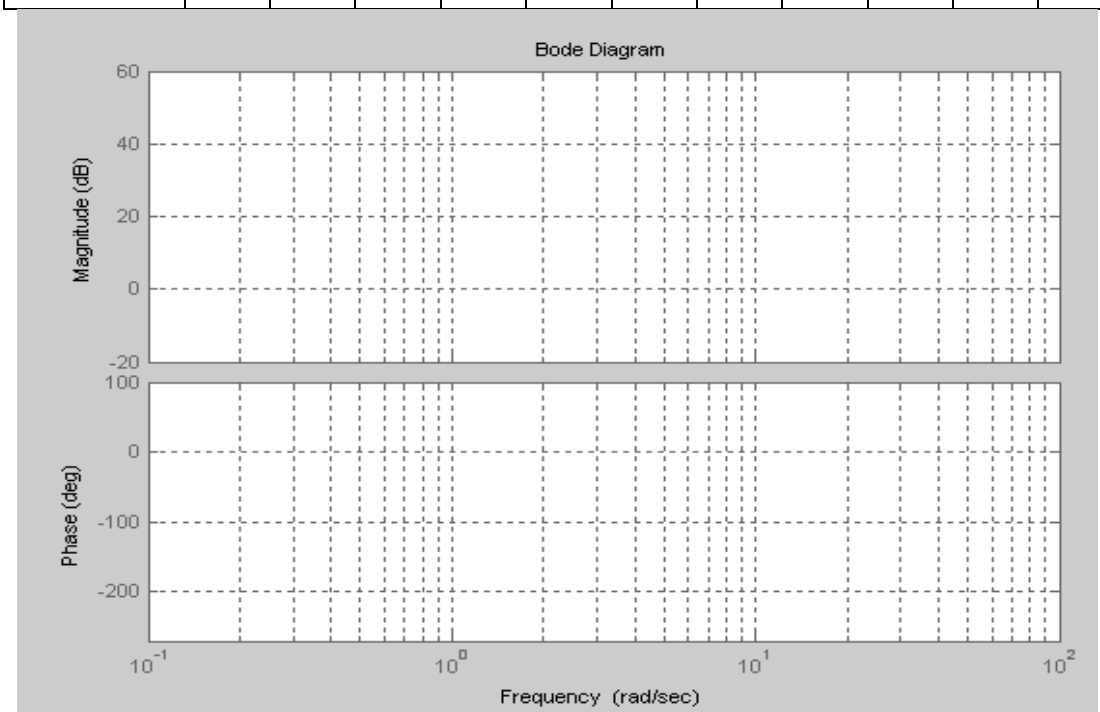


Table 9.4.a first order element: K= 1; T=0.1

$\omega[rad/\sec]$												
Lm [dB]												
ϕ[deg]												

Table 9.4.b first order element: K= 1; T=. 01

ω[rad / sec]												
Lm [dB]												
ϕ[deg]												

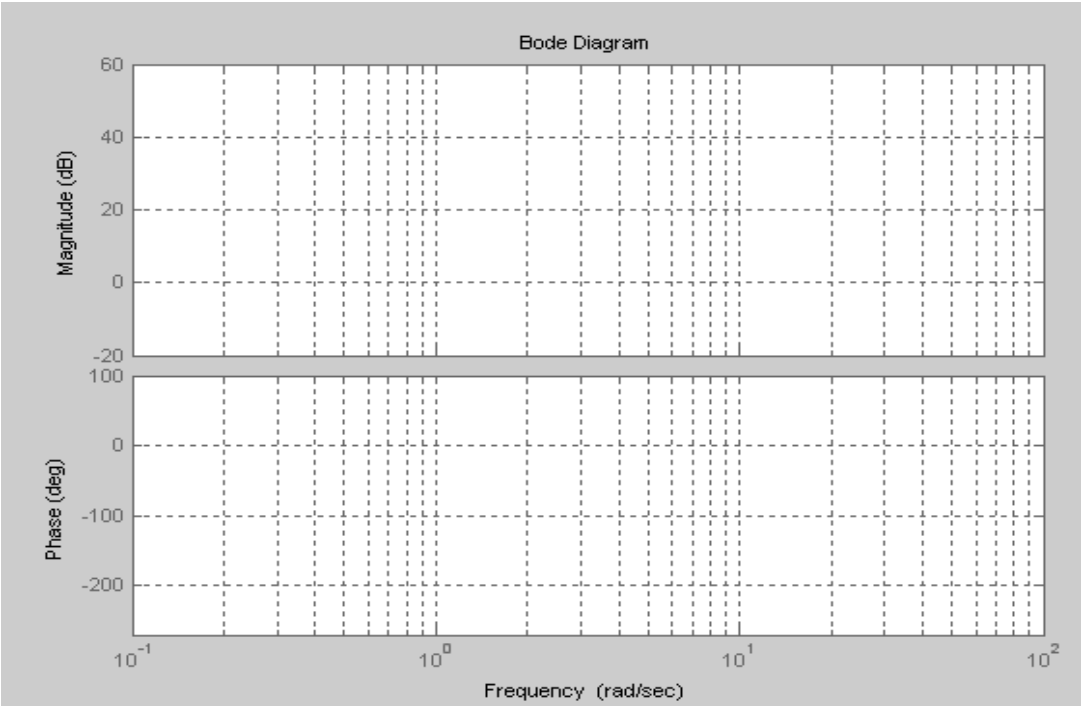


Fig.

Comments

- frequency-response of a second order system

Introduction:

A second order is given as: $\frac{C(s)}{R(s)} = \frac{\omega_n{}^2}{s^2 + 2\zeta\omega_n + \omega_n{}^2}$

Or the sinusoidal transfer function is: $\frac{C(j\omega)}{R(j\omega)} = \frac{1}{1-\{\frac{\omega}{\omega_n}\}^2 + j2\zeta\frac{\omega}{\omega_n}}$

And Lm = -20 log $\sqrt{\{1-\frac{\omega^2}{\omega^2{}_n}\}^2 + (2\zeta\frac{\omega}{\omega_n})^2}$

$$\phi = -\tan^{-1}\{\frac{2\zeta\frac{\omega}{\omega_n}}{1-\frac{\omega^2}{\omega_n{}^2}}\}$$

Procedure:

1-Simulate a second order system and record as shown in the following tables

Table 9-5.a : second order element: $\zeta = 0.1$; $\omega_n = 1$

$\omega[rad/\sec]$												
Lm [dB]												
ϕ[deg]												

Table 9-5.b: second order element: $\zeta = 0.5$; $\omega_n = 1$

$\omega[rad/\sec]$												
Lm [dB]												
ϕ[deg]												

Table 9-5.c: second order element: $\zeta = 2;\ \omega_n = 1$

$\omega[rad/\sec]$												
Lm [dB]												
ϕ[deg]												

2- Sketch the magnitude versus frequency and the phase angle versus frequency from table 9.5 on the same graph

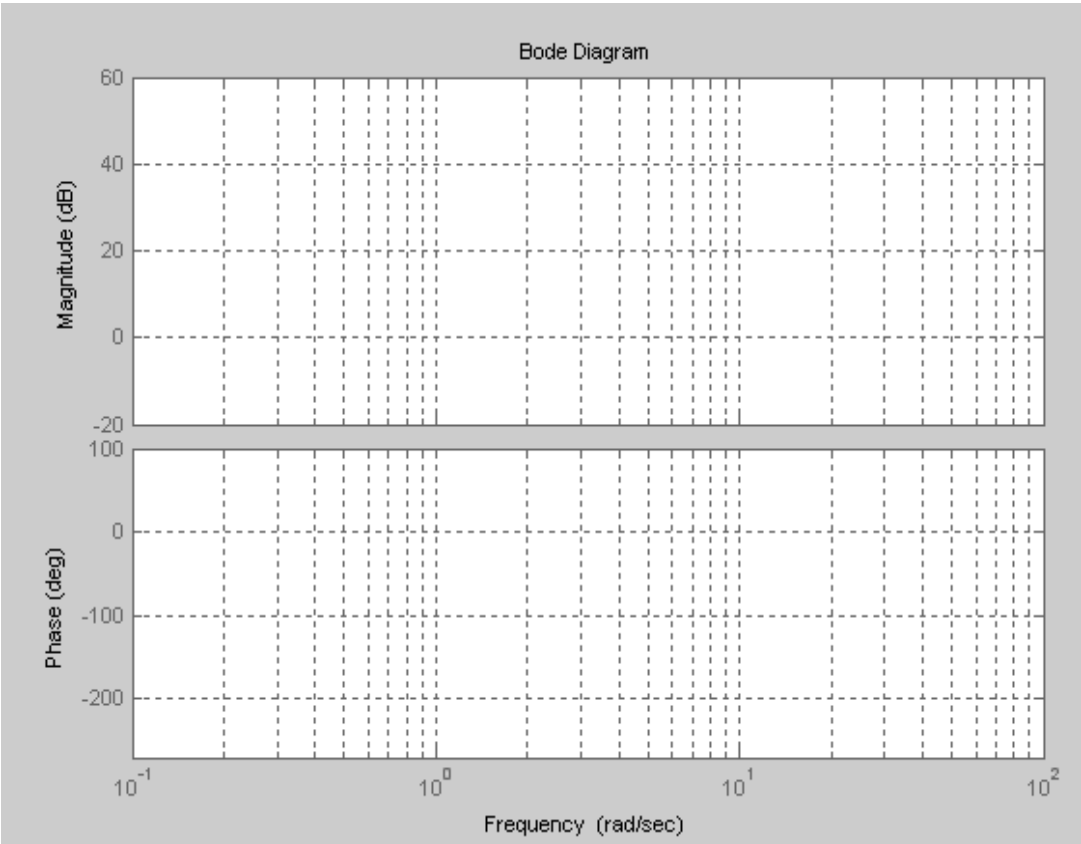


Fig.

Comments:

Analogue Servo Fundamentals Familiarization

Objective

When you have completed this assignment you will know:

- Realize that the 33-002 equipment comprises sub-systems which may be combined various ways to make control systems.
- Be familiar with two of the sub-systems, the Mechanical Unit and the Analogue Unit.

Equipment Required

Qty	Apparatus
1	33-110 Analogue Unit
1	33-100 Mechanical Unit
1	Power Supply ±*15V* dc, *1.5A*; *+5V* dc, *0.5A* (eg Feedback 01-100)
1	Oscilloscope, storage or long-persistence, preferably with X-Y facility. (eg Feedback 1810- 01229)

Knowledge Level

Before you start this assignment you should:

- Have some experience of using an electric motor.
- Have some experience of handling electronic circuits.
- Know how to use an oscilloscope

Introduction

The 33-002 **Servo Fundamentals Trainer** is intended to provide students with a sound introduction to the principles of analogue servomechanisms, and by extension to those of closed-loop systems more generally.

The 33-002 consists of 2 units:

- Analogue Unit 33-110
- Mechanical Unit 33-100

Which are connected as in fig 1.1, where dotted boxes represent essential additional items?

A *Digital unit* the 33-120 is available which allows the Mechanical unit, 33-100, to be used in conjunction with a computer. Feedback ***Discovery II*** Software package 33-911 accompanies the Digital unit.

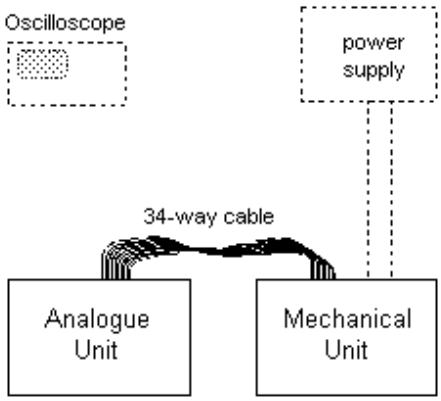


Fig 1.1 - Principal System Interconnections

Descriptions - Mechanical Unit (33-100)

Contains a power amplifier to drive the motor from an analogue or switched input. The motor drives the output shaft through a 32:1 belt reduction. The motor shaft also carries a magnetic brake disc and an analogue speed transducer (tachogenerator). A two-phase pulse train for digital speed and direction sensing is also derived from tracks on the brake disc.

The output shaft carries analogue (potentiometer) and digital (64 location Gray code) angle transducers.

The unit contains a simple signal generator to provide low frequency test signals; sine, square and triangular waves, and requires an external power supply providing:

+15V, *0*, *-15V* at *1.5A*

+5V, *0*, at *0.5A*

The Feedback Power Supply 01-100 is suitable.

Initial Mechanical and Analogue Unit check

Initially examine the **Mechanical** and **Analogue Units** without making any connections and identify all items mentioned in figs 1.9 and 1.10. The digital facilities in the Mechanical Unit, speed and output tracks will not be used with the Analogue Unit.

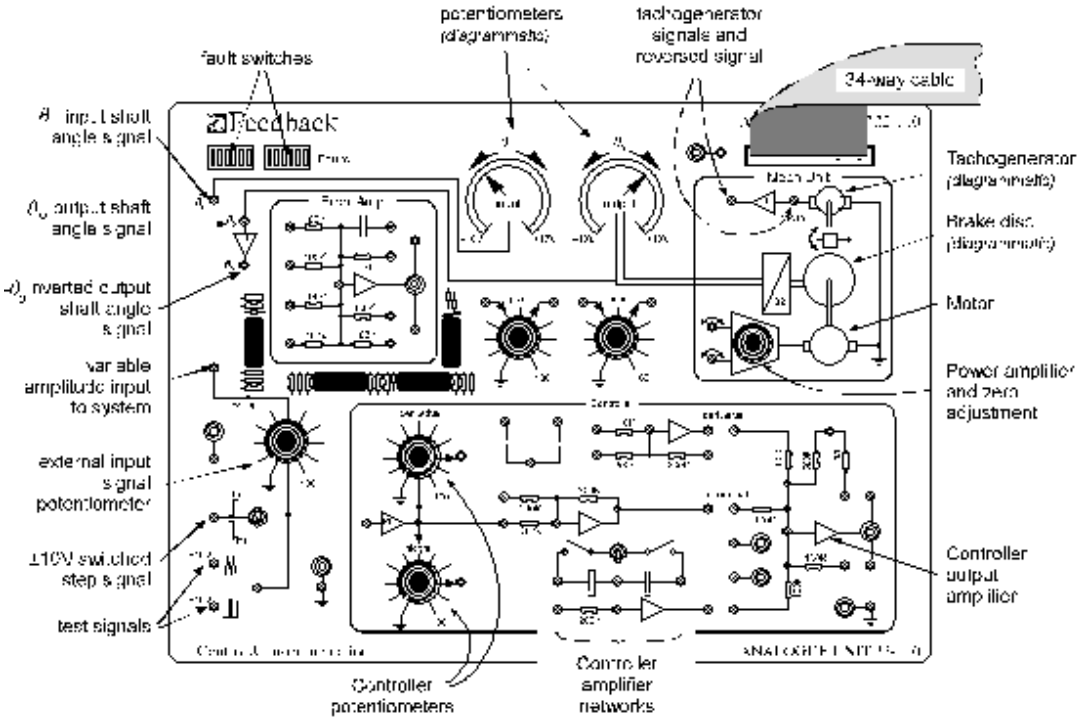


Fig 1.9 - The Analogue Unit

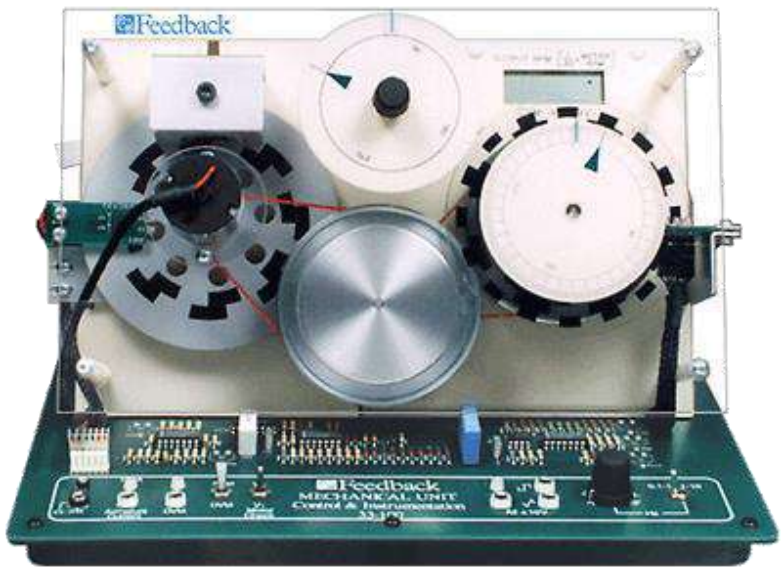


Fig 1.10 - The Mechanical Unit

With the power supply switched *off*, connect its outputs to the Mechanical Unit.

Set the brake fully upwards.

The Analogue Unit should not be connected.

Switch the power supply *on*.

The motor should remain stationary - there may be a slight movement when the supply is actually switched.

The output shaft speed display should show:

0.00

This indicates that the *5V* supply is operating.

Hold the ***motor check switch*** to the *right* and the motor should run clockwise and the output speed display should indicate *15* to *25 rpm*.

Hold the switch *left* and the motor should run anti-clockwise with approximately the same speed.

This test indicates that the *±15V* supplies are operating.

Hold the ***check switch*** to one side and gradually lower the brake to maximum. The motor should slow down.

These tests indicate that Power Supply and Mechanical Unit are operating correctly.

Switch the power *off*.

Connect the Analogue Unit to the Mechanical Unit by the 34-way cable.

Raise the brake fully.

Switch the power *on*.

Rotating the ***power amplifier zero adjustment*** should enable the motor to be driven in both directions up to about the same speed as before.

Zero the amplifier to stop the motor.

Overall the tests indicate that the system is working correctly.

To Display the Waveforms

It is assumed that a suitable oscilloscope is available with:

Either

- A single Y channel or preferably two Y channels when used in conjunction with a time base, with ...
- External sync input for the time base.

Or

- A facility for X-Y operation with X and Y both able to operate with a d.c input.

Test Waveforms

Connect the oscilloscope to the test signals using either the 4mm sockets in the Mechanical Unit or the 2mm terminals in the Analogue Unit.

Observe that the frequency may be varied between *0.1* and *1Hz* or *1* and *10Hz*.

System Waveforms

The system waveforms may be observed either from an externally triggered display against a time base as shown in fig 1.11(a) or from an X-Y display as shown in fig 1.11(b). Signal source sockets are provided on the Mechanical Unit (4mm) and the Analogue Unit (2mm).

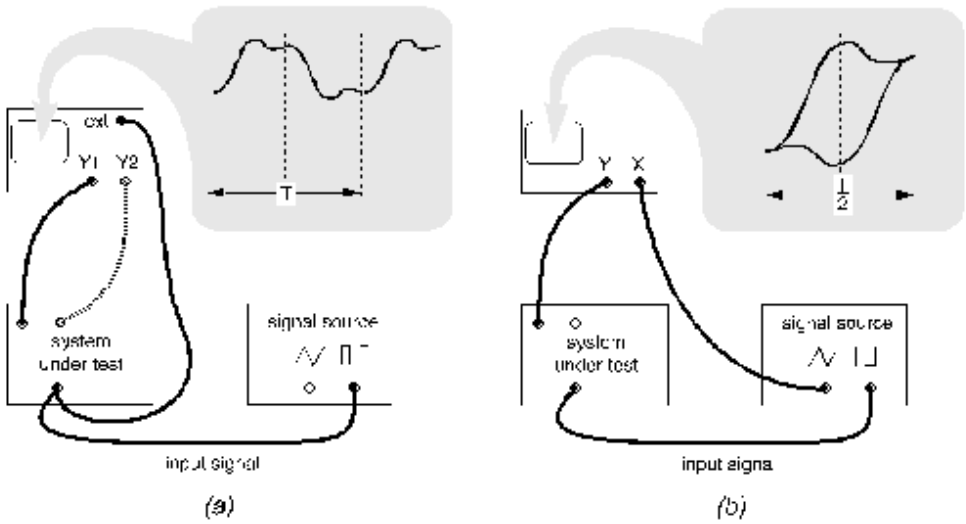


Fig 1.11 - Oscilloscope Connections and Display

Experiment No 10
Steady state and transient response of DC motor

[assignment 3]

Objective

When you have completed this assignment you will know:

- That the steady speed of the motor is ideally proportional to the applied voltage, less an amount proportional to load torque.
- That a d.c tachogenerator provides a signal representing speed, independent of motor loading.
- That the response of the motor to a change of input is not immediate, but may be expressed as a time constant.

Equipment Required

Qty	Apparatus
1	33-110 Analogue Unit
1	33-100 Mechanical Unit
1	Power Supply ±*15V* dc, *1.5A*; *+5V* dc, *0.5A* (eg Feedback 01-100)
1	Oscilloscope, storage or long-persistence, preferably with X-Y facility. (eg Feedback 1810- 01229)

Knowledge Level

Before you start this assignment you should:

- Have completed the Familiarisation assignment.
- Understand simple applications of operational amplifiers.

Introduction

The **motor** and **tachogenerator** were used to display the speed response characteristics as part of the general outline in the Familiarisation assignment. In the present assignment a more detailed and wider investigation is carried out.

The motor is a permanent magnet type and can be represented in idealised form as in fig 3.1(a), where R_a is the armature resistance and T_1, T_2 are the actual motor terminals.

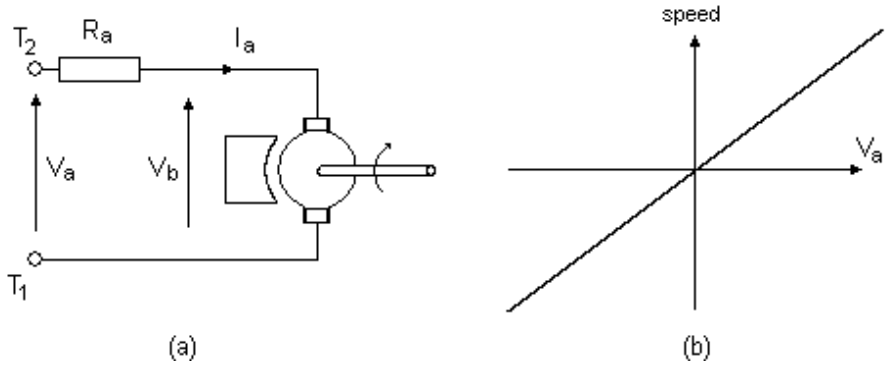


Fig 3.1 - Representation of a Motor in terms of an Ideal Motor.

If the motor is stationary and a voltage V_a is applied, a current I_a flows which causes the motor to rotate. As the motor rotates a back emf V_b is generated. As the motor speeds up the back emf increases and I_a falls. In an ideal (loss free) motor, the armature current falls to substantially zero and V_b approximately equals V_a. Thus if V_a is varied slowly in either polarity, the motor speed is proportional to V_a, and a plot of motor speed against V_a would have the form of fig 3.1(b).

In the 33-100 the armature voltage V_a is provided by a **power amplifier**. A power amplifier is necessary, because although the voltages in the error channel may be of the same order as V_a, the motor current may be up to *1A*, while the error channel operates with currents of less than 1mA and could not drive the motor directly. The amplifier has two input sockets, enabling the motor rotation direction to be reversed for a given input.

The **tachogenerator** is a small permanent magnet machine and hence when rotated produces an emf proportional to speed which can be used as a measure of the rotation speed.

Brake Characteristics

The **magnetic brake** consists of a permanent magnet which can be swung over an aluminium disc. When the disc is rotated eddy currents circulate in the area of the disc within the magnet gap, and these react with the magnet field to produce a torque which opposes rotation. This gives an adjustable torque speed relation of the form of fig 3.2, and provides a very convenient load for the motor.

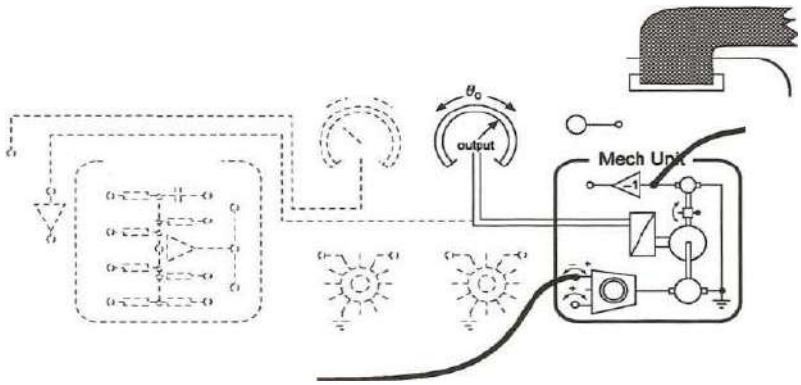


Fig. 3.3

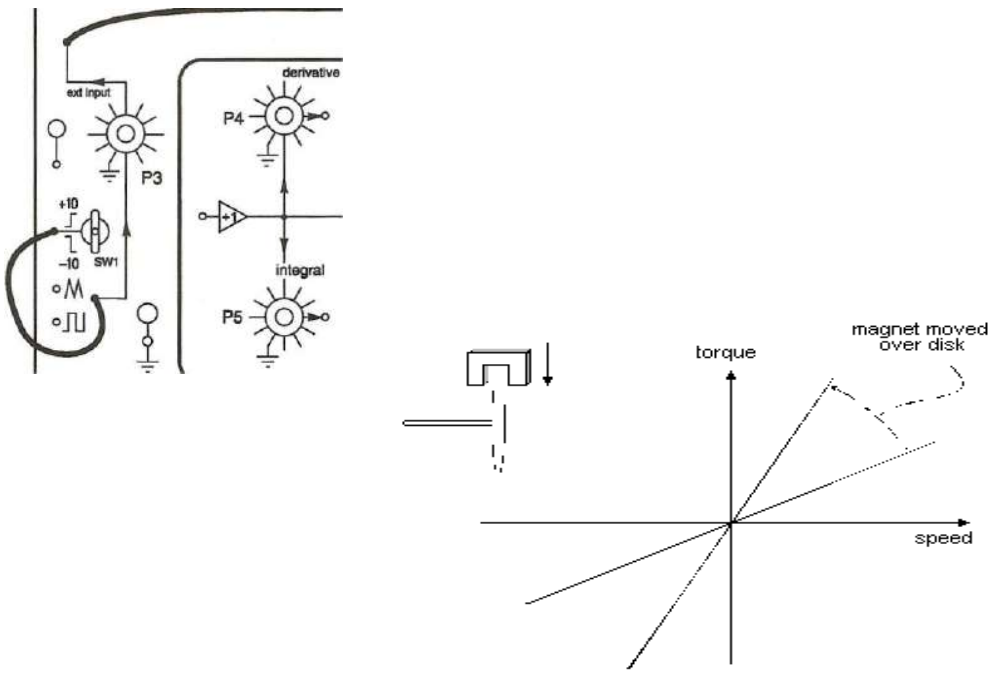


Fig 3.2 - Characteristic of Magnetic Brake

The overall characteristics of a motor may be considered from two aspects, both of which can be related to the idealised representation of fig 3.1(a). These aspects are:

- **Steady-state**, which are concerned with constant or very slowly changing operating conditions, and
- **Transient**, corresponding with sudden changes.

Both are important in control system applications.

Patching diagram A

Practical 3.1 Steady-State Characteristics

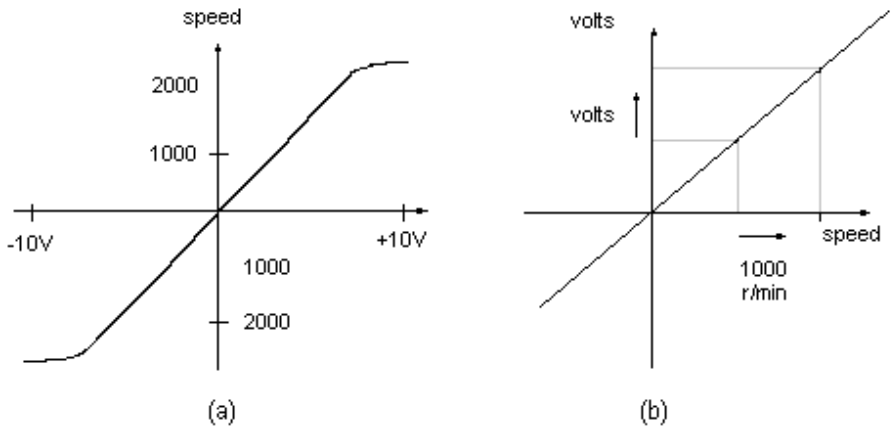


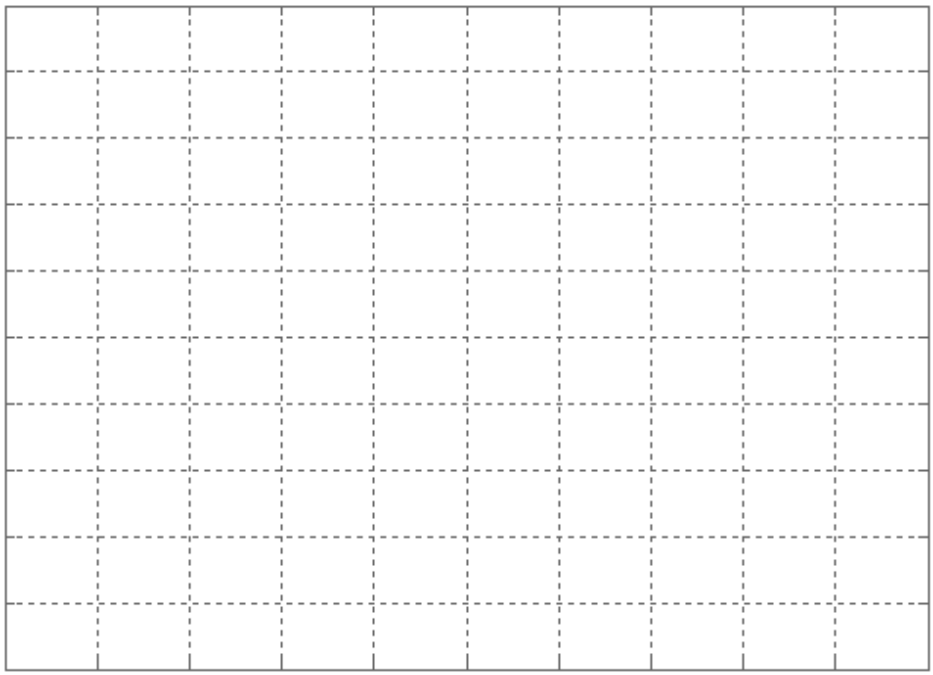
Fig 3.4 - Characteristics of Motor (a) and Tachogenerator (b)

In this practical the motor is operated in a range of steady-state conditions.

Arrange the system as shown in Fig. 3.3 ,where ***P_3*** enables a voltage in the range ±*10V* to be applied to the power amplifier.

Use the DVM on the 33-100 for voltage measurements. For each measurement set up the required steady state then switch between DVM and RPM.

By setting ***SW1*** and varying ***P_3***, make a plot of motor speed against amplifier input, in the range ±*10V*, scaling the vertical axis in units of *1000 r/min*. The plot should have the general shape of fig 3.4(a).



Plot 1:

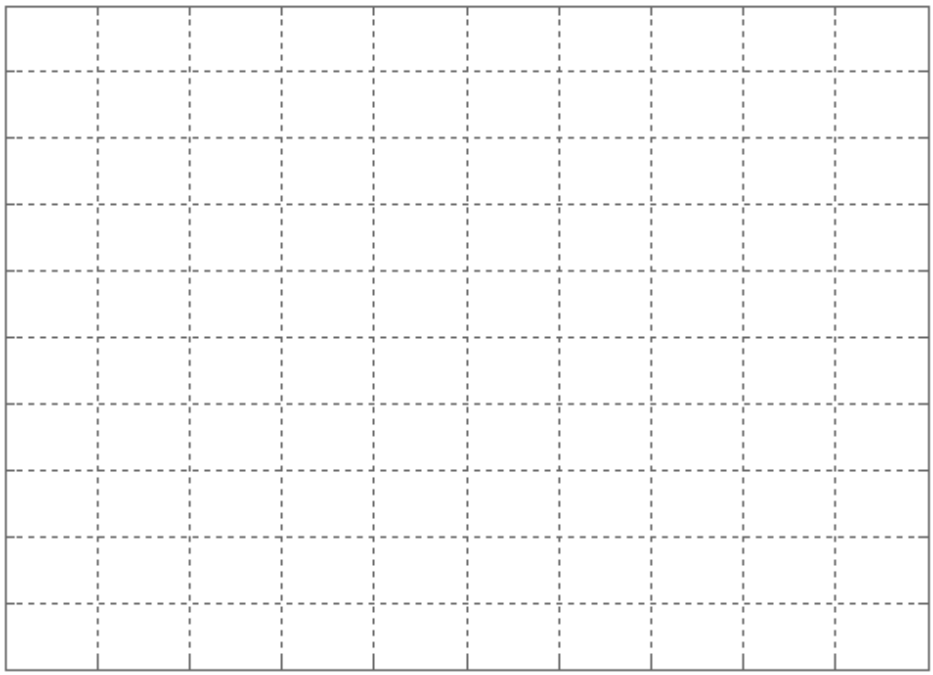
Note Since the reduction to the output shaft is 32:1, the motor speed is calculated by multiplying the r/min readng by 32. eg a reading of 31.25 = a motor speed of *1000 r/min*.

Initially the motor speed increases substantially linearly with the voltage to the amplifier because the motor back emf V_b, see fig 3.1(a), approximately equals the amplifier output, but finally the amplifier limits before the full *±10V* input is reached.

Tachogenerator

The tachogenerator provides a voltage proportional to speed, which is required for various aspects of control system operation.

Plot the tachogenerator characteristics by setting the motor speed to various values by ***P_3*** and measuring the generated voltage.



Plot 2:

An important parameter in the use of tachogenerators is the tachogenerator factor in volts per *1000 r/min*.

Determine the tachogenerator factor by measuring the change in generated volts for a speed change of *1000 r/min*.

The factor should be approximately *2.5V* per *1000 r/min*.

Practical 3.3 Transient Response of Motor

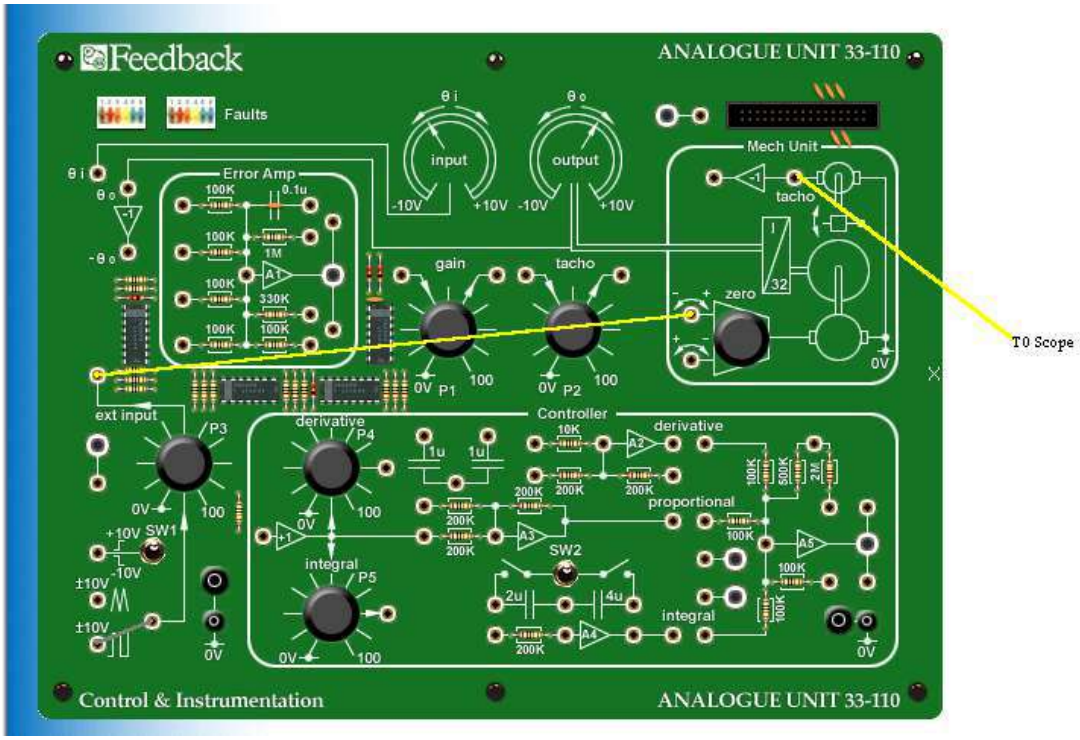


Fig. 3.6

The motor cannot change speed instantly due to the **inertia** of the armature and any additional rotating load (the brake disc in the 33-002). This effect has very important consequences for control system design.

If V_a for an ideal motor has a step form as in fig 3.7(a), initially a large current will flow, limited only by the armature resistance. As the motor rotates and speeds up the back emf increases and the current is reduced to nearly zero in an ideal motor. This is shown in the left portion of fig 3.7 (b). If V_a is suddenly reduced to zero the back emf still exists, since the motor continues to rotate, and drives a current in the reverse direction dissipating energy and slowing the motor. This is illustrated in the right-hand portion of (b).

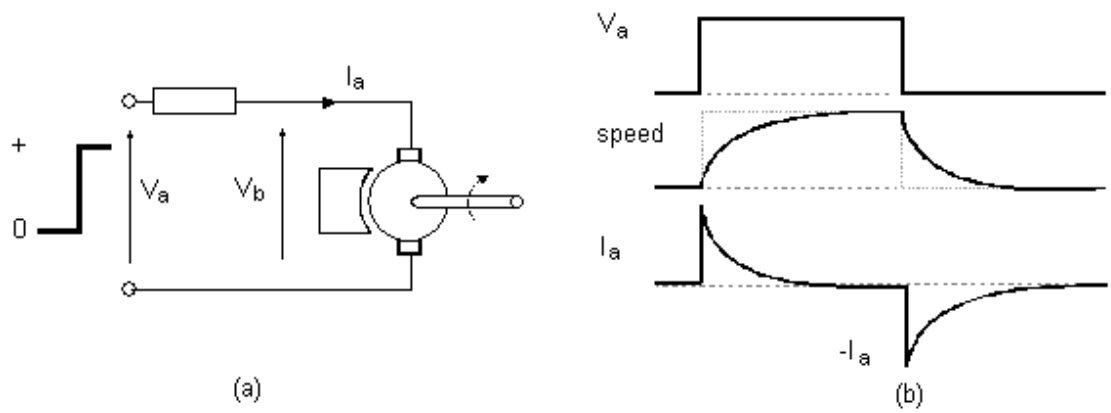


Fig 3.7 - Transient Characteristics of Motor

The 33-001 motor shows a speed characteristic approximating to fig 3.7(b), but the power amplifier is arranged to limit the maximum armature current which does not show the ideal pulse characteristic.

Connect the system as shown in Fig. 3.6, which enables the motor to be driven from the test square-wave, and allows the speed to be displayed on the Y axis of an oscilloscope. It is convenient to use an X-Y display.

Set ***P3*** to *zero* and the test signal frequency to *0.2Hz*.

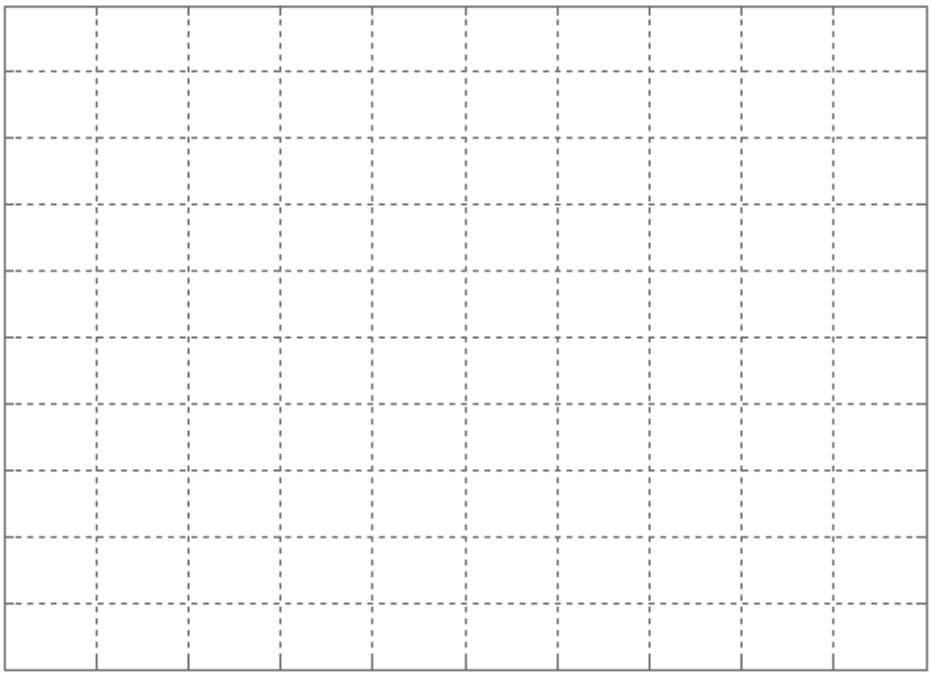
Set the ***power amplifier zero adjustment*** to run the motor at maximum speed in one direction.

Turn up ***P3*** and the square-wave signal will speed up and slow down the motor.

Adjust ***P3*** until the motor is stationary for one half cycle. This corresponds with V_a in fig 3.7(b).

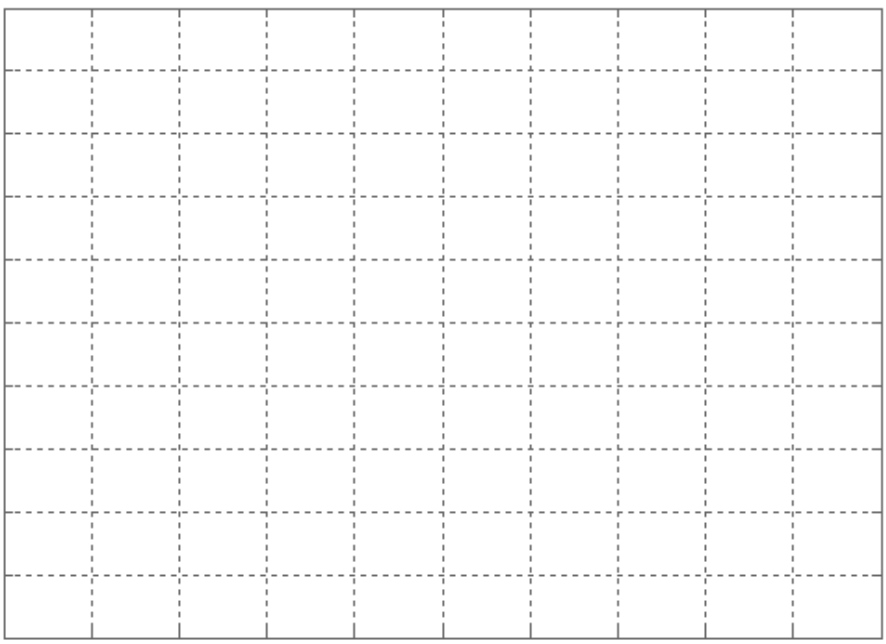
The oscilloscope will now display the speed corresponding with V_a in (b).

Plot the speed vs. time for two values of the input signal:



Plot 3:

Plot the current vs. time for two values of the input signal:



Plot 4:

Summary:

Experiment No 11

Speed Control System [assignment 9]

Objective

When you have completed this assignment you will know:

- Velocity feedback can be used (without position feedback) to enable a speed to be closely regulated.The polarity of the feedback is important (as for position feedback).
- The effectiveness of the control depends mainly on the gain employed.

Equipment Required

Qty	Apparatus
1	33-110 Analogue Unit
1	33-100 Mechanical Unit
1	Power Supply *±15V* dc, *1.5A*; *+5V* dc, *0.5A* (eg Feedback 01-100)
1	Oscilloscope, storage or long-persistence, preferably with X-Y facility. (eg Feedback 1810- 01229)

Knowledge Level

Before you start this assignment you should:

- Be familiar with the equipment.

Introduction

The previous assignments have been concerned with position control, but an important aspect of closed-loop control is speed control, which has many industrial applications, varying from heavy industrial, such as paper mills or steel rolling mills, to tape or video transport mechanisms.

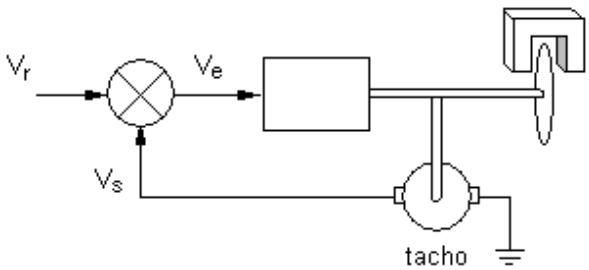


Fig 9.1 - Essential features of a *Closed Loop Speed Control*

The essential principle of closed-loop speed control is similar to position control, as illustrated in fig 9.1, except that the feedback signal is now an output velocity signal V_s, normally from a tachogenerator, which is compared with a reference voltage V_r to give an error

$$V_e = V_r - V_s$$

In operation the reference is set to a required value, which drives the motor to generate V_s, which reduces the error until the system reaches a steady speed.

If the motor is loaded, eg with the magnetic brake on the 33-100, the speed falls; this tends to increase the error, increasing the motor drive and thus reducing the speed fall for a given load. Note that this implies negative feedback around the loop.

The speed fall with load, sometimes termed droop is a very important characteristic in speed control systems.

The rotation direction can be reversed by reversing the reference voltage, though many industrial speed control systems are required to operate in one direction only.

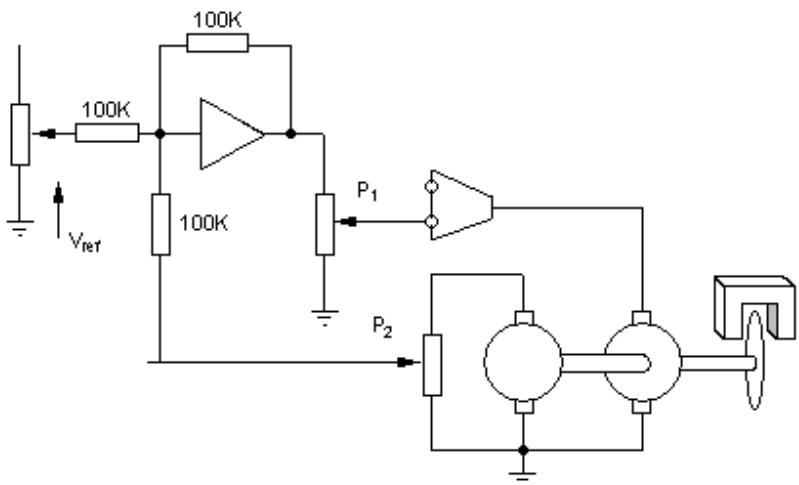


Fig 9.2 - Speed Control System

Practical 9.1 Closed-loop Speed Control

A speed control system which can be made with the 33-002 corresponding to fig 9.1, is shown in fig 9.2. Connect the board as shown in the Fig.9.3 .

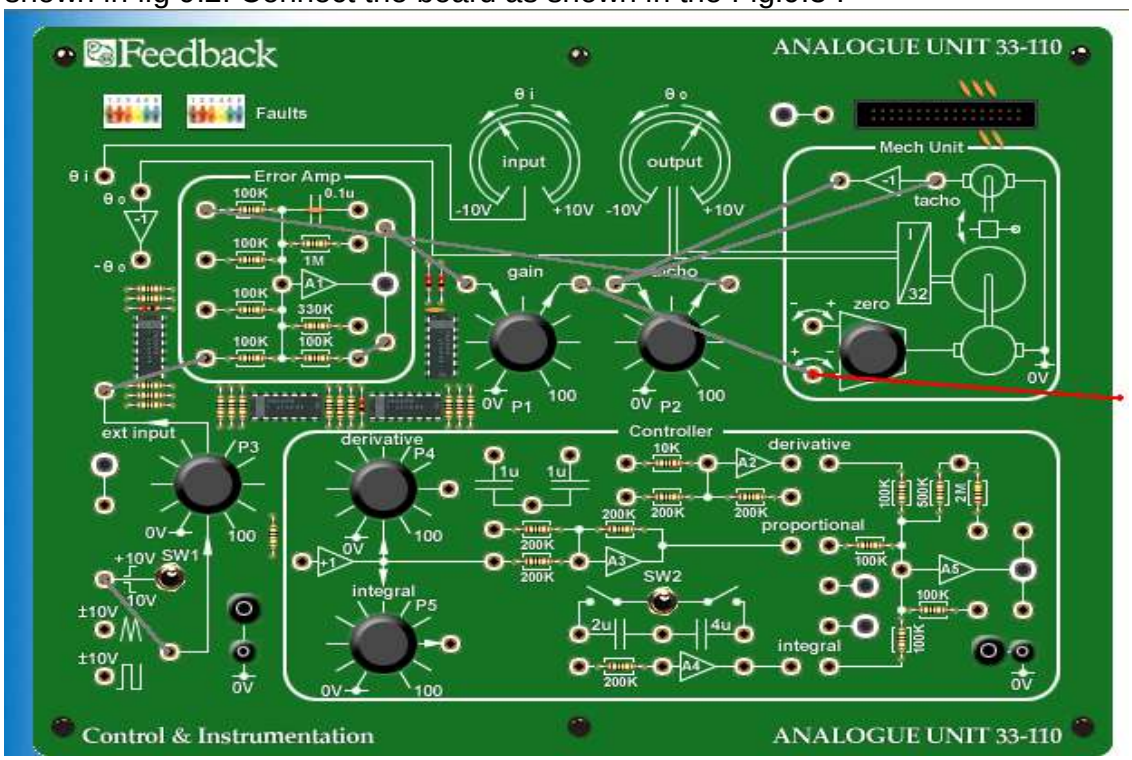


fig 9.3.

I-a Set ***P₂*** (tacho) to *zero* and set the amplifier feedback resistor to *100K* Ω, this gives G = 1. Set ***P₁*** to *100*. Set ***SW1*** up to +10 and adjust ***P₃*** to run the motor at *1000 r/min* (*31.25 r/min* at output).

Turn up ***P₂*** slightly, if the speed decreases the loop feedback is negative as required. If the speed increases use the other tachogenerator polarity.

Note that if the system has negative feedback and both the tachogenerator polarity and the power amplifier input are reversed, the system still has negative feedback, but the motor runs in the opposite direction.

I-b Set ***P₂*** to *zero* measure and plot the speed response for zero load and full load.

Time[ ]								
Speed[no load]								
Speed[Full load]								



Plot 1:

2- Set ***P_2*** to *100* and readjust ***P_3*** to give *1000 r/min* with the brake off. Measure and plot the speed response for zero load and full load.

Time[ ]								
Speed[no load]								
Speed[Full load]								



Plot 2:

3-Change the feedback resistor to *330K* Ω(G = 3.3), adjust ***P3*** to give *1000 r/min* with no load, measure and plot the speed response for zero load and full load.

Time[ ]								
Speed[no load]								
Speed[Full load]								



Plot 3:

Note:The tachogenerator output contains a ripple component which will be amplified in the forward path, and with high gain could saturate the power amplifier.

The ripple can be reduced by connecting a capacitor across the error amplifier output resistor, which introduces a time-constant,

Summary:

Experiment No 12

closed-loop position control [assignment 6]

Objective

When you have completed this assignment you will know:

- Introducing position feedback into a closed-loop position control can make it more stable.
- The improved stability may enable more gain to be used.
-

Equipment Required

Qty	Apparatus
1	33-110 Analogue Unit
1	33-100 Mechanical Unit
1	Power Supply ±*15V* dc, *1.5A*; *+5V* dc, *0.5A* (eg Feedback 01-100)
1	Oscilloscope, storage or long-persistence, preferably with X-Y facility. (eg Feedback 1810- 01229)

Knowledge Level

Before you start this assignment you should:

- Be familiar with the equipment.
- .

Introduction

Increasing the *gain* increases the *oscillatory tendency* in the response, which is undesirable. There are good practical reasons for using a high gain. An important one is that, due to the brushes and other factors, all practical motors have a constant friction (called coulomb friction). Also usually an increased amount of friction force (called *stiction*) has to be overcome to start the motor from rest. Therefore a minimum voltage has to be applied to the motor before rotation starts. This means that there is a minimum input below which the system will not respond; this is termed dead-band. If the gain is high the dead-band is reduced, which is advantageous, but the system may display unwanted oscillation in the response.

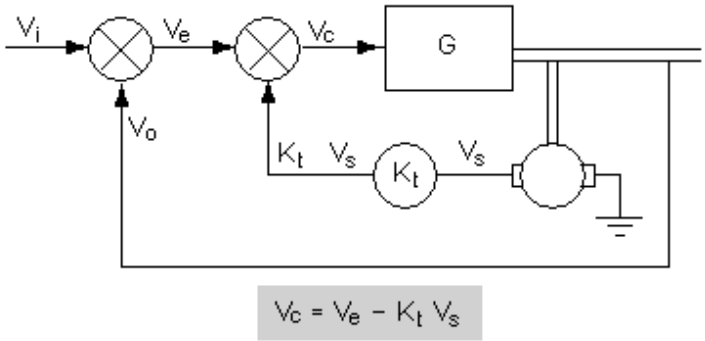


Fig 6.1a - Velocity Feedback

The form of the system response with high gain can be much improved by applying a feedback signal to the input proportional to the output shaft position. This arrangement is termed position feedback, and is illustrated in fig 6.1a and fig6.1b.

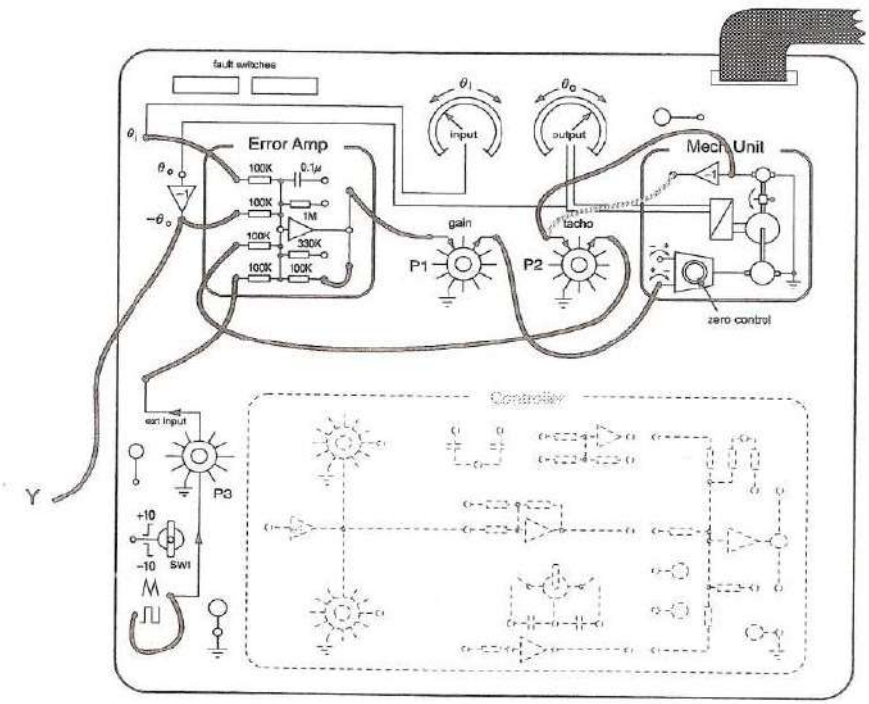


Figure 7-6-2: Connections for Practical 6.1

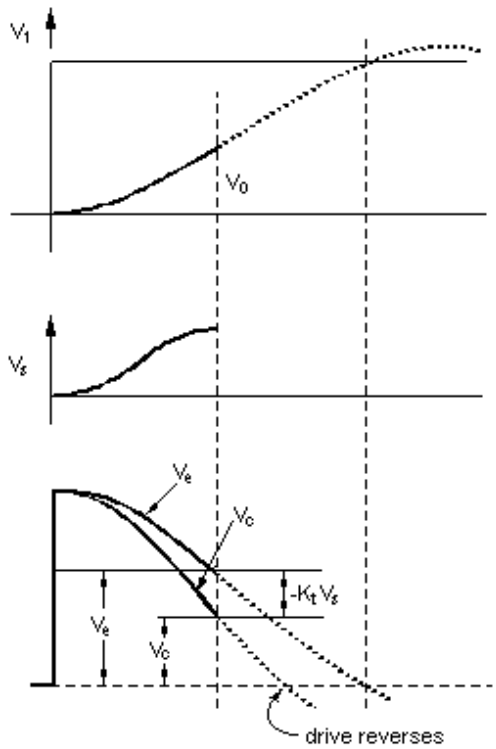


Fig 6.1b - Velocity Feedback

In the first diagram it is assumed that the input and output signals are available as voltages and a voltage V_s proportional to output shaft speed is available from a tachogenerator. A fraction of this voltage $K_t V_s$ is subtracted from V_e to give V_c, which is the control voltage applied to the power amplifier giving:

$$V_c = V_e - K_t V_s$$

If a step input is applied to the system the error will initially *equal* the input step and decrease as the motor speeds up. As the motor speed increases the velocity signal V_s increases and subtracts from V_e to give a drive voltage V_c, which is less than the error.

This is illustrated in the second diagram, and the motor drive goes to zero and reverses before the error goes to zero. This means that the motor begins to slow up before the initial alignment, greatly reducing or even preventing any overshoot.

Simple Position Feedback

1]Connect the system as in Fig.7. 6.2, initially omitting the connection from ***P₂*** to the tachogenerator and set ***P₁*** to *zero*.Connect the oscilloscope for an Y display as indicated. Set the error amplifier feedback resistor to *100K*, giving G = 1, and set P_3 to give a V_i of *±5V* at *0.1Hz*. Turn ***P₁*** to *100* and a slightly oscillatory response should be obtained. Set p3 to zero and set θi to 45

Measure and plot the position vs. time .

Time[ ]								
Position[no load]								
Position[Full load]								



Plot !:

2]Connect ***P2*** to one tachogenerator output and increase its setting; if the overshoot decreases the velocity feedback polarity is correct. If the overshoot increases connect to the other socket.

When adjusting the tachogenerator feedback a dead beat response will be obtained when the system aligns in the least possible time, but with no overshoot as in fig 6.3. Additional velocity feedback will cause an over-damped response, in which the system slowly moves into alignment.

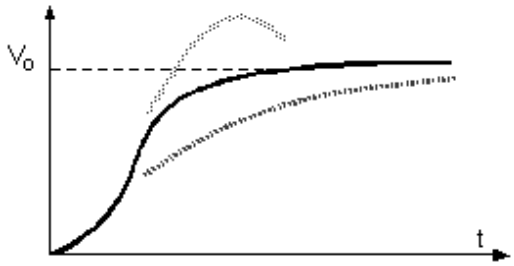


Fig 6.3 - Dead-beat Response

Increase ***P_2*** until a deadbeat response is obtained, this occurs when the motor just does not reverse. Measure and plot the position vs. time for θi -= 45

Time[ ]								
Position[no load]								
Position[Full load]								



Plot 2:

3]Increase the feedback resistors to 200*K*, *G* = *2*; and repeat step 2.

Time[ ]								
Position[no load]								
Position[Full load]								



Plot 3:

Summary:

Experiment No 13

Introduction to 3-Term Control

Objective

When you have completed this assignment you will know that:

- A very versatile control signal can be obtained by combining components depending on the error, the derivative of the error, and on the integral of the error.
- With capacitor input, an operational amplifier acts as a differentiator.
- The input must be modified, giving partial differentiation, for practical reasons.
- With a capacitor in the feedback path, an operational amplifier can act as an integrator.

Equipment Required

Qty	Apparatus
1	33-110 Analogue Unit
1	33-100 Mechanical Unit
1	Power Supply ±*15V* dc, *1.5A*; *+5V* dc, *0.5A* (eg Feedback 01-100)
1	Oscilloscope, storage or long persistence, preferably with X-Y facility (eg Feedback 1810-01229)

Knowledge Level

Before you start this assignment you should:

- Understand the basic working of an operational amplifier, including the concept of 'virtual earth'.
- Preferably have completed the Operational Amplifier Characteristics assignment.

Differentiation by Operational Amplifier

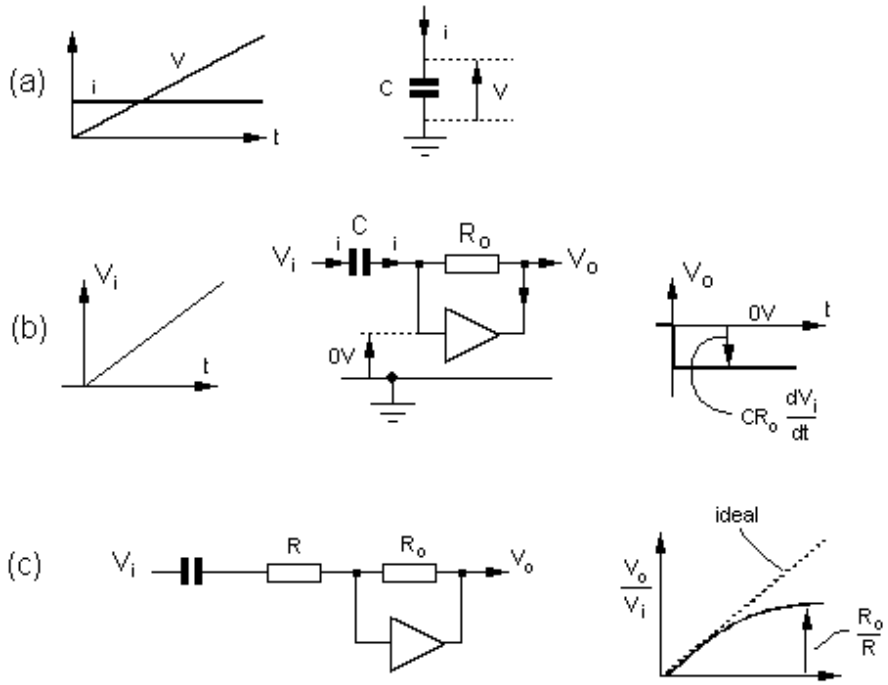


Fig 10.4 - Differentiation by Operational Amplifier

Suppose that a ramp voltage is applied to a capacitor as in fig 10.4(a). Since the voltage across the capacitor rises steadily a constant current must flow into the capacitor. The current is proportional to the capacitance C, being given by the relation

$$i = C\frac{dv}{dt}$$

If the capacitor is used as the input element for an operational amplifier as in (b), the amplifier input will be a virtual earth point (see the Operational Amplifier Characteristics assignment and fig 2.3) and the amplifier output will be given by

$$V_o = iR_o$$

and since $$i = C\frac{dv}{dt}$$

finally $$V_o = -CR_o\frac{dv}{dt},$$

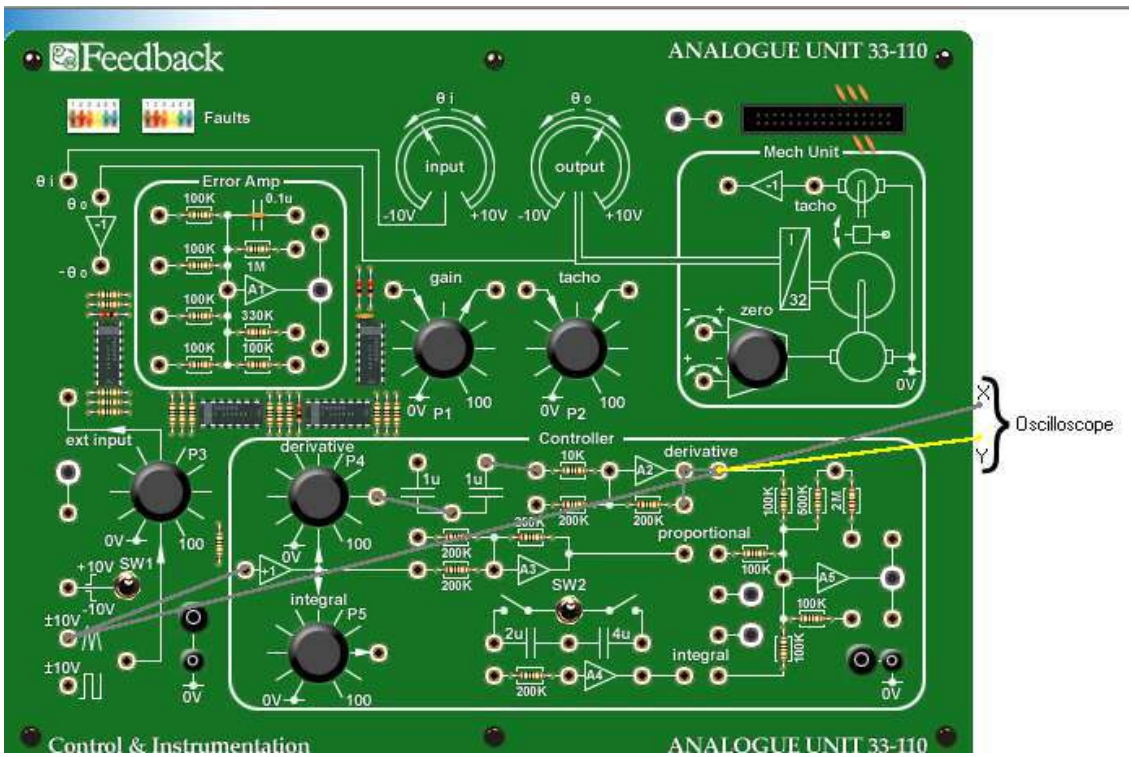
giving the constant output voltage of (b). This indicates the derivative of the input with a scaling factor $-CR_o$.

Although this circuit in principle measures the *derivative*, there is a limitation in practical application. If the input signal contains noise or disturbance components which are small but rapidly changing, these may cause currents in the capacitor comparable to those of slower changing signals for which the derivative is required. These unwanted

components in the input are emphasised and may even saturate the amplifier or some later stage in the amplifying system.

The effect of unwanted rapidly changing high frequency components can be limited by a resistor in series with the capacitor as in fig 10.4(c). If the input is changing slowly the input current is largely determined by the capacitor, but if the input is changing fast the current is limited by the resistor giving an overall gain of R_o/R.

In frequency response terms the gain of the ideal differentiator in (c) rises continuously with increasing frequency, noise corresponding with high frequency components. The introduction of an input resistor, called a limited derivative, gives a gain initially rising with frequency, representing correct derivative action, but finally becoming constant preventing emphasis of high frequency components.



[patching diagram](#) a

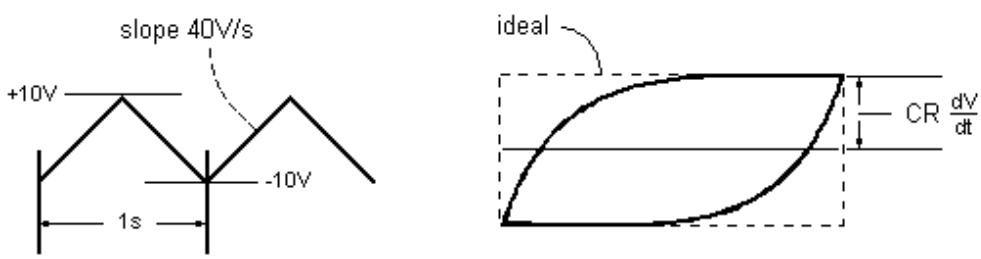


Fig 10.5(b) - X-Y Display

The upper amplifier in the Controller is intended for use as a differentiator. The two *1µF* capacitors can be used in series to give *0.5µF*, singly to give *1µF* or in parallel to give *2µF*.

Arrange the circuit as displayed within the patching diagram a, where the upper amplifier is connected as a limited differentiator with *1µF* capacitor and adjustable input from the triangle test waveform.

Set the test frequency to about *1Hz*, and turn up ***P4*** to *100*.Measure and plot the output of the controller

Time[ ]								
Vo{R0=100}								
Vo{R0=200}								



Plot 1:

Intergration by Operational Amplifier

Conversely to fig 10.4(a), the voltage across a capacitor is given by the integral of the current through the capacitor:

$$v = \frac{1}{C}\int i\,dt$$

This is illustrated in fig 10.6(a). A constant current gives a steady increase of voltage, the voltage representing the area under the current /time plot.

The illustrated current pulse waveform, comprising a positive pulse, a negative pulse and then zero current, gives a final voltage value, which is the overall time integral of the current.

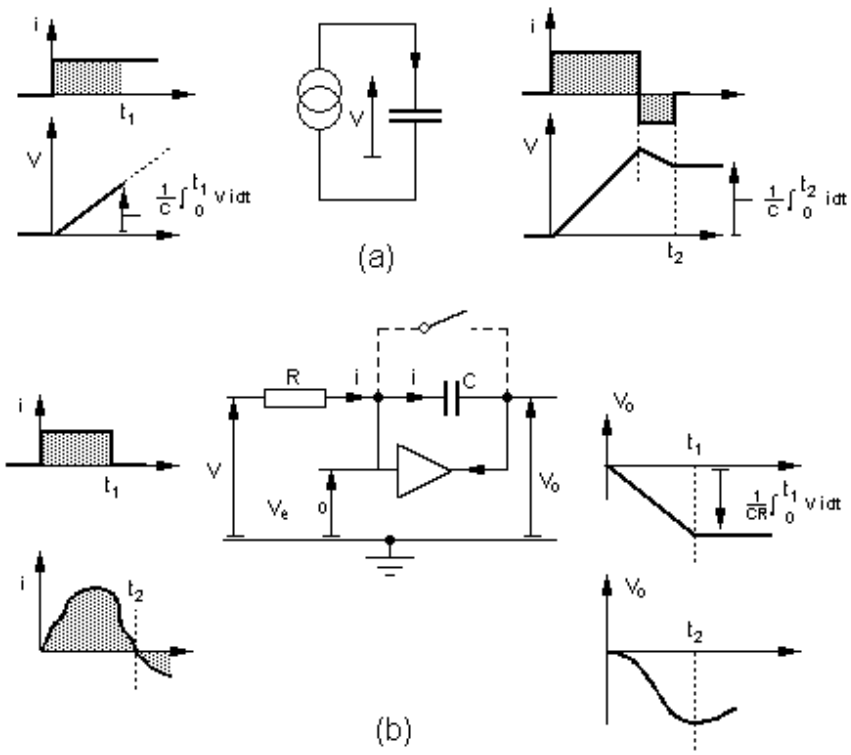


Fig 10.6 - Operational Amplifier as Integrator

Integration can be obtained by an operational amplifier if the feedback resistor is replaced by a capacitor as in fig 10.6(b). If a voltage V is applied an input current i will flow through R and the amplifier output will change to hold the amplifier input point (virtual earth point) substantially at zero. This means that the amplifier output will steadily increase to maintain the current i through the capacitor. The input current is

$$i = \frac{1}{C}.$$

The voltage across the capacitor will be

$$V_c = \frac{V}{R}\int V\,dt$$

giving the amplifier output as

$$V_o = \frac{1}{CR}\int V\,dt$$

which is the negative scaled integral of the input.

If the applied voltage *V* becomes *zero*, the current *i* is also *zero* and the integrator holds indefinitely whatever output has been obtained, since the amplifier does not ideally draw any current at the virtual earth point. If the input is a general waveform the output is correspondingly scaled integral.

It is often required to set the output of an integrator to *zero* before the start of integration, and this can be arranged by a switch, mechanical or electronic, connected across the capacitor, as shown dotted, which discharges the capacitor. Integration does not start until the switch is open, irrespective of a possible input signal.

An ideal integrator will hold an accumulated signal indefinitely, however an operational amplifier may draw a very small current at the virtual earth input, which will cause the output to drift very slowly. The drift may or may not be important and depends on the amplifier used.

Operational Amplifier Intergrator

The lower amplifier in the Controller is intended for use as an integrator. The capacitors are *2µF* and *4µF* and the switch short-circuits both capacitors when set towards the amplifier.

Arrange the integrator amplifier with *2µF* as in the patching diagram b. Set ***SW1*** down to *-10* and set ***P_3*** to supply about *-3V* to the controller input. ***P_5*** can then be used to obtain a smaller voltage for the integrator input resistor.

Short circuit the integrator capacitor by the switch and adjust the integrator input voltage to *-1V*.

Connect the voltmeter to the amplifier output and open the capacitor switch. The output should move positively and can be reset to *zero* by closing the switch.

If the input voltage is *1V* and the input resistor is *200k* Ω, then the input current *i* is

$$\frac{1V}{200k\Omega} = 5\mu A$$

If the capacitor is *1μF*, *5μA* gives 5 volts per second

2μF, 5μA gives 2.5 volts per second

hence the output should take about four seconds to move from *0V* to *+10V*.

The *+10*, *0*, *-10* switch can be used to make the integrator operate in both directions or **hold** with the switch at *0V*.

Measure and plot the out put of the controller

Time[ ]								
Vo{C=2*μF* }								
Vo{ C=4*μF* }								



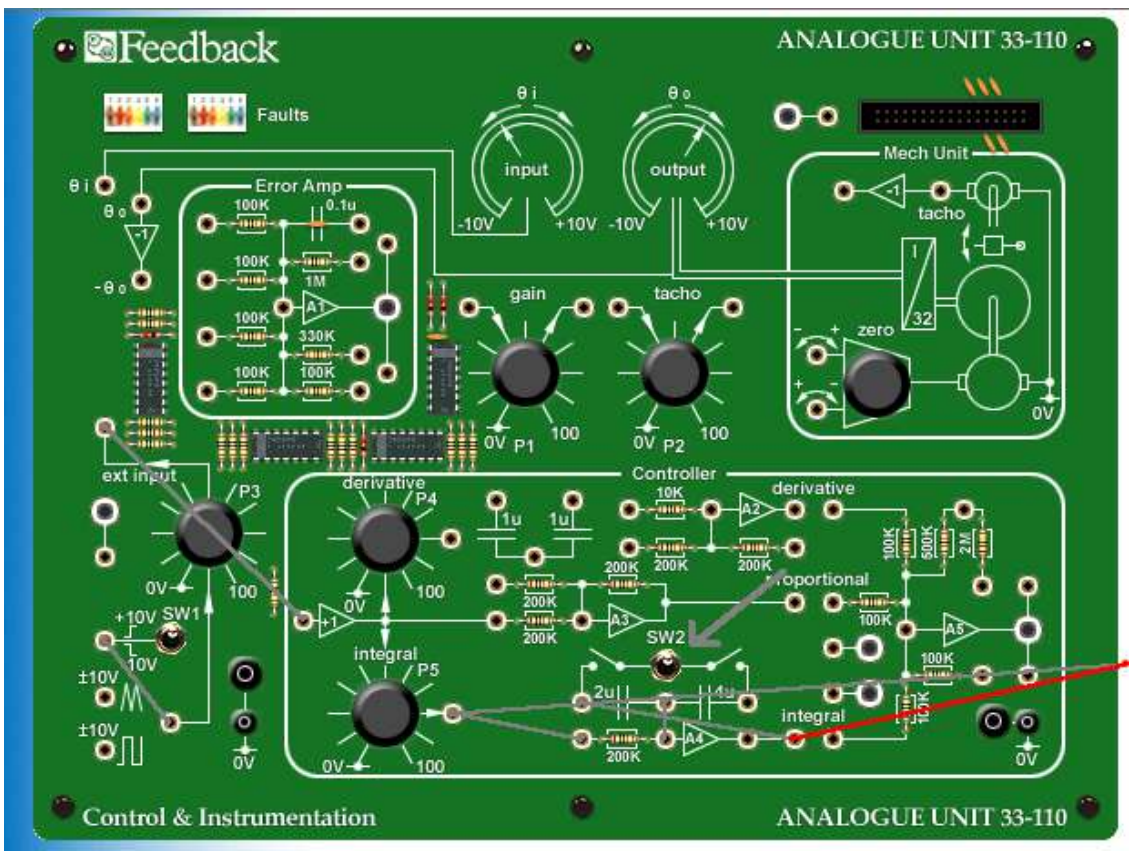
Plot 2:

Disconnect the input from ***P5***. Now any current drawn at the virtual earth point will pass through the capacitor and be integrated.

Discharge the capacitor by the switch and then open the switch and estimate the time required for the output to reach *10V* - it may be some minutes.

The input current can then be determined, since *1µA* gives

0.5 volt/second with *2µF*. The value should be much less than *1µA*.



patching diagram b

Three-Term Controller Realisation

The operational amplifier circuits providing integration and differentiation have been investigated and can be combined together with a proportional component to give the 3-Term Controller of fig 10.8(a).

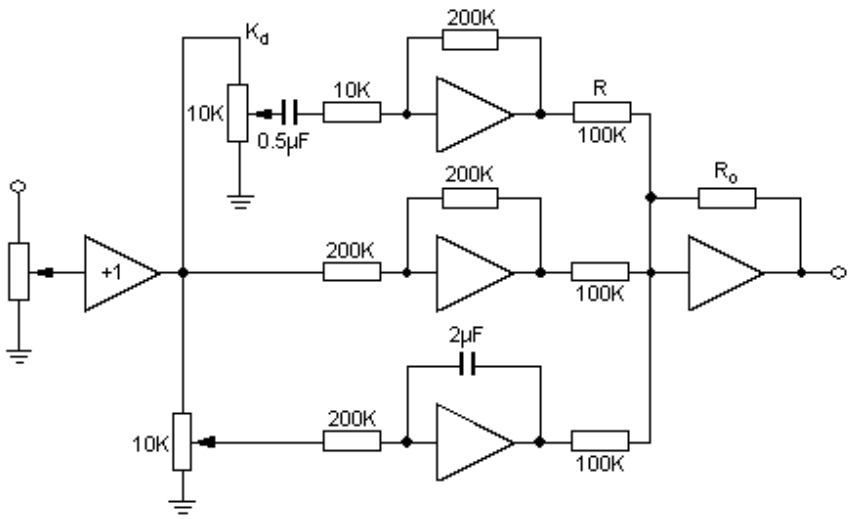


Fig 10.8(a) - Operational Amplifier Realisation of 3-Term Controller

The component values have been chosen so that the circuit can be plugged up on the controller section of the analogue unit as in the patching diagram. Individual adjustment of K_d(P4) and K_i(P5) is given by the two potentiometers and a fixed overall gain of K is obtained in the output summer by R_o/R.

A very important point in the operational amplifier realisation is that both the integrator and differentiator have a negative sign, hence the proportional component **must** also have a negative sign conveniently provided by an inverting unit in the proportional path. If the individual components are combined in a summer, which introduces an additional sign reversal, there is then no overall sign reversal.

The input is passed through a +1 buffer unit, which provides a high input resistance so that the circuit can be operated from another potentiometer to provide an adjustment on the overall gain K.

Practical 10.3 3-Term Controller Test

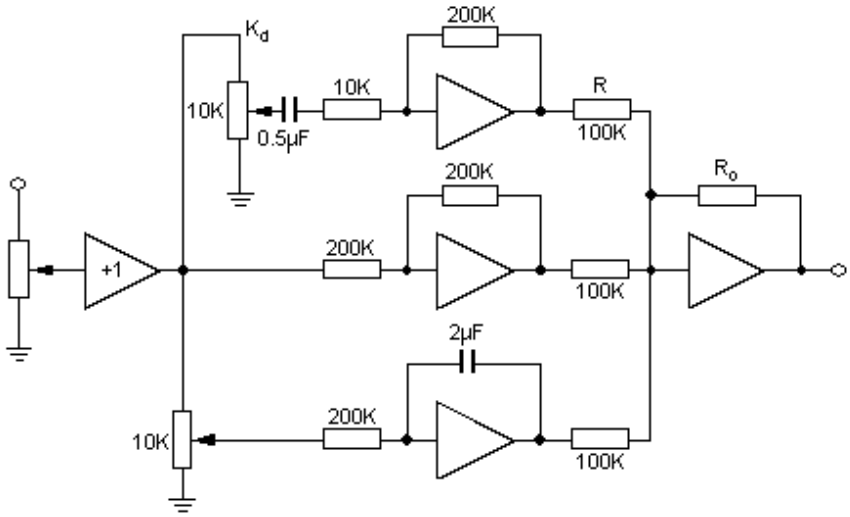


Fig 10.8(a) - Operational Amplifier Realisation of 3-Term Controller

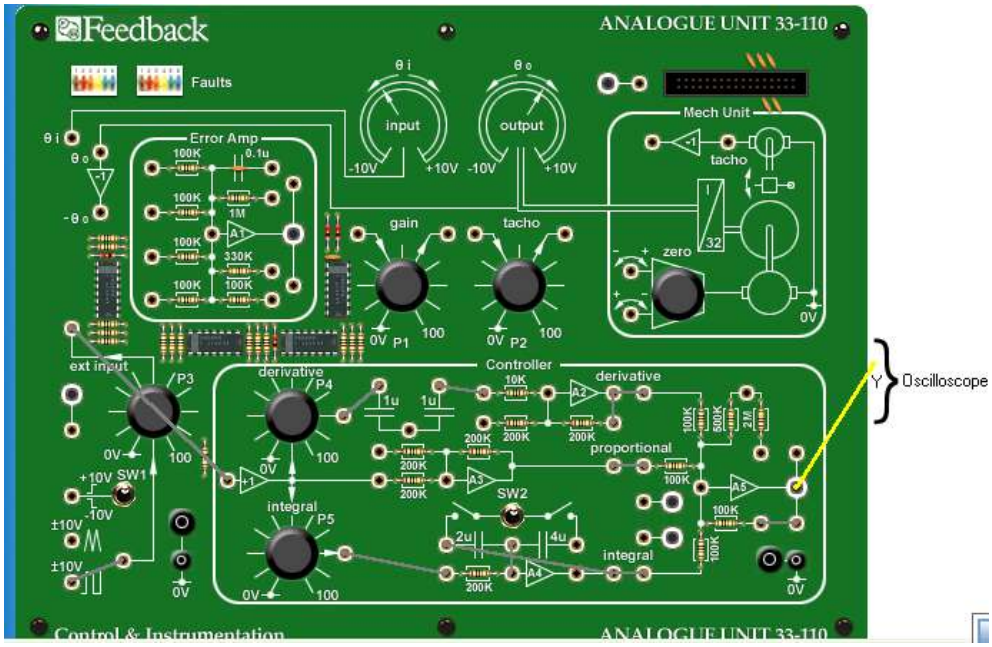
Connect up the circuit as shown in the patching diagram C with R_o *100K* Ω, and apply a square wave ±*5V* at about *1Hz*. Set ***P4*** and ***P5*** to *zero*.

The proportional path has an effective gain of +1 and hence should provide a direct square wave output of ±*5V*.

Unplug the proportional component from the summer and set ***P4*** to about *20*. The output component will be a pulse as shown in fig 10.9. The ideal derivative of a square wave is infinite and of zero duration when the wave changes sign and zero when the wave is constant. A practical differentiator gives a finite pulse as shown.

Set up ***P4*** to *zero* and ***P5*** to *100*. Discharge the capacitor by ***SW2***. When ***SW2*** is opened the integrator output will be a triangular waveform of about *6V* total swing.

Finally combine all components together to give a waveform generally as in fig 10.9.



patching diagram C

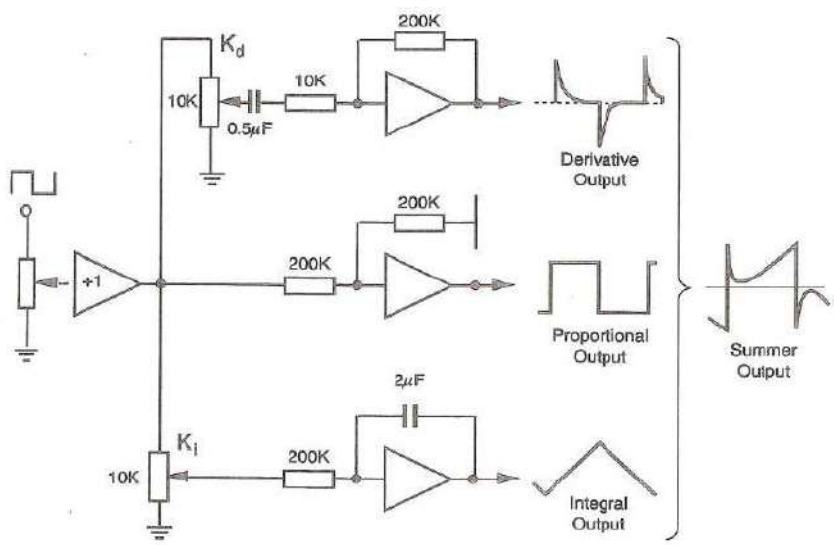


Figure 7-10.9: P I D Output Waveforms
(sign reversal omitted for components)

Summary

Experiment no 14
Applications of 3-Term Control

Objective When you have completed this assignment you will know that:

- Proportional with Derivative control improves transient response
- Proportional with Integral control eliminates steady errors but may give a slow response
- The combination of proportional, derivative and integral gives the best response.

Equipment Required

Qty Apparatus

1 33-110 Analogue Unit

1 33-100 Mechanical Unit

1 Power Supply ±*15V* dc, *1.5A*;
+5V dc, *0.5A*
(eg Feedback 01-100)

1 Oscilloscope, storage or long persistence, preferably with X-Y facility
(eg Feedback 1810-01229)

Knowledge Level

Before you start this assignment you should:

- Be familiar with 3-Term control principles, preferably from completing the Introduction to 3-term Control assignment.

Practical 11.4 Elimination of Following Error

A circuit to investigate following error is given in fig 11.3(a) with the connection diagram as illustrated given in Fig.11.3b.

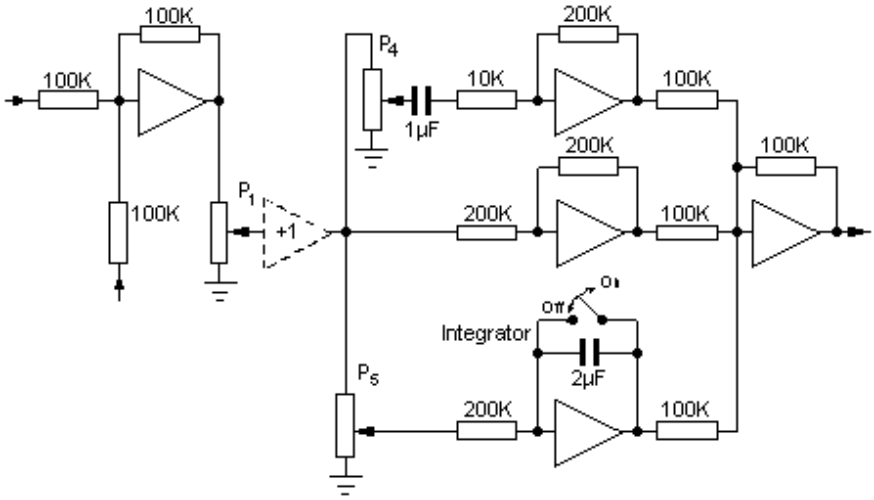


Fig 11.3.a - Circuit diagram

With the integrator switched *Off* (SW2 down), ***P4*** set to *zero*, the 3-term summing amplifier feedback resistor set to *100K*, and ***P1*** to *100*, the system has a forward gain *G = 1*. Apply a triangular input of about ±*5V* at *0.1Hz* from ***P3***.

Make an X-Y display of the error, which should give a constant steady value.

Set the integrator input ***P5*** to about *40* and open the integrator short circuit switch.

The error should be "integrated out", though the response may be slow.

Experiment with ***P5*** adjustment. Note that increasing ***P5*** eliminates the initial error more quickly but gives a more oscillatory response.

These tests have shown the general property of integral control. An improved transient performance can be obtained by introducing derivative control and increasing the forward path gain.

Change the feedback resistor in the 3-term summing amplifier to *500K*, this gives *G = 5*, but the direct error is still available at the error operational amplifier output.

With triangle input as before, adjust ***P4*** and ***P5*** by trial and error to a possible 'best' transient: perhaps just no overshoot. It is useful to increase the input to *0.2Hz*.

Display the integrator output to show the output reaching the value required to supply the motor drive.

The adjustment of integral and derivative controls that gives the 'best' transient for following does not give the best step response transient, and some acceptable compromise may be necessary.

Compare the error transients with triangle and step inputs and note that different adjustments ***of P4*** and ***P5*** give a slightly better response for each input.

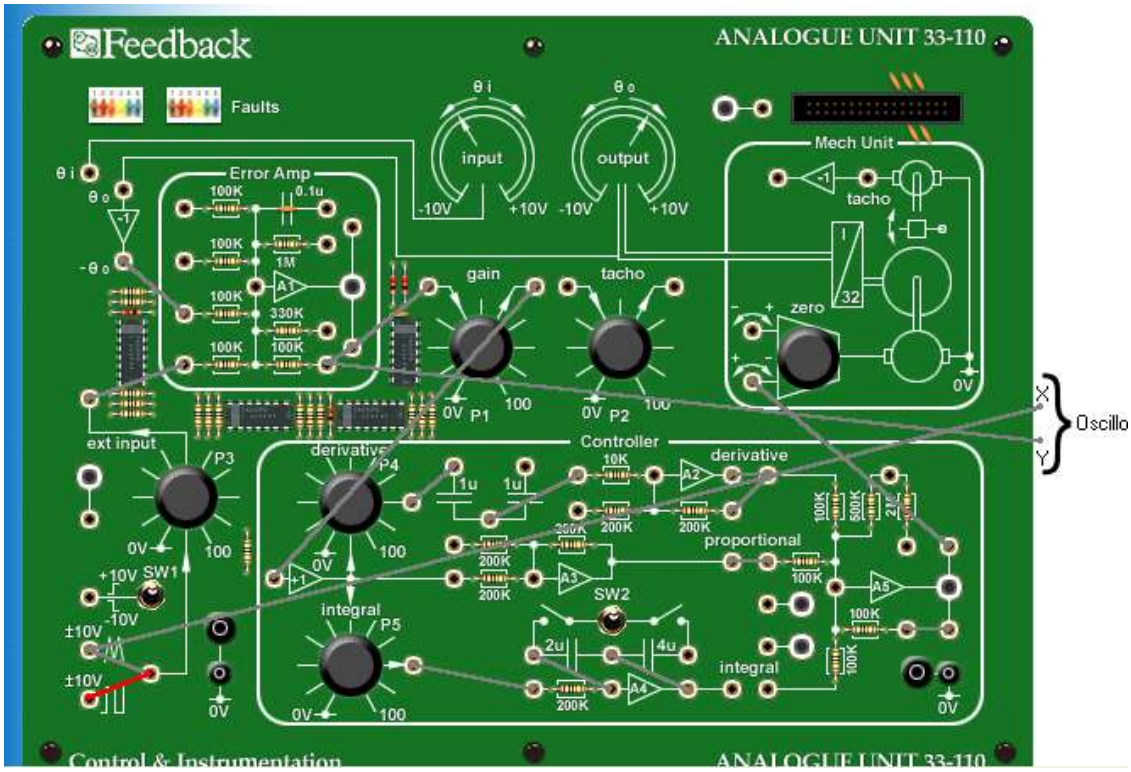


Fig.11.3b.

Measure and plot the time response

Time[ ]								
Speed[ ] Gc(s)=1								
Speed[ ] Gc(s)=5								



Plot 1:

Practical 11.5 Elimination of Disturbance

Using the same procedure as in practical 11.4 , measure and plot the output as in the following table

Time[ ]								
Speed[ ] T_I=0								
Speed[ ] T_I=max								



Plot 2:

Summary: